Explore more resources
Altera® Design Hub

Altera® FPGA Software Installation and Licensing

Updated for Quartus® Prime Design Suite: **25.3**

Answers to Top FAQs:

Q ***Where can I find hardware and disk requirements?***
A Determining Hardware Requirements on page 17

Q ***How to download Altera FPGA software?***
A Downloading Altera FPGA Software on page 59

Q ***Where can I find device support files?***
A Downloading Device Support on page 15

Q ***How do I get Questa* Altera FPGA license?***
A Questa Altera FPGA Edition License on page 95

Q ***How to get Altera FPGA software or IP license?***
A Requesting a License from the SSLC on page 56

Q ***How to manage Altera FPGA software licenses?***
A Using the Altera FPGA SSLC on page 6

Q ***How to troubleshoot software license files?***
A Altera FPGA Software License Troubleshooter

Q ***How to get Installation and Licensing support?***
A Altera FPGA Licensing Support Center

Online Version
Send Feedback

683472
2026.02.27

Contents

1. Answers to Top FAQs..... 5

2. Introduction to Altera FPGA Software Installation and Licensing..... 6

2.1. Quartus Prime Design Suite Overview.....6
2.2. Altera FPGA Software Download Center..... 8
2.3. Altera FPGA Self-Service Licensing Center.....8
2.4. Before You Begin.....8
2.4.1. Navigate Content Through Tasks.....8
2.4.2. Prerequisites.....9
2.4.3. Acronyms..... 9

3. Determining System Requirements.....11

3.1. Reviewing the Quartus Prime Software Release Notes..... 11
3.2. Determining Hardware Requirements.....11
3.2.1. Disk Space and Memory Requirements.....12
3.2.2. Cable and Port Requirements.....12
3.3. Determining Software Requirements.....13
3.3.1. Operating System Requirements..... 13
3.3.2. Questa* Altera FPGA Edition Software Requirements 13
3.3.3. Other Altera FPGA Software Requirements.....14
3.3.4. Third-Party Software Requirements..... 14

4. Downloading Altera FPGA Software..... 15

4.1. Software Available in the FPGA Software Download Center..... 15
4.2. Downloading Design-Specific Software Components 16
4.3. Determining the Download Type.....17
4.3.1. Downloading Software Using the Quartus Prime Installer.....17
4.3.2. Downloading Software Packages Manually..... 25
4.4. Downloading Other Altera FPGA Software.....27
4.5. Downloading Docker Containers for Quartus Prime FPGA Design Software..... 28

5. Installing Altera FPGA Software.....29

5.1. Installing Altera FPGA Software Through Quartus Prime Installer.....29
5.2. Installing the Altera FPGA Software Manually.....30
5.2.1. Selecting the Installation Path..... 30
5.2.2. Installing Altera FPGA Software at the Command Prompt.....31
5.2.3. Installing from .tar Files.....32
5.2.4. Installing Device Support to Existing Installation..... 33
5.2.5. Installing Programming Cable Drivers..... 33
5.2.6. Installing and Configuring a JTAG Server.....34
5.3. Using the Same Installation Files on Multiple Systems.....36
5.4. Managing Multiple Versions and Copies of Altera FPGA Software..... 36
5.5. Setting Quartus Prime Environment Variables..... 36
5.6. Starting the Quartus Prime Software..... 37
5.7. Updating Altera FPGA Software.....38
5.8. Installing and Uninstalling a Software Patch.....39
5.9. Uninstalling Altera FPGA Software..... 40
5.9.1. Uninstalling on Windows*..... 40

5.9.2. Uninstalling on Linux 40
5.10. Troubleshooting Installation Issues 40

6. Licensing Altera FPGA Software 42
6.1. Summary of Altera FPGA Software Licenses Required 42
6.1.1. Quartus Prime Software License 45
6.1.2. Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License 45
6.1.3. Intellectual Property (IP) Cores Licenses 49
6.1.4. Siemens EDA* AXI Verification IP Suite License (Altera FPGA Edition) 49
6.2. Evaluating the Quartus Prime Software 49
6.3. Acquiring Free, No-Cost Licenses 51
6.3.1. Refreshing License Server Token Automatically 56
6.4. Licensing Altera FPGA Software Walkthrough 58
6.4.1. Obtaining Necessary Hardware Information 58
6.4.2. Requesting a License File from the Altera FPGA Self-Service Licensing Center 59
6.4.3. Setting up a Fixed License 60
6.4.4. Setting up a License in a Network License Server 61
6.5. Getting Hardware Information for License 64
6.5.1. Identifying Host's NIC ID 64
6.5.2. Identifying the UNIX Host ID 67
6.5.3. Locating Your Hard-Disk Serial Number 67
6.5.4. Identifying the USB Software Guard ID 67
6.6. Using the Altera FPGA Self-Service Licensing Center 67
6.6.1. Using the License Assistant 68
6.6.2. Getting a License File with Your New Purchase 69
6.6.3. Viewing Licenses 70
6.6.4. Filtering Your Licenses 71
6.6.5. Viewing and Generating a Legacy License 72
6.6.6. Viewing Licenses or License Files Associated with a Computer 73
6.6.7. Regenerating a License File 74
6.6.8. Generating a Temporary Checkout License 75
6.6.9. Generating a Companion License 76
6.6.10. Renewing Your License 78
6.6.11. Managing a Computer Profile 79
6.6.12. Rehosting a License on a Different Computer 80
6.6.13. Rehosting All Licenses from Current Computer to a Different Computer 81
6.6.14. Rehosting Multiple/Partial Licenses from Current Computer to a Different Computer 82
6.6.15. Sharing a License with Another User 83
6.6.16. Adding Delegate Administrators to Your Licenses and Computers 84
6.6.17. Splitting Seats on Your License and Generating Licenses 84
6.6.18. Merging or Adding Seats on Your License and Generating License 85
6.6.19. Adding Floating Seats 86
6.6.20. Signing Up for an Evaluation or No-Cost License 86
6.6.21. Signing Up For an Employee License 87
6.7. About Altera FPGA Software License Files 88
6.7.1. License File Components 88
6.7.2. Using a FLEXlm Options File 91
6.7.3. License.dat Example Files 92
6.7.4. Syntax of license.dat License File 93

6.8. Checking the IP License Status..... 94
6.9. Viewing IP Core License Data..... 95
6.10. Configuring the License Manager Server.....96
6.10.1. Installing the FLEXlm License Manager Server Software on Another License Server.....96
6.10.2. Upgrading the FLEXlm License Manager Server Software..... 97
6.10.3. (Windows Only) Starting and Stopping the License Server..... 98
6.10.4. (Windows Only) Starting the License Server Automatically.....98
6.10.5. Rereading an Existing License File on a License Server..... 99
6.11. Troubleshooting License File Issues.....99

7. Next Steps After Installing and Licensing the Software..... 101
7.1. Additional Resources..... 101
7.2. Training..... 102

A. Altera FPGA Software Installation and Licensing Archives.....104

B. Document Revision History for Altera FPGA Software Installation and Licensing.....105

1. Answers to Top FAQs

Q Where can I find hardware and disk requirements?	A Determining Hardware Requirements on page 17
Q How to download Altera FPGA software?	A Downloading Altera FPGA Software on page 59
Q Where can I find device support files?	A Downloading Device Support on page 15
Q How do I get Questa* Altera FPGA license?	A Questa Altera FPGA Edition License on page 95
Q How to get Altera FPGA software or IP license?	A Requesting a License from the SSLC on page 56
Q How to manage Altera FPGA software licenses?	A Using the Altera FPGA SSLC on page 6
Q How to troubleshoot software license files?	A Altera FPGA Software License Troubleshooter
Q How to get Installation and Licensing support?	A Altera FPGA Licensing Support Center

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

2. Introduction to Altera FPGA Software Installation and Licensing

This manual provides comprehensive information for downloading, installing, and licensing Altera FPGA software.

If you are unsure what an FPGA is, review FPGA Basics to help you get started.

2.1. Quartus Prime Design Suite Overview

Quartus Prime Software Editions

The Quartus Prime software is available in three editions based on your design requirements:

Table 1. Quartus Prime Software Editions

	Pro Edition	Standard Edition	Lite Edition
	Optimized to support the advanced features in Altera FPGAs and SoCs with the following device families: • Agilex™ 3 • Agilex 5 • Agilex 7 • Stratix® 10 • Arria® 10 • Cyclone® 10 GX	Includes extensive support for earlier device families in addition to the following device families: • Cyclone 10 LP • MAX® 10	An ideal entry point to Altera's high-volume device families and is available as a free download with no license file required.

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

Supported Features

The following is the Quartus Prime feature support matrix:

Figure 1. Quartus Prime Feature Support Matrix

Software Features	Quartus® Prime Pro Edition	Quartus® Prime Standard Edition
Agilex™ 3 Device Support	✔	
Agilex™ 5 Device Support	✔	
Agilex™ 7 Device Support	✔	
Stratix® 10 Device Support	✔	
New Hybrid Placer & Global Router	✔	✔
New Timing Analyzer	✔	✔
New Physical Synthesis	✔	✔
Platform Designer (formerly Qsys)	✔	✔
Partial Reconfiguration	✔	
Block-Based (Hierarchical) Design Flows	✔	
Interface Planner (formerly BluePrint)	✔	
Incremental Fitter Optimization	✔	

Supported Altera FPGA Developmental Tools

The Quartus Prime software suite supports the following Altera FPGA development tools:

- **Questa* Altera FPGA Edition**: Simulates FPGA designs using Altera-specific simulation libraries. It includes all features of Siemens EDA Questa* Core, including behavioral simulation, HDL test benches, and Tcl scripting.
- **Advanced Link Analyzer**: Analyzes jitter/noise and evaluates high-speed serial link performance. It is an ideal predesign tool supporting Altera FPGA IBIS-AMI standards and enhanced models to help you understand how Altera FPGA solutions can fit your system requirements.
- **Altera SoC FPGA Embedded Development Suite (SoC EDS)**: A comprehensive tool suite for embedded software development on Altera SoC FPGAs.
- **Ashling* RiscFree* IDE for Altera**: Integrated development environment for creating embedded applications on the RISC-V-based Nios® V soft processors and the Arm*-based hard processor system.
- **DSP Builder**: Supports a model-based design flow from algorithms to hardware in a common environment.

- **oneAPI Base Toolkit**: Enables you to target FPGAs for heterogeneous acceleration and simulate entire system flows by abstracting some parts of the hardware.
- **Simics Simulator for Altera FPGAs**: A full-system simulator that supports defining, developing, and deploying virtual platforms.
- **FPGA AI Suite**: Provides several components to help in enabling Artificial Intelligence (AI) and creating optimized FPGA AI platforms efficiently.
- **Power Thermal Analyzer**: Estimates your design's power consumption and provides thermal design parameters for Altera FPGA devices, such as Agilex 3, Agilex 5, Agilex 7, and Stratix 10.
- **Quartus Prime Pro Embedded Edition**: provides the Quartus Prime Pro Edition Programmer, Platform Designer, the Nios V processor, and IP board files to support your embedded development.
- **Quartus Prime driver installer**: A lightweight installer that installs the drivers only.

2.2. Altera FPGA Software Download Center

The Altera FPGA Software Download Center contains resources to download Altera FPGA software and IP cores.

Related Information

- Downloading Altera FPGA Software on page 15
- Installing Altera FPGA Software on page 29
- Altera FPGA Software Download Center

2.3. Altera FPGA Self-Service Licensing Center

The Altera FPGA Self-Service Licensing Center provides support for licensing Altera FPGA software. For detailed instructions about obtaining and managing licenses in the Altera FPGA Self-Service Licensing Center, refer to Using the Altera FPGA Self-Service Licensing Center on page 67.

Related Information

- Licensing Altera FPGA Software on page 42
- Summary of Altera FPGA Software Licenses Required on page 42
- Altera FPGA Self-Service Licensing Center (SSLC)

2.4. Before You Begin

Before you begin installing the Altera FPGA software, review the following topics to help you navigate and understand this guide:

2.4.1. Navigate Content Through Tasks

Use the following table to navigate this guide through user-tasks:

Table 2. Navigate Content Through Tasks

A	B	C	D	E
Determine Hardware and Software Requirements	Download Altera FPGA Software through the installer or manually	Install Altera FPGA Software	• Evaluate the Quartus Prime Software • Obtain License from SSLC	Review Next Steps

2.4.2. Prerequisites

Using the Quartus Prime software to create a basic FPGA design requires the following prerequisite knowledge:

Knowledge

- Basic knowledge of digital logic design.
- Basic knowledge of how to describe a hardware design using VHDL, Verilog HDL, SystemVerilog, or EDA schematic tools.

Training

If you are new to FPGA or the Quartus Prime software, you can review the following training modules:

- Read Me First!
- Basics of Programmable Logic: History of Digital Logic Design
- Basics of Programmable Logic: FPGA Architecture
- Beginner Workshop for Altera FPGAs
- University Self-Guided Lab: Introduction to FPGAs and the Quartus Prime Software

2.4.3. Acronyms

This document uses the following acronyms throughout:

Table 3. Acronyms Used in this Document

Acronym	Meaning
AI	Artificial Intelligence
CLI	Command-line Interface
CPLD	Complex Programmable Logic Device
DSP	Digital Signal Processing
EDS	Embedded Design Suite
EULA	End-User License Agreement
FPGA	Field Programmable Gate Arrays

continued...

Acronym	Meaning
GUI	Graphical User Interface
IP	Intellectual Property
RAM	Random Access Memory
RHEL	Red Hat Enterprise Linux
SDK	Software Development Kit
SoC	System-on-a-Chip
SSLC	Self-service Software Licensing Center
SLES	SUSE Linux Enterprise Server
WSL	Windows* Subsystem for Linux*

3. Determining System Requirements

Your system must meet minimum hardware and software requirements to download, install, and run Altera FPGA software. Refer to the following sections for further details:

Navigate Content Through Tasks

Use the following table to navigate this guide through user-tasks:

A	B	C	D	E
Determine Hardware and Software Requirements	Download Altera FPGA Software through the installer or manually	Install Altera FPGA Software	• Evaluate the Quartus Prime Software • Obtain License from SSLC	Review Next Steps

3.1. Reviewing the Quartus Prime Software Release Notes

Verify the Quartus Prime software edition and release version you are about to install supports the platform where you intend to use it. Review relevant Quartus Prime software release notes for new features, enhancements, changes to software behavior, device support status, software issues, and software patches:

- Quartus Prime Pro Edition Software and Device Support Release Notes
- Quartus Prime Standard Edition Software and Device Support Release Notes

3.2. Determining Hardware Requirements

Before installing the Quartus Prime software, ensure your hardware conforms to these requirements.

The Quartus Prime software requires:

- A Windows* PC or Linux workstation.
- A minimum CPU of 64-bit Intel® Nehalem (2008) or an AMD Bulldozer (2011) microarchitecture processor with SSE4.2 instruction set or later.
- A monitor capable of at least 1024 x 768 display resolution.

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

3.2.1. Disk Space and Memory Requirements

Disk Space Requirements

If you are using the Quartus Prime Installer, it displays the disk space needed to successfully install the software.

If you choose to download and install the software manually, your system must have at least 80 GB of free disk space to hold both zipped tar files and uncompressed installation files. The disk space may be significantly more based on the device families included in the installation. After successful installation, delete the downloaded zipped files and uncompressed zip files to release the disk space.

For disk space requirements of individual software components or FPGA IP cores, follow these instructions:

1. Visit the FPGA Software Download Center.
2. Select the desired software product from the collection.
3. Within the selected software product landing page, refer to the **Detailed Description ➤ System Requirements** section.

Memory Requirements

For memory recommendations, refer to the relevant Quartus Prime release notes listed in the following:

- Quartus Prime Pro Edition Version Software and Device Support Release Notes
- Quartus Prime Standard Edition Version Software and Device Support Release Notes

Altera recommends that your system be configured to provide virtual memory equal in size or larger than the recommended physical RAM size that is required to process your design. This additional virtual memory effectively doubles the total effective memory available to process your design.

Important: Peak virtual memory might exceed these recommendations. These recommendations are based on the amount of physical memory required to achieve runtime within 10% of that achieved on hardware with an infinite amount of RAM.

3.2.2. Cable and Port Requirements

Using an Altera FPGA download cable or programming unit to program devices with the Quartus Prime software requires the following minimum hardware:

- USB port for connecting a supported Altera FPGA download cable.

  For information about the FPGA Download Cable II, refer to the *Altera FPGA Download Cable II (formerly USB Blaster II) User Guide*.
- 10/100 Mb Ethernet connection for connecting a supported Altera FPGA Ethernet download cable..

  For more information, refer to the *EthernetBlaster II Communications Cable User Guide*.

Related Information

Installing Programming Cable Drivers on page 33

3.3. Determining Software Requirements

Consider the following requirements when installing the Altera FPGA software:

3.3.1. Operating System Requirements

For the most up-to-date FPGA software operating system requirements, refer to the Operating System Support page.

The Quartus Prime software requires:

Table 4. Prerequisites For All Altera FPGA Software

OS	Prerequisite
Any Linux OS distribution	Altera has tested using **GNOME** desktop environment and MobaXTerm for remote desktop use. Other tools may work but have not been tested.
	Native XServer software
Red Hat Enterprise Linux 8 and 9	Install the following RPM packages: • **X Windows libraries**: `make`, `libX11.i686`, `libXau.i686`, `libXdmcp.i686`, `libXext.i686`, `libXft-devel.i686`, `libXft.i686`, `libXrender.i686`, `libXt.i686`, and `libXtst.i686` • **For RHEL 8 and 9**: `ncurses-compat-libs` • **GIMP toolkit**: GTK+2
SUSE Linux Enterprise 12	Install the 64-bit `libpng12.so.0` library. Obtain equivalent or substitute packages listed for Red Hat Enterprise Linux or above.
Ubuntu 22.04	Install `libglib2.0-0`, `libxext6`, `libfontconfig1`, and `libsm6` libraries.
Windows	• Microsoft Visual C++ 2015-2022 Redistributable Package (x64) *Note:* The installation of Quartus Prime software installs the Microsoft Visual C++ package automatically if not already installed. • Install any of the unzipping tools, such as WinZip*, 7-Zip*, or WinRAR*, to extract the tar files if you download combined-files packages.

Related Information

Operating System Support

3.3.2. Questa* Altera FPGA Edition Software Requirements

The prerequisites to install Questa* Altera FPGA Edition depend on your OS:

Table 5. Questa* Altera FPGA Edition Software Requirements

OS	Requisite
Windows	Microsoft Visual C++ 2019 Redistributable Packages *Note:* The installation of Questa* Altera FPGA Edition installs these packages automatically.

Related Information

Operating System Support

3.3.3. Other Altera FPGA Software Requirements

The following table provides references to the system requirements of other development tools that Quartus Prime supports:

Table 6. System Requirements of Other Software

Software	References
Advanced Link Analyzer	Advanced Link Analyzer System Requirements
DSP Builder	DSP Builder System Requirements
oneAPI Base Toolkit	oneAPI Base Toolkit System Requirements oneAPI DPC++/C++ Compiler System Requirements
Altera SoC FPGA Embedded Development Suite (SoC EDS)	Arm* Development Studio Requirements
FPGA AI Suite	Operating System Requirements Device Support Intel Distribution of OpenVINO™ Toolkit Requirement
Ashling* RiscFree* IDE for Altera	Ashling* RiscFree* IDE for Altera Device Support Quartus Prime Software Support for Ashling* RiscFree* IDE for Altera
Simics Simulator for Altera FPGAs	Simics Simulator for Altera FPGAs User Guide

3.3.4. Third-Party Software Requirements

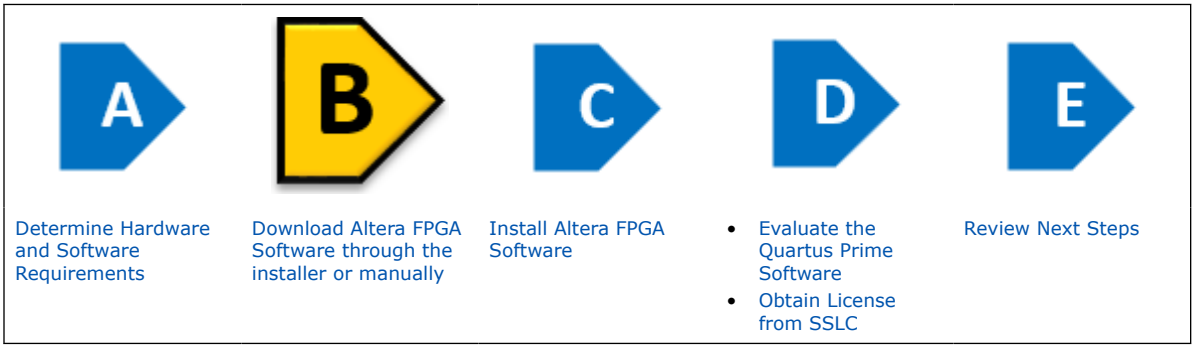
- One of the following browsers with an Internet connection for Quartus Prime software Internet resources:
  - ☐ Mozilla Firefox version 3.6 or later
  - ☐ Microsoft Edge
  - ☐ Google Chrome
- If you are running the Bitdefender antivirus software, ensure to temporarily disable the software when downloading and installing the Quartus Prime software.

4. Downloading Altera FPGA Software

Before downloading Altera FPGA software, ensure that your system complies with all requirements described in *System Requirements & Prerequisites*.

Select and visit the desired software landing page from the FPGA Software Download Center to download software. You have several options for downloading software. Choose depending on download speed, design requirements, and installation method.

Navigating Content Through Tasks



4.1. Software Available in the FPGA Software Download Center

- Quartus Prime Pro, Standard, and Lite Edition software[1] (includes the IP Library)
- Quartus Prime Help
- Questa* Altera FPGA Edition software[2]
- DSP Builder
- FLEXlm license server software
- Advanced Link Analyzer

(1) This product includes software derived from the RSA Data Security, Inc. Secure Hash Algorithm-1 (SHA-1).

(2) Download this product using the instructions in Downloading Software and Device Support Files on page 25.

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

- Stand-alone Quartus Prime Programmer and Tools[1], which includes:
  - ☐ Signal Tap Logic Analyzer
  - ☐ System Console
  - ☐ Logic Analyzer Interface Editor
  - ☐ In-System Sources and Probes Editor
  - ☐ In-System Memory Content Editor
- Power Thermal Analyzer
- Altera SoC FPGA Embedded Development Suite (SoC EDS)
- Ashling* RiscFree* IDE for Altera[3]
- FPGA AI Suite
- Simics Simulator for Altera FPGAs
- Quartus Prime Pro Embedded Edition
- Quartus Prime software driver installer

You can install the supported device files either from the Quartus Prime installer or FPGA Software Download Center. This release supports Agilex 3, Agilex 5, Agilex 7, Stratix, Arria, and Cyclone, or MAX device families as part of the Quartus Prime installation.

Note: You can also obtain older versions of software and legacy software from the FPGA Download Center. However, 16.1 and older releases are discontinued as stated in Custom Advisory ADV2011 and Custom Advisory ADV2135.

Related Information

- Determining System Requirements on page 11
- Altera FPGA Software Download Center
- Operating System Support
- KDB Article: Is the Nios® II Software Build Tools (SBT) for Eclipse included in the full installation of the Quartus Prime Pro Edition software starting from version 19.1?

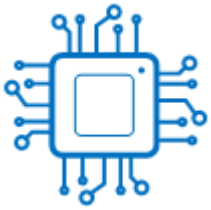
4.2. Downloading Design-Specific Software Components

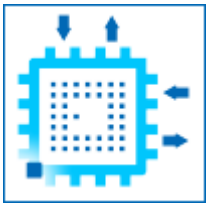
Altera provides a complete suite of development tools for every stage of your design for Field Programmable Gate Arrays (FPGAs), Complex Programmable Logic Devices (CPLDs), and System-on-a-Chip (SoC) FPGAs. You do not need to rerun the Quartus Prime installation to install these tools separately. Using the Altera FPGA suite of development tools, you can create a complex FPGA design as a hardware engineer, write software for an embedded processor as a software developer, model a digital signal processing (DSP) algorithm, or focus on system design.

[3] Ashling* RiscFree* IDE for Altera supports debugging for Nios V and HPS.

Use the following table to explore design tools for each stage of your specific design journey:

Table 7. Software for each Stage of the Design Journey

System Architecture Planning	Board Development	FPGA Interface Designing
		
Explore the Altera FPGA devices and the design software through the Quartus Prime Installer.	Download the following tools using the Quartus Prime Installer: • Power Thermal Analyzer • Advanced Link Analyzer	Download the following tools using the Quartus Prime Installer: • Quartus Prime software • Questa* Altera FPGA Starter Edition

FPGA Application/Interconnect Designing	Embedded Software Development	High-level Design Development
		
Download the following tools using the Quartus Prime Installer: • Quartus Prime software • Questa* Altera FPGA Starter Edition	Download the following tools: • Ashling* RiscFree* IDE for Altera using the Quartus Prime Installer. • Simics Simulator for Altera FPGAs. See download instructions. • Quartus Prime Pro Embedded Edition	Download the following tools: • oneAPI Base Toolkit. See download instructions. • FPGA AI Suite. See download instructions. • DSP Builder using the Quartus Prime Installer.

4.3. Determining the Download Type

Starting with the Quartus Prime Pro Edition software version 23.3, you have the option to either download the software through the Quartus Prime Installer or obtain packages manually.

The following topics provide more information about each of these options:

4.3.1. Downloading Software Using the Quartus Prime Installer

The Quartus Prime Installer is a light-weight application that allows you to download and install various development tools and FPGA device files that the Quartus Prime software supports for a fresh installation, or add components or devices to an existing installation.

Benefits of the Quartus Prime Installer

- Saves disk space. The installer eliminates the need to download large `.tar` files and unzip them locally.
- Allows dynamically selecting files to download or using the default options.
- Supports pausing and resuming software downloads as desired.
- Automatically retries or resumes interrupted software download when it detects the Internet connection.
- Handles different proxy configurations.
- Supports downloading multiple software in parallel.
- Works on Windows and Linux systems.
- Works in the GUI and CLI modes.
- Supports automatic installation upon download completion.

Review the following topics for more information:

Supported Use Cases

- Perform a fresh install of the selected components and devices.
- Update an existing Quartus Prime installation with additional components and devices. In this case, the already installed components and devices are grayed out, and you can select only those components not installed on your system.
- Add devices to your existing installation.
- Download, install, and add components through the command-line interface in the unattended mode.
- Install on lab or server machines that are not connected to the Internet.

  In this case, you can download the necessary components from a machine connected to the Internet, transfer the downloaded files and Quartus Prime Installer to the machine without the Internet connection, and run the Quartus Prime Installer to install the components.

Related Information

- Quartus Prime Installer (GUI Mode)
- Quartus Prime Installer (CLI Mode)

4.3.1.1. Accessing the Quartus Prime Installer

Follow these steps:

1. Visit the FPGA Software Download Center page.
2. Using the left-hand filter pane, perform the following steps to refine the search results:

a. Select the **Quartus Prime Design Software** option. This displays three Quartus Prime software editions (Pro, Standard, or Lite).

b. Select Quartus Prime Pro Edition software. This displays a list of supported software versions.

c. Select 23.3 or later software version.

3. In the refined list of pages, click the Quartus Prime Pro Edition download page. The product download page appears.
4. Click the **Installer** tab.
5. Click the **Download** button.
   - For GUI mode, use the `.run` (on Linux) or `.exe` (on Windows) file. It automatically extracts the files to your `TEMP` directory and launches the installer.

     If you run the installer on a Linux system without a display, the installer defaults to command-line (CLI) mode.
   - For CLI mode, perform these steps:

     a. Run the `.run` file with the `-- --target` *<path>* option on Linux or `.exe` file with the `-sp"--target` *<path>*" option on Windows to extract the files to the <*path*> location. For example: `qinst-windows-25.3-107.exe -sp"--target c:\quartus_installer"`

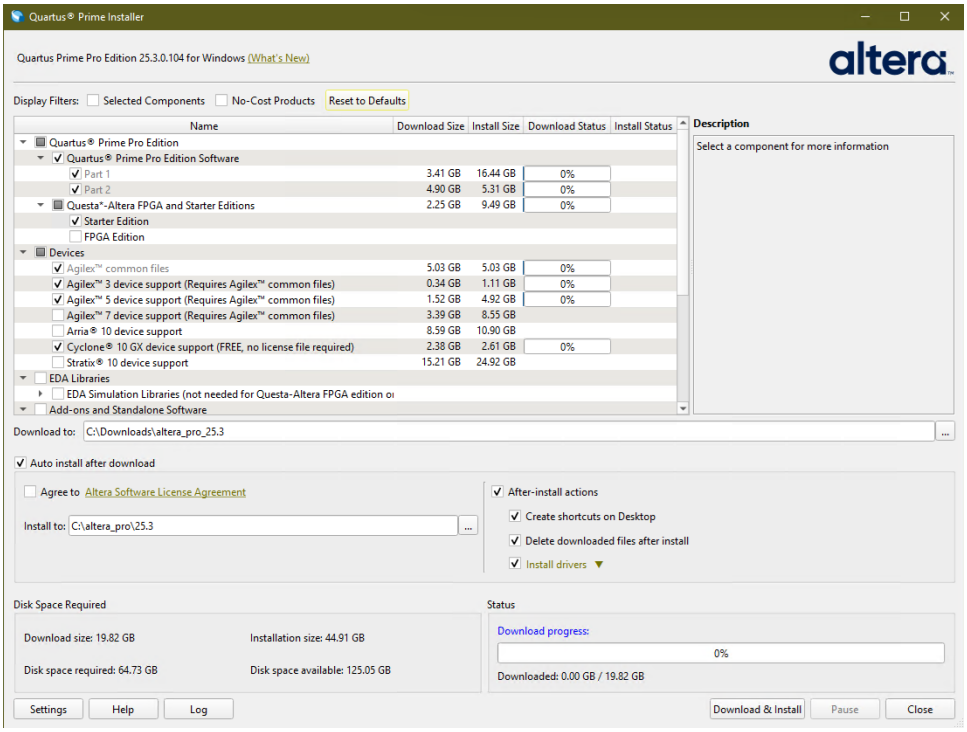
     Note: On Linux, the extra `--` is needed.

     b. Run the `qinst.sh --cli` command on Linux or `qinst_cli.bat --cli` command on Windows. If you need help, run the command with `--help` to view the help manual. For more information about CLI commands, refer to Using CLI Commands on page 23.

4.3.1.2. Getting Started With the Quartus Prime Installer GUI

The top section of the GUI lists all downloadable components and their relevant information in a hierarchical format. You can select and download the desired components, as shown in the following image:

Figure 2. Quartus Prime Installer GUI



- The default download location is *<user_home>/Downloads* and installation location is `C:\altera_pro\<ver>` on Windows and `/<user_home>/altera_pro/<ver>` on Linux. However, you can select a different location using the browse button and selecting the desired location. Space is not allowed in both download and installation paths.
- By default, the **Auto install after download** and **After-install actions** options install supplementary software and programming hardware drivers automatically following Quartus Prime installation. To disable automatic installation of one or more supporting software or drivers, turn off or change the value of the **Install drivers** option. If installation of any supplementary software or driver files fails, the installer prompts you to correct the issue before continuing.
- The **Agree to the Altera Software License Agreement** is disabled by default. You must enable it to proceed with software download and installation.
  - ☐ If you want to review the license agreement, click the **Altera Software License Agreement** link to view it in your default browser.
  - ☐ If your installation directory contains a matched version of Quartus Prime software, the installer changes into the update mode, allowing you to install additional components or uninstall components. If the installation directory does not contain a matched version of the Quartus Prime software, the installer displays an error when you click the **Download & Install** button.
  - ☐ If you are downloading for a platform (Linux or Windows) other than the one your system is running on, the **Auto install after download** option is disabled, and the tooltip shows why it is disabled.
- The bottom section of the GUI displays the download size, installation size, disk space required for download and installation, disk space available, and download status. The disk space is highlighted if the available space is less than the disk space needed. If you proceed with the download and installation, you are prompted with a warning message for insufficient disk space. The download status dynamically changes when the download is in progress.
- If you want to modify the installer settings, click the **Settings** button. Refer to Modifying the Installer Settings on page 21 for more information.
- After selecting the components and reviewing all settings, click the **Download & Install** button. Use the **Pause** button if you want to interrupt the download.
- If you encounter any error throughout the download process, use the **Log** button to invoke the Log Viewer. The log file is saved to your download directory if you only downloaded the files. However, enabling the **Auto Install** option saves the log file to your installation directory.

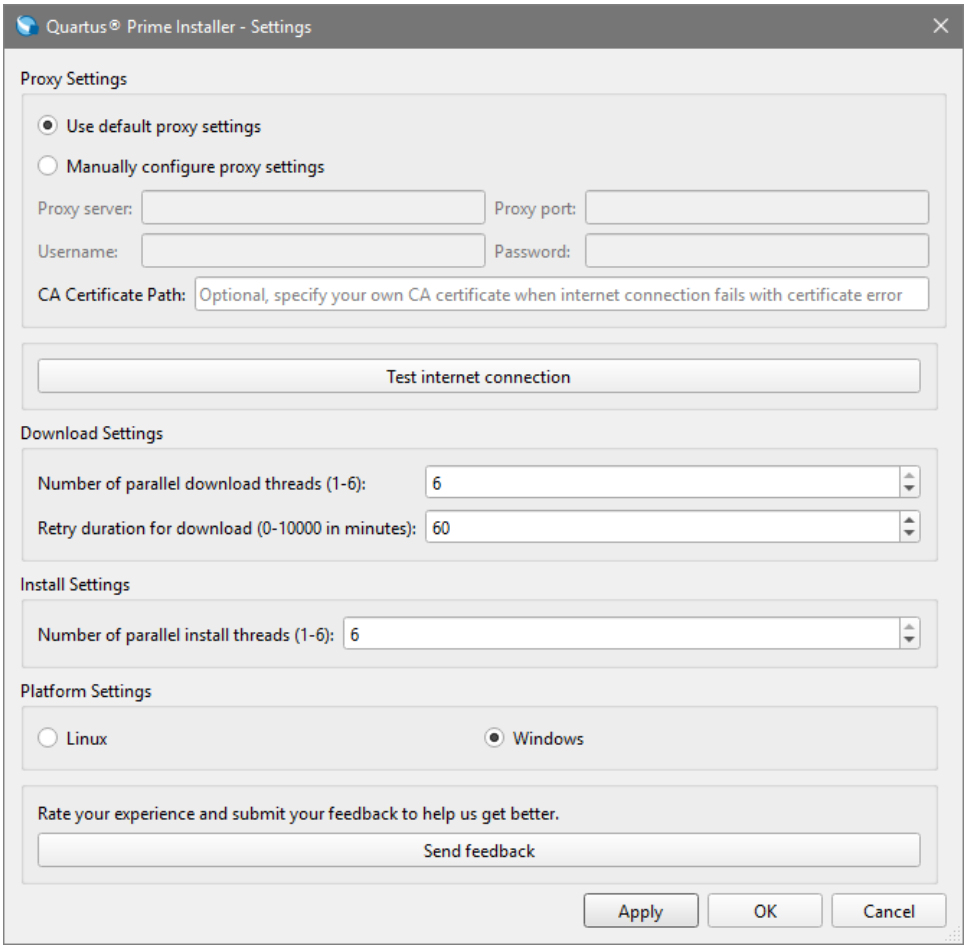
Important: If **Delete downloaded files after install** is enabled, downloaded files corresponding to the installed components are deleted automatically after a successful installation. If the option is disabled, you need to manually delete the downloaded files if you no longer need them.

4.3.1.3. Modifying the Installer Settings

The **Settings** dialog allows you to overwrite the default installer settings. It is set to the following default values:

- **Use default proxy settings** is enabled to retrieve your system's proxy settings.
- **CA Certificate Path** is an optional setting to specify your own CA certificate. Set this only when the Internet connection fails with certificate error. You can pass in either a file of a CA certificate or a directory that holds CA certificates.
- **Number of parallel download threads (1-6)** is set to 4.
- **Retry duration (0-10000 in minutes)** is set to 60 minutes.
- **Installer Platforms** defaults to the platform your system is currently running on.

Figure 3. Quartus Prime Installer Settings GUI



Follow these instructions to modify the installer settings:

- You can use the **Test Internet Connection** button to confirm if your system is connected to an active Internet connection.

  If the installer cannot detect your Internet connection, select the **Manually configure proxy setting** option to set up your proxy settings and establish an Internet connection manually. The installer cannot progress until it detects the Internet connection. For information about the accepted values for the proxy server, refer to https://curl.se/libcurl/c/CURLOPT_PROXY.html.

  Note: If you are behind a firewall, ensure that you **do not** have `no_proxy` variable set in your system.

- In the **Number of parallel download threads (1-6)** field, enter a value between 1 and 6 to modify the number of threads.
- In the **Retry duration (0 - 10000 in minutes)** field, you can customize the retry duration if your Internet connection interrupts or is unstable. You can enter a value between 0 and 10000 minutes. You can stop the retries within the retry period.
- You can select Linux or Windows platform, and the installer downloads files relevant to the chosen platform.

  Note: If you are downloading for a platform other than the one your system is running on, the **Auto install after download** option in the main GUI is disabled with a tooltip that shows why it is disabled.

4.3.1.4. Using CLI Commands

You can run the Quartus Prime installer application entirely from the CLI and access the same functionality as the GUI interface. In this case, you must pass all the necessary parameters through the CLI only. No input is accepted through the console.

Table 8. Command-line Options

Option	Description
`--help`	Displays the help.
`--cli`	Runs the installer in the command-line mode. *Note:* If you pass other options without passing the `-cli` option, the following error message is displayed: `Please use -cli if you want to run this app on command-line, otherwise, remove all options passed in.`
`-view-eula`	Allows you to view the End-User License Agreement (EULA). The EULA launches in your default browser if your system has a display. Otherwise, it displays the following message: `You can view Altera License Agreement in <install_dir>/licenses.`
`--cacert arg`	Allows you to specify your own CA certificate if the certificate that comes with the installer does not work with your firewall. You can pass in either a CA certificate file or a directory that holds CA certificates.
`--proxy <server:port>` `--proxy-user <user:password>`	Allows you to specify your proxy settings manually if the installer cannot detect the Internet connection. For information about the accepted values for the proxy server, refer to https://curl.se/libcurl/c/CURLOPT_PROXY.html.

continued...

Option	Description

`--test-download`	`--proxy` and `–proxy-user` options are optional. If the installer is able to accept all other options and process the command, it runs the `--test_download` option first.
	• If `--test-download` fails and you have not specified the proxy settings, the following error message is displayed:
	`Internet connection cannot be established, if you're behind a firewall, please enter the proxy settings and try again.`
	• If you have already passed the proxy settings but the Internet connection still fails, the following error message is displayed:
	`Internet connection cannot be established, please recheck your internet connection or proxy settings.`
	Note: If you are behind a firewall, ensure that you **do not** have `no_proxy` variable set in your system.

`--download-dir <dir>`	Specifies where to download the installer files.
`--install-dir <dir>`	Specifies where to install the application. • If you specify the `--install-dir` option, the `--auto-install` option defaults to on. • If your installation directory already has an existing installation, then passing `--install-dir` option displays the following error message: `This application does not support updating an existing installation. You must run the installer manually after download.` • If you specify only the `--download-dir` and not the `--install-dir` option, `--auto-install` is set to off.
`--auto-install`	Specifies whether to install software or drivers automatically after download completes. *Important:* On Windows, you must run this command option as an administrator. If you execute it without the administrator rights, then passing the `--auto-install` option displays an error message appropriately.
`--bypass`	Specifies whether to bypass the disk space availability check.
`--accept-eula`	Specifies whether to accept the Quartus Prime software EULA. By passing the `--accept-eula` option, you are acknowledging to accept the EULA. *Note:* If you pass only the `--install-dir` option without the `–accept-eula` option, the following error message gets displayed: `You must agree to the Altera license agreement to install the software.`
`--delete-downloads`	After successful installation, it only deletes the files that are installed.
`--parallel-downloads <arg>`	Specifies the number of parallel downloads. Supports values in the range of 1 to 6. Default is 4.
`--retry-duration <arg>`	Specifies the retry duration (in minutes) if the Internet connection gets interrupted. Supports values in the ranges of 0 to 10000 minutes. Default is 60 minutes.
`--platform <linux|windows>`	Specifies the platform (Linux or Windows) for which the installer files are downloaded. Platform defaults to the one your system is currently running on.
`--components <quartus,agilex3, agilex5,agilex7,stratix10,all>`	Specifies components to download or install with comma-separated values. If you are updating an existing installation, the installer installs only those components that are not installed on the system yet and ignores the components already installed. Dependent components are automatically included.

Option	Description
	• Default components are Quartus Prime and Questa* Altera FPGA Starter Edition. • The default devices are Cyclone 10 GX, Agilex 3, and Agilex 5. The remaining devices, such as Agilex 7, Stratix 10, and Arria 10, are optional. • Optional components are Questa* Altera FPGA Edition, EDA Libraries, Advanced Link Analyzer (Windows only), Ashling* RiscFree* IDE for Altera, DSP Builder, FPGA AI Suite, FLEXlm license server, Power Thermal Analyzer, Quartus Prime Driver Installer, Quartus Prime Help, Quartus Prime Programmer and Tools, and Quartus Prime Embedded Edition. Specify `all` to install all components.

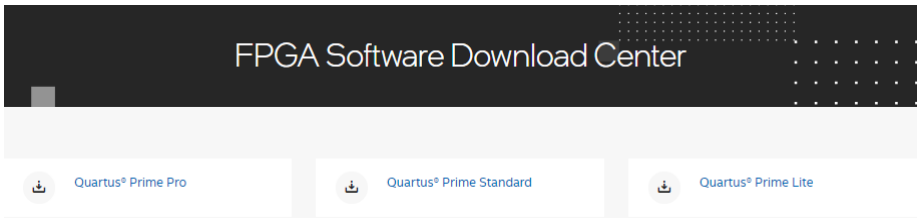
4.3.2. Downloading Software Packages Manually

You can download software packages manually using one of the following options:

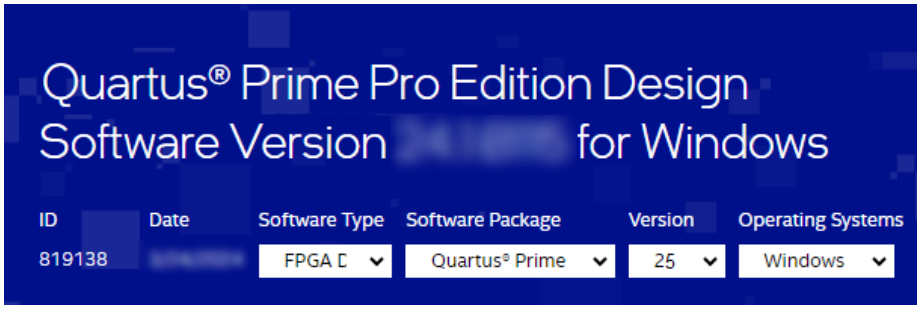
4.3.2.1. Downloading Software and Device Support Files

To download Altera FPGA software with individual executable and device support files, follow these steps:

1. Visit the FPGA Software Download Center page.
2. Click the desired **Quartus Prime Design Software** option.



3. In the displayed page, refine your selection using the **Software Type**, **Software Package**, **Version**, and **Operating Systems** drop-down options to download the desired software.



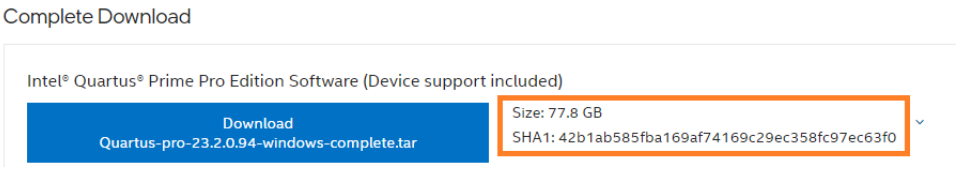
4. If you are proceeding to download the latest edition of the Quartus Prime design software, scroll down the page, select the **Individual Files** tab and locate one of the following sections to download the desired files:
   - ☐ **Complete Download** to download the complete package with the device support and add-on software files.
   - ☐ **Individual Files** to download the packages in parts and Questa* Altera FPGA Starter Edition software.
   - ☐ **Devices** to download the device support files.

- ☐ **Add-On and Stand-Alone Software** to download individual executable files of the add-on software.

5. Click **Download *<file_name>*** to download the desired software.
6. Click **Accept** to accept the Software License Agreement.

File download starts automatically.

Important: Wait for all the files to download completely before beginning installation. Altera recommends verifying the `sha1sum <filename>` to ensure the file downloaded completely. The `sha1` value must match with that mentioned on the webpage. Example SHA1 shown in the following image:



Verify the checksum in the following method:

- **Linux OS**: `sha1sum <filename>`
- **Windows OS**: In the Command prompt window, type `certutil -hashfile <filename>`

Related Information

Selecting the Installation Path on page 30

4.3.2.2. Downloading Copyleft Licensed Software

The Altera FPGA Download Center also supports downloading some of the copyleft licensed software that are required for FPGA development. Follow these steps to download copyleft-licensed software:

1. Access the product download page and ensure that you have installed the Quartus Prime software. If not, then download the Quartus Prime software installation files and device files as described in Downloading Software and Device Support Files on page 25.
2. Select the **Copyleft Licensed Source** tab.
3. Download the desired software you want to install.

   Attention: Wait for all the files to download completely before beginning installation. Altera recommends verifying the `sha1sum <filename>` to ensure the file downloaded completely. The `sha1` value should match with that mentioned on the webpage.

To install the files, refer to Installing from .tar Files on page 32.

Related Information

- What is a Copyleft License?
- copyleft.org

4.4. Downloading Other Altera FPGA Software

Downloading SoC EDS

Follow these steps:

1. Visit the FPGA Software Download Center.
2. In the left-hand filter pane, select **Altera SoC FPGA Embedded Development Suite** under **Additional Software**.
3. Select **Altera SoC FPGA Embedded Development Suite** Pro or
4. Standard Edition. This displays a list of product download pages.
5. Click the desired page.
6. Under the **Downloads** section, select the **Software Download** tab and download the software.

Downloading Arm* Development Studio

Follow these steps:

1. Visit the FPGA Software Download Center.
2. In the left-hand filter pane, select **Arm* Development Studio**.
3. Click the desired page.
4. Under the **Downloads** section, select the desired platform tab (Linux or Windows) and download the software.

Downloading Simics Simulator for Altera FPGAs

Follow these steps:

1. Visit the FPGA Software Download Center.
2. Select **Simics Simulator for Altera FPGAs** in the left-hand filter pane.
3. Click the desired page and download the software.

Downloading FPGA AI Suite and Distribution of OpenVINO Toolkit

Follow these steps:

1. Visit the FPGA AI Suite web page.
2. Under the **Get Started Today** section, use relevant download options to download the FPGA AI Suite and **Distribution of OpenVINO Toolkit**.

Downloading oneAPI Base Toolkit and FPGA Add-on for oneAPI Base Toolkit

Follow these steps:

1. Visit oneAPI Base Toolkit web page.
2. Select the desired operating system.
3. Select the distribution method.

4. You can either sign up and download the software, or use **Continue without signing up (download starts immediately)** link to start downloading directly without signing up.
5. Visit FPGA Add-on for oneAPI Base Toolkit to download FPGA add-ons for oneAPI Base Toolkit.
6. Select the FPGA platform.

4.5. Downloading Docker Containers for Quartus Prime FPGA Design Software

Docker* is an open platform for developing, shipping, and running applications. If you are new to Docker*, refer to Get Docker to understand how Docker can help with running your FPGA applications and get started.

To retrieve containers for the Quartus Prime Pro and Standard Edition FPGA design software, perform these steps:

1. Visit Docker* Hub's Quartus repositories.
2. Select the desired Quartus repository.
3. Follow the instructions on the page to retrieve the image.

5. Installing Altera FPGA Software

Note: Before you install the software, ensure that your system complies with all requirements described in Determining System Requirements on page 11.

To download software, refer to Downloading Altera FPGA Software on page 15.

Depending on the download type, you can install the software automatically after downloading through the Quartus Prime Installer or manually.

Navigating Content Through Tasks

Use the following table to navigate this guide through user-tasks:

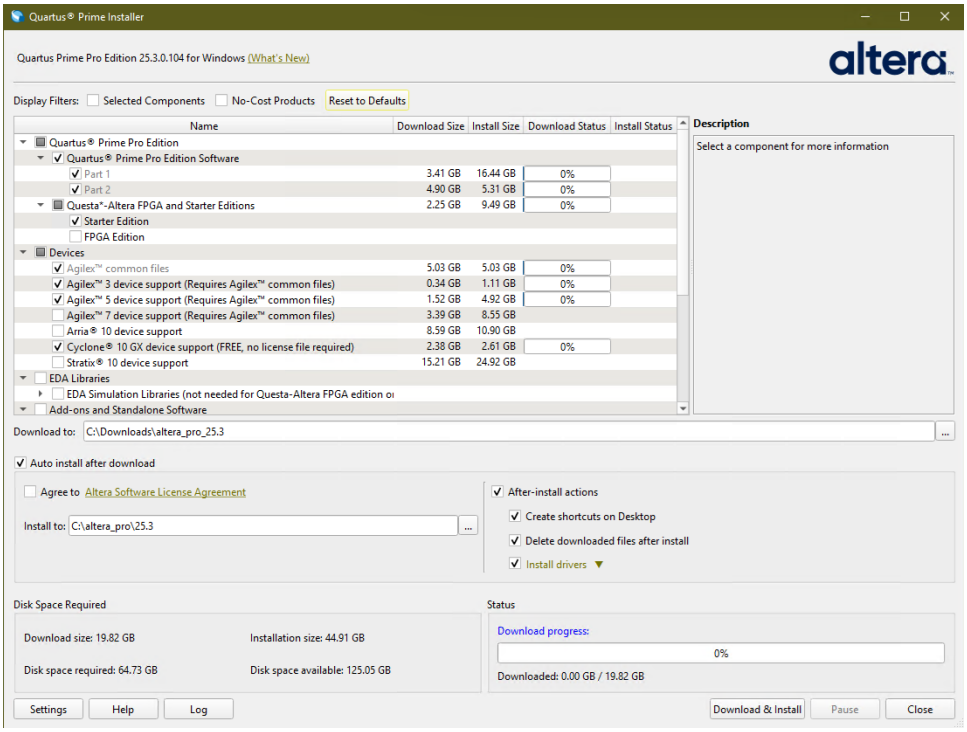


5.1. Installing Altera FPGA Software Through Quartus Prime Installer

The Quartus Prime Installer is a light-weight application that allows you to download and install various development tools and FPGA device files that the Quartus Prime software supports for a fresh installation, or add components or devices to an existing installation. For more information, refer to Accessing the Quartus Prime Installer on page 18.

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

Figure 4. Quartus Prime Installer GUI



- By default, the **Auto install after download** and **After-install actions** options install supplementary software and programming hardware drivers automatically following Quartus Prime installation. To disable automatic installation of one or more supporting software or drivers, turn off or change the value of the **Install drivers** option. If installation of any supplementary software or driver files fails, the installer prompts you to correct the issue before continuing.

Related Information

Installing Drivers on RHEL 8/Ubuntu 18 and Above Versions on page 34

5.2. Installing the Altera FPGA Software Manually

If you downloaded the package manually using one of the options in Downloading Software Packages Manually on page 25, use instructions in this section to install those packages.

5.2.1. Selecting the Installation Path

The installation path must satisfy the following requirements:

- Contain only alphanumeric characters
- No special characters or symbols, such as !$%@^&*<>,
- Only English characters
- No spaces

Default Installation Path

Linux	*<home directory>/<edition>/<version number>*	*<home directory>* is the default path of the Linux workstation, or as set by the system administrator
Windows	*<drive>:\<edition>\<version number>*	

where *<edition>*:

- `altera_lite`—for Lite Edition
- `altera_std`—for Standard Edition
- `altera_pro`—for Pro Edition

If you use a different path, substitute the appropriate name for *<installation-directory>* in the installation steps.

5.2.2. Installing Altera FPGA Software at the Command Prompt

You can install Altera FPGA software for Windows* or Linux using command-line arguments. You can also group installation instructions into a custom installation script.

- To run the installer executable, use:
  - ☐ Linux:

    ```
    <product>-<version>-<os>.run
    ```

  - ☐ Windows*:

    ```
    <product>-<version>-<os>.exe
    ```

  The installation wizard appears to guide you through the installation process.

  Note: The Quartus Prime software installer is the main installer. You must have administrator privileges to install it on Windows. It installs other software, such as DSP Builder, if that installer exists in the same location. If a standalone product is not installed by the Quartus Prime software installer, then you must install it separately.

Example 1. Command-line Installer Examples

To see the installer's Help in Linux:

```
<product>-<version>-<os>.run --help
```

To run the installation with minimal or no interaction with the installer:

```
<product>-<version>-<os>.exe --mode unattended
                             --unattendedmodeui minimal
                             --installdir <installation directory>
                             --accept_eula 1
```

To install in console mode (no X display) for Linux:

```
<product>-<version>-<os>.run --mode text
                             --installdir <installation directory>
```

5.2.2.1. Command-Line Options

The available command-line arguments for the installer are:

Option	Description
`--help`	Displays the installer help.
`--version`	Displays product information.
`--unattendedmodeui <unattendedmodeui>`	Specifies whether the installation requires user interaction. This argument is useful with unattended mode. Options are: • `none`—(Default) Installation does not require user interaction, and no output appears. • `minimal`—Installation does not require user interaction. A dialog box shows the installation progress. • `minimalWithDialogs`—In addition to the installation progress bar, other dialog boxes appear. This mode may require user interaction. *Note:* If you install in unattended mode, you must also include the `--accept_eula 1` option.
`--mode <mode>`	Specifies the installation mode. For Linux, the available modes are: • `qt` (Default) • `gtk` • `xwindow` • `text` • `unattended` For Windows*, the available modes are: • `qt` (Default) • `win32` • `unattended`
`--accept_eula 1`	States that you accept the Altera FPGA End User License Agreement. This option is mandatory when you use unattended mode. *Note:* To view the license agreement files prior to installation, run the installer with the `--install-lic` command-line option.
`--disable-components <disable-components>`	Specifies which components you want to disable, in a comma-separated list.
`--installdir <installdir>`	Specifies the target installation directory for Altera FPGA software. Refer to *Download and Installation Prerequisites* for the default installation directories.
`--product_copy_name <product_copy_name>`	If you install multiple copies of the same software with the same version on Windows, it specifies the installation name. Default=none.
`--install-lic <install_lic>`	Specifies the target directory for the license agreement files.

Related Information

Selecting the Installation Path on page 30

5.2.3. Installing from .tar Files

To install Altera FPGA software with `.tar` files:

1. Extract the installation files into the same directory.
   The `components` subdirectory stores software and device installation files.
2. If you use Windows OS, extract the installation files by unzipping the `.tar` files.

   WinZip*, 7-Zip*, and WinRAR* are examples of unzipping tools. Do not use unzip program shortcuts, such as **1-Click Unzip**.
3. Run one of the following scripts to begin installation:

Table 9. Installation Scripts

Quartus Prime Edition	OS	
	Linux	Windows*
Lite and Standard Editions	`setup.sh`	`setup.bat`
Pro Edition	`setup_pro.sh`	`setup_pro.bat`

The wizard appears, and guides you through the installation process.

5.2.4. Installing Device Support to Existing Installation

Once you complete downloading device support, follow these steps to install it to an existing Altera FPGA software package:

1. Ensure the files are downloaded completely before beginning the installation process.
2. Start the Quartus Prime software.
3. Click **Tools ➤ Install Devices** to open the Installation Wizard. If you do not see the **Install Devices** option, then on the Windows* Start menu, go to **Altera FPGA *<version number>* *<Lite/Standard/Pro>* Edition ➤ Device Installer**.
4. Install in the same location as the Quartus Prime software.

5.2.5. Installing Programming Cable Drivers

By default, the **Auto install after download** and **After-install actions** options install supplementary software and programming hardware drivers automatically following Quartus Prime installation. To disable automatic installation of one or more supporting software or drivers, turn off or change the value of the **Install drivers** option. If installation of any supplementary software or driver files fails, the installer prompts you to correct the issue before continuing.

- On Windows, locate the appropriate drivers in the following location and install the driver:

  ```
  <installation-directory>\quartus\drivers
  ```
- For Linux, refer to Installing Drivers on RHEL 8/Ubuntu 18 and Above Versions on page 34:

For detailed information on installing drivers and setting up cables, refer to the corresponding cable user guide.

Related Information

- FPGA Download Cable (formerly USB Blaster) User Guide

- FPGA Download Cable (formerly USB Blaster II) User Guide
- EthernetBlaster II Communications Cable User Guide (PDF)
- Cable and Adapter Drivers Information

5.2.5.1. Installing Drivers on RHEL 8/Ubuntu 18 and Above Versions

To allow communication between the host computer and a development board, you must install the drivers for USB download cables, even for development kits with embedded Altera FPGA Download Cable circuits. Configuring these options requires administrator (root) privileges.

Note: The USB-Blaster is now the FPGA Download Cable. The USB-Blaster II is now the FPGA Download Cable II.

The Quartus Prime software accesses the FPGA Download Cable, FPGA Download Cable II, or FPGA Download Cable III through the USB file system (`usbfs`). To program devices, change the permissions on the ports.

Follow these steps to install on Red Hat Enterprise Linux 8 and above and Ubuntu versions 18 and 20 operating systems:

1. Create a file named `/etc/udev/rules.d/92-usbblaster.rules`.
2. In the `/etc/udev/rules.d/92-usbblaster.rules` file, add the following lines:

```
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6001",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6002",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6003",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6010",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6810",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6020",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6022",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6024",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6025",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6026",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602C",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602D",
MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602E",
MODE="0666"
```

5.2.6. Installing and Configuring a JTAG Server

A JTAG Server communicates with the hardware and allows multiple programs to use JTAG resources at the same time. `jtagd` is the Linux version of `jtagserver`.

Related Information

JTAG Settings tab
In *Quartus Prime Help*

5.2.6.1. Installing and Configuring jtagserver on Windows

If you install the Quartus Prime software for Windows* on a network drive, and have a JTAG Server, then you must set up the Quartus Prime software to allow the JTAG Server to run JTAG services.

Altera recommends enabling the **Install JTAG Server as a Windows service and run automatically when Windows starts** option on the final page of the installation wizard to automatically install and run the `jtagserver` as a service upon system start up.

Note: For additional information about operating JTAG Server in a more secure manner, refer to the application note AN 939: JTAG Connections Over SSH.

If you choose not to install the JTAG Server as a Windows service automatically, then when necessary, you can install the JTAG Server manually using the command line option, `jtagserver.exe --install` as an Administrator. The `jtagserver.exe` is available within your Quartus Prime software installation directory (`<path>/quartus/bin64`). The installed service does not run as an Administrator.

Tip:

- After the installation, you can either use the **Task Manager ➤ Services** tab or the command-line option `--status` to verify if the `jtagserver` service is running.
- You can control the `jtagserver` service using the `--start` and `--stop` command-line options or using the Task Manager.

If there is no active JTAG Server process running, client applications such as Programmer automatically launches an instance of the non-persistent JTAG Server (without it running as a service) and exists after two minutes of idleness.

5.2.6.2. Installing and Configuring a Local JTAG Server (jtagd) on Linux

`jtagd` is the Linux version of `jtagserver`.

In a typical operation, the client application (for example, Quartus Prime Programmer) starts the `jtagd`, which runs under your user account that is running the Altera FPGA software. In this mode, `jtagd` exits after two minutes of idleness.

If you want to connect to the JTAG Server remotely, then manually run `jtagd` as a process by running the command `<path>/quartus/linux64/jtagd` with no argument, followed by running the command `jtagconfig --enableremote <password>` to make JTAG Server accept connection from remote clients. This allows you to log off without terminating the `jtagd` process.

Note: For additional information about operating JTAG Server in a more secure manner, refer to the application note AN 939: JTAG Connections Over SSH.

5.3. Using the Same Installation Files on Multiple Systems

You can install Altera FPGA software on multiple systems using the same installation files that you download once. This method reduces overall download and installation time.

To use the same installation files for multiple systems:

- Download and save the installation files to a location that is accessible by each system on which you want to install Altera FPGA software.
- To perform the installation, you can:
  - ☐ Launch the installation wizard from each system.
  - ☐ Write and run a custom installation script using the available command-line arguments.

5.4. Managing Multiple Versions and Copies of Altera FPGA Software

You can install multiple copies of the same version of software on the same computer.

- To specify which version you want to open by default, setting the `QUARTUS_ROOTDIR_OVERRIDE` environment variable.
- To preserve project compilation databases from an earlier version of the software before installing a newer version, open the design in the earlier version of the software and export a version-compatible database for the design.

Related Information

Setting Quartus Prime Environment Variables on page 36

5.5. Setting Quartus Prime Environment Variables

The Quartus Prime software installation process initializes environment variables. In specific cases, you can take advantage of modifying these variables.

QUARTUS_ROOTDIR_OVERRIDE

Use this optional environment variable when you have multiple copies of the Quartus Prime software on the same computer and you want to determine which version to open by default.

Note: If you have multiple copies of the same version of the Quartus Prime software, the system displays an error message indicating that Quartus Prime software files cannot be found when you start the program. The solution is setting `QUARTUS_ROOTDIR_OVERRIDE` to one of the copies.

QUARTUS_LIBRARY_PATHS

The `QUARTUS_LIBRARY_PATHS` environment variable specifies user-defined library paths.

You can use this environment variable to define multiple library paths at the same time, instead of individually adding each path to the user library.

PATH

You must add `$QSYS_ROOTDIR` to the `PATH` variable. The `$QSYS_ROOTDIR` locates the directory at `${installdir}/qsys/bin`.

In general, Altera recommends adding all `${installdir}/bin` to the `PATH` variable to allow running a software without adding the full path.

LM_LICENSE_FILE

The `LM_LICENSE_FILE` environment variable specifies the location of the license file. However, the Quartus Prime software overrides the value of the environment variable with the location that you specify in the **License Setup** dialog box.

Note: Separate multiple license servers and node locking license files with ":" (Linux) or ";" (Windows).

LC_ALL

You must ensure that your operating system locale is set up correctly. In particular, the locale you specify in `LC_ALL` environment variable must match the locale settings (such as `LANG`). If there is a mismatch, the following Perl warning might be generated during IP generation:

```
perl: warning: Setting locale failed.
```

This warning results in IP generation failure.

Note: You can also set the environment variable `PERL_BADLANG=0` to avoid the Perl warning.

Sample Setup Script

Instead of setting up the above listed environment variables individually, you can also create a script to set up the environment for a particular version in Linux as shown in the following sample:

```
Sample setup script for Quartus:
quartus_x.x.bash (executable)
#Setup pointers to version X.X
export QUARTUS_ROOTDIR="<quartus-install-path>/quartus"
export QSYS_ROOTDIR="$QUARTUS_ROOTDIR/qsys/bin"
export ALTERAOCLSDKROOT=<some_specific_release>
export INTELFPGAOCLSDKROOT=<some_specific_release>
export PATH="$QUARTUS_ROOTDIR/bin:$QSYS_ROOTDIR:$PATH"
# Adding  any /bin under $ALTERAOCLSDKROOT or $INTELFPGAOCLSDKROOT to $PATH if
applicable
export LM_LICENSE_FILE=<path_to_license_file>
```

Related Information

Selecting the Installation Path on page 30

5.6. Starting the Quartus Prime Software

You can run the Quartus Prime software on Windows* and Linux.

For versions that you purchase, the Quartus Prime software prompts you for license information if it does not detect a license upon startup. You are prompted to evaluate, buy, or retrieve a license. The Quartus Prime Pro Edition software provides the additional option of running the software in free mode for Cyclone 10 GX devices only.

Note: You can evaluate the Quartus Prime software at no cost for a period of time.

Starting the Altera FPGA Software on Windows

Use any of the following methods to start Altera FPGA software on Windows:

- On the Windows* desktop, double-click the Altera FPGA software icon.
- On the Windows* Start menu, click **Altera FPGA <*version number*> <*Lite/Standard/Pro*> Edition ➤ Quartus**.
- At a command prompt, type:

```
<installation-directory>\bin64\quartus
```

Starting the Altera FPGA Software on Linux

Use the following method to start Altera FPGA software on Linux:

- Type the following at the command prompt:

```
<installation-directory>/quartus/bin/quartus
```

Note: Starting the Quartus Prime software in a platform-specific directory on Linux (for example, `<installation-directory>/linux64/quartus/`), can cause problems running the software.

Related Information

Evaluating the Quartus Prime Software on page 49

5.7. Updating Altera FPGA Software

To download available Altera FPGA Software updates:

1. Verify that you are using the latest version of the Altera FPGA software.
2. Visit the FPGA Software Download Center page.
3. Using the left-hand filter pane, perform the following steps to refine the search results:
   a. Select the **Quartus Prime Design Software** option. This displays three Quartus Prime software editions (Pro, Standard, or Lite).
   b. Select the desired Quartus Prime software edition. This displays a list of supported software versions.
   c. Select the desired Quartus Prime software release version.
   d. Select the operating system (Linux or Microsoft Windows*).
4. In the refined list of pages, click the desired page to download the software.
5. On the version-specific download page, click **Updates**.
6. Download by clicking the **Download** button under each device.

Wait for all the files to download completely before beginning the installation process.

7. Run the executable files to launch the installation wizard, which guides you through the installation process. On Windows*, it automatically finds the existing installation to be updated. On Linux*, you need to input the installation path of an existing installation to be updated.

   Note: For information about the latest updates, refer to the update release notes on www.altera.com.

5.8. Installing and Uninstalling a Software Patch

Altera provides patches for the Quartus Prime and standalone Programmer software to address critical issues and protect your system against vulnerabilities.

Installing a Software Patch

Once you have downloaded the patch you want to install, perform the OS-specific steps as follows:

On Linux Systems

1. Modify the patch installer file permission by running the `chmod +x <file_name>` command, which makes the file executable.
2. Run the patch installer file and proceed.
3. Accept the license agreement.
4. Specify the installation directory to update.
5. Proceed and complete the installation.

On Windows Systems

1. Run the patch installer file. The **Patch Setup Wizard** appears.
2. Click **Next**. The **License Agreement** screen appears.
3. Accept the license agreement and proceed. The **Installation Directory** screen appears.

   The installer automatically detects the existing installation directory to update.
4. Verify if the installation directory is the one you want to update. If not, then indicate the correct installation directory.
5. Proceed and complete the installation.

Note:

- If a patch applies to both Quartus Prime and the standalone Programmer software, select the **Install patch to standalone Programmer and Tools installation**. Otherwise, by default, the patch is applied only to the Quartus Prime software installation.
- The **Allow patches to be uninstalled** option is on by default. It allows uninstalling the patches. However, it consumes additional disk space and slows down the installation process. You can choose to turn it off.

Uninstalling the Patch

To uninstall a patch, use the **Uninstall** option from the **Start** menu shortcut on Windows or launch the uninstaller from the *<installation_directory>/uninstall* directory on Linux systems.

Note: If you have more than one patch installed, you can uninstall patches only in the reverse order in which you installed them.

5.9. Uninstalling Altera FPGA Software

5.9.1. Uninstalling on Windows*

1. Right-click on the Windows* Start menu and click **Apps and Feature**.
2. Point to ***<software name> <version number>*** ➤ **Uninstall *<software name> <version number>***.

   The installation wizard appears and guides you through the uninstallation process.

5.9.2. Uninstalling on Linux

1. Ensure that none of your project files are contained in the Altera FPGA software installation directory
2. Delete the software directory.

5.10. Troubleshooting Installation Issues

If your installation of Altera FPGA software hangs, use the following steps to troubleshoot:

- Ensure your system meets the hardware and software requirements.
  - ☐ Ensure that your operating system is supported. Refer to Operating System Requirements on page 13 for more information.
  - ☐ Check if the disk space is sufficient for the Altera software installation.
  - ☐ If you are running any antivirus software, temporarily disable it during the Quartus Prime software download and installation process.
- Install with *Administrator* privileges.
- Verify the checksum values of the installation files to detect any file corruption.
- Retry the download and installation process with the latest software packages.

The following articles carry solutions to some of the frequently asked questions:

- Why does the Quartus Prime installer fail on some Linux platforms?
- Why does the installation of the Quartus Prime Pro Edition Software hang during silent installation with System Center Configuration Manager (SCCM)?
- FLEXlm software error: Version of vendor daemon is too old (11.16.1.0)
- How do I install Cygwin for SoC EDS?
- How can I improve the security of my Quartus Prime Software installation?

- Why do my tar packages fail to unpack successfully?
- Which version of Altera SoC FPGA Embedded Development Suite (SoC EDS) version should I install for Quartus Prime edition software version 20.2?
- Why does Windows 10 crash when Nios EDS tools are used under WSL?
- Why is there a new installation file for the Quartus Prime Pro Edition version 18.1 Update 2 software for Linux?
- Flexlm software error: version of vendor daemon is too old
- Why does the Quartus Prime Pro and Standard Edition version 18.1 fail to run on Ubuntu 18.04 LTS?
- Why does get an error when hosting an Altera FPGA Software License and a Mentor Graphics license on the same server?
- What extra libraries do I need to install to use the Quartus Prime software and related tools on Linux?

Tip: If you are facing any issue not listed in this topic, explore Altera FPGA Knowledge Base articles or post your query on Altera FPGA Software Installation and Licensing forum.

6. Licensing Altera FPGA Software

Most Altera FPGA software requires using a license to enable the software. Altera provides this license in a file named `license.dat`. The Quartus Prime Lite Edition is free and does not require a license. Questa* Altera FPGA Starter Edition is also free, but requires zero-cost licenses. The Quartus Prime Pro Edition software requires no license for Cyclone 10 GX devices only.

When you purchase a Quartus Prime software subscription, the license file supports:

- The version of the Quartus Prime software you purchase
- All previous supported versions
- All versions released within a year of the purchase date

Most license entitlements for Quartus Prime software and IP cores are perpetual. Any exceptions for products or complementary tools such as Questa* Altera FPGA Edition are noted in the license file and on the Self Service License Center as having a license expiration date.

Altera also uses licenses to enable additional software, such as purchased IP cores. For more information about licensing conditions of Altera FPGA software, refer to Summary of Altera FPGA Software Licenses Required on page 42.

Navigating Content Through Tasks

Use the following table to navigate this guide through user-tasks:

A	B	C	D	E
Determine Hardware and Software Requirements	Download Altera FPGA Software through the installer or manually	Install Altera FPGA Software	• Evaluate the Quartus Prime Software • Obtain License from SSLC	Review Next Steps

Related Information

Altera FPGA Self-Service Licensing Center on page 8

6.1. Summary of Altera FPGA Software Licenses Required

Altera provides licenses for purchased versions of the Quartus Prime software, the Questa* Altera FPGA Edition software, purchased IP cores, development kits, and university program software.

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

Table 10. Summary of Altera FPGA Software Licenses Required

Software/IP	License Key Required?	Licensing Cost Involved?	Description
Quartus Prime Lite Edition	**No**	**No**	Supports fixed node and floating license. For more information, refer to Quartus Prime Software License on page 45.
Quartus Prime Pro Edition Quartus Prime Standard Edition	**Yes**	**Yes**	Supports fixed-node and floating licenses. You must choose one of them. *Note:* There is a single ordering code for Quartus Prime Software, which includes licenses for both Pro and Standard Editions. Buy subscription from one of the distributors under Distributor Partner Directory. For more information, refer to Quartus Prime Software License on page 45.
Questa* Altera FPGA Starter Edition	**Yes**	**No**	Supports fixed-node and floating license. The Questa* Altera FPGA Starter Edition Software does not support companion licenses. For additional information, refer to Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License on page 45.
Questa* Altera FPGA Edition	**Yes**	**Yes**	Free with the purchase of the Quartus Prime software. Supports fixed-node and floating licenses. Questa* Altera FPGA Edition does not support companion licenses. You can also purchase the license separately from one of the distributors under Distributor Partner Directory. License fee includes 12 months of maintenance renewable annually. For additional information, refer to Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License on page 45.
Simics Simulator for Altera FPGAs	**No**	**No**	Needs no additional license. You just need to accept the EULA when you install the software.
FPGA AI Suite	**Yes**	**Yes** (only for unlimited inferences)	You can download and run limited number of inferences for free. To run unlimited inferences on hardware, you must purchase FPGA AI Suite software from an Altera authorized distributor. For additional information, refer to FPGA AI Suite: Getting Started Guide.
DSP Builder	**Yes**	**Yes**	A DSP Builder supports fixed-node and floating license. • A 30-day evaluation license is free. For more details, refer to DSP Builder. • After 30-day evaluation license expires, you must purchase the license from an Altera authorized distributor. For additional information, refer to DSP Builder (Advanced Blockset): Handbook.
Advanced Link Analyzer	**No**	**No**	
Power Thermal Analyzer	**No**	**No**	
oneAPI Base Toolkit	**No**	**No**	
Altera SoC FPGA Embedded Development Suite (SoC EDS)	**Yes**	**Yes**	There are no licensing differences between the Pro and Standard SoC EDS editions. None of the components included with the SoC EDS require a license. Only the Arm* Development Studio for SoC FPGA Edition requires a license.

continued...

Software/IP	License Key Required?	Licensing Cost Involved?	Description
			Arm* Development Studio Community Edition supports fixed-node license. • A 30-day evaluation license is free. For more information, refer to Arm* Development Studio for SoC FPGA. • After 30-day evaluation license expires, you must purchase the license from an Altera authorized distributor. For additional information, refer to Altera SoC FPGA Embedded Development Suite (SoC EDS) User Guide.
IP Core	**Varies**	**Varies**	All IPs are enabled for evaluation (at a minimum) upon installation of the combined Quartus Prime software and IP release. Production licenses for the IP Base Suite are provided at no extra charge with each seat license of the Quartus Prime Pro Edition and Quartus Prime Standard Edition design software. Contact your local sales Altera authorized distributor for support. For more details, refer to IP Evaluation and Purchase and Intellectual Property (IP) Cores Licenses on page 49.
Siemens EDA* AXI Verification IP Suite	**Yes**	**Yes**	To access it with the Quartus Prime Lite Edition software, you must upgrade to version 13.1 or higher and purchase seat licenses by contacting your Altera sales representative. For additional information, refer to Siemens EDA* AXI Verification IP Suite License (Altera FPGA Edition) on page 49 and Siemens EDA* AXI Verification IP Suite webpage.

Table 11. Other Licenses

License	Description
Development Kits Containing the Quartus Prime Software	Development kits that include the Quartus Prime software include instructions for obtaining a license for that software. You can purchase development kits from the Development Kits, Daughter Cards & Programming Hardware website. For more information, refer to Altera FPGA Development Kits.
University Program Software Licenses	The University Program offers licensed and unlicensed FPGA software to participating universities. For use in teaching, the University Program recommends the Quartus Prime Lite Edition software, which does not require a license. The licensed commercial versions of the Quartus Prime Standard and Pro Edition software are available for installation in university laboratory facilities. Refer to the University Program website for more information. For more information, refer to Altera FPGA Academic Program Teaching Materials.

Quartus Prime Software License on page 45

Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License on page 45

Intellectual Property (IP) Cores Licenses on page 49

Siemens EDA* AXI Verification IP Suite License (Altera FPGA Edition) on page 49

Related Information

- Development Kits, Daughter Cards and Programming Hardware
- University Program

6.1.1. Quartus Prime Software License

Quartus Prime Lite Edition Software License

Altera offers the entry-level Quartus Prime Lite Edition software. This software supports selected devices, provides limited feature support, and **does not** require license files.

Quartus Prime Pro and Standard Edition License

Quartus Prime Pro Edition software and Quartus Prime Standard Edition software require a paid license (fixed or floating). However, before you purchase the license, you can obtain a 90-day no-cost evaluation license from the Altera FPGA Self-Service Licensing Center that allows using the Quartus Prime software for 90 days for evaluation, including generating bitstream.

Note: The Quartus Prime Pro Edition software provides a free mode which supports Cyclone 10 GX devices only.

When you purchase an Quartus Prime subscription from one of the distributors, you choose between a single-user license that is fixed to a specific computer, or a multi-user floating license.

- *Fixed license*—A stand-alone (node-locked, single-user) license is tied to the network interface card (NIC) ID of the computer on which you installed the software.
- *Floating license*—A floating network (multi-user) license is for users running the Quartus Prime software on multiple computers connected on a network. A license server issues licenses to computers on demand. Floating licenses are not specific to an operating system. If you want to run the FPGA software on additional computers, you can purchase additional seats to add to your floating license after its original purchase.

A subscription provides access to the Quartus Prime Pro Edition software, Quartus Prime Standard Edition, and the Questa* Altera FPGA Edition with one year of maintenance support. Once you have bought the subscription from one of the distributors, your order gets processed, and license entitlements are set up and managed in the Altera FPGA Self Service Licensing Center (SSLC), where you can generate a `license.dat` file. This file enables you to use the software or IP product licensed, all previous versions, and all versions released within one year of the purchase date. For additional information, refer to Using the Altera FPGA Self-Service Licensing Center on page 67.

Related Information

License.dat Example Files on page 92

6.1.2. Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License

Both Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition require a valid software license. This license expires 12 months after the date of purchase. If you are looking for instructions to download and install Questa* Altera FPGA Edition

and Questa* Altera FPGA Starter Edition, follow the instructions provided in *"Downloading Individual Executable Files"* section to obtain the software and *"Installing Altera FPGA Software at the Command Prompt"* section to install.

Generating the License

The Quartus Prime software requires a paid license (fixed or floating license subscription). A paid subscription provides access to the Quartus Prime Pro Edition software, Quartus Prime Standard Edition, and the Questa* Altera FPGA Edition with one year of maintenance support. Once you have bought the subscription from one of the distributors, your order gets processed, and license entitlements are set up and managed in the Altera FPGA Self Service Licensing Center (SSLC), where you can generate a `license.dat` file. This file enables you to use the software or IP product licensed, all previous versions, and all versions released within one year of the purchase date.

Questa* Altera FPGA Starter Edition license is free and you can directly obtain it from the Quartus Prime GUI or Altera FPGA Self Service Licensing Center (SSLC).

If you do not have access to SSLC, you must first complete registering to SSLC and create an account by visiting Register for Altera FPGA Self Service Licensing Center (SSLC).

Note:

- The Questa* Altera FPGA Edition software does not support companion licenses. You must purchase additional licenses if you want to run the software on other computers.
- The Questa* Altera FPGA Edition software license does not support Remote Desktop access with node-locked, uncounted licenses.

Follow these steps to generate the license:

1. Go to the Altera FPGA Self-Service Licensing Center (SSLC).
2. Select the **Sign up for Evaluation or Free Licenses** option on the menu bar.
3. In the list of products displayed, select the desired option:
   - For Questa* Altera FPGA Starter Edition license, select the **Questa* Altera FPGA Starter Edition(License: SW-QUESTA)** option.
   - For Questa* Altera FPGA Edition license, select **Quartus Prime Software 90-Day Evaluation (Standard and Pro Editions) (License: EVALUATION-LIC)** option.
4. Under the **# of Seats** column, enter the number of seats you require.
5. Read the license terms of use.
6. Select the "I have read and agree to the terms of use of this license as listed below" check box.
7. Click **Get License**. A dialog box prompts you to specify the computer for license assignment. You can use one of the following options:

- **Option 1**: Click **Create a New Computer** if you want to assign the license to a new computer. You must provide information about the required hardware and license type. For information about the license type, refer to Altera FPGA Software License Types. For information about how to extract information about your computer hardware, refer to Hardware Information Required When You Request a License.
- **Option 2**: Click **Assign an Existing Computer** and search for the computer name/NIC ID that you have created previously in your **My Altera** account. To view your list of computers, use any of the following options:
  - ☐ Visit the **License Assistant** and select **Regenerate License by Primary Computer ➤ View all computers and select**
  - ☐ On the SSLC menu bar, click **Computers and License Files** and select the desired option.

8. Click **Generate**. You receive an email with the license attached to your registered email address.
9. Save the `license.dat` file on your computer (for example, `~/altera_pro/LR-xxxxxx_License.dat`).

Note: Before using Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition software, you must set an environment variable to point to the location of the license.

Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition licenses appear as a `FEATURE` line inside the Quartus Prime `license.dat` file.

Setting Up the Questa* Altera FPGA Starter Edition Software License

`LM_LICENSE_FILE` and `MGLS_LICENSE_FILE` are deprecated in this release. Future versions of Questa* Altera FPGA Edition will not read them. Set the license through `SALT_LICENSE_SERVER`. The format for `SALT_LICENSE_SERVER` is different from the `LM_LICENSE_FILE` on Windows or `MGLS_LICENSE_FILE` on Linux. The separator between license server locations must be semicolon (;) instead of a colon (:).

After you receive and save the `license.dat` file on your computer, follow these instructions:

Note: The Questa* Altera FPGA Edition software license does not support Remote Desktop access with node-locked, uncounted licenses.

On Windows System

1. Go to **This PC**, right-click, and select **Properties**.
2. Click **Advanced System Setting**.
3. In the **Advanced** tab, select **Environment Variable**.
4. Under **System variables**, create a new variable with the name as `SALT_LICENSE_SERVER` and value as `<license.dat file path>`.
5. Click **OK** and restart the Questa* Altera FPGA Edition software.

Alternatively, open a command prompt and run the following command to set up the `SALT_LICENSE_SERVER` environment variable:

```
setx SALT_LICENSE_SERVER <path_to_license_file>;%SALT_LICENSE_SERVER%
```

For example: `setx SALT_LICENSE_SERVER C:\intelFPGA\license.dat;%SALT_LICENSE_SERVER%`

You can also set up the Questa* Altera FPGA Edition software license using the Siemens EDA license daemon `saltd`, which you can download from the FPGA Software Download Center. Before starting the Questa* Altera FPGA Edition software, set the `SALT_LICENSE_SERVER` environment variable to the location and file name of the Questa* Altera FPGA Edition license file. For example:

```
SALT_LICENSE_SERVER (<Questa installation directory>/licenses/eda/license.dat)
```

or with the *<port>@<hostname>* notation, where *<port>* is the license port number and *<hostname>* is the license server's host name. For example, `1900@my_lic_server`.

Warning: In the Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition version 2024.1, you might see errors if you are using the *mgcld* daemon version 11.16.4. For solution, refer to https://www.intel.com/content/www/us/en/support/programmable/articles/000099465.html

On Linux System

Run one of the following commands in a command prompt window:

```
export SALT_LICENSE_SERVER=<path to license>:$SALT_LICENSE_SERVER
```

```
setenv SALT_LICENSE_SERVER "<path_to_license_file>"
```

You can also set up the Questa* Altera FPGA Edition software license using the Siemens EDA license daemon `saltd`, which you can find in the *<Questa – Altera FPGA Edition installation directory>/* directory. Before starting the Questa* Altera FPGA Edition software, set the `SALT_LICENSE_SERVER` environment variable to the location and file name of the Questa* Altera FPGA Edition license file. For example:

```
SALT_LICENSE_SERVER (<Questa installation directory>/licenses/eda/license.dat)
```

or with the *<port>@<hostname>* notation, where *<port>* is the license port number and *<hostname>* is the license server's host name. For example, `1900@my_lic_server`.

Warning: In the Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition version 2024.1, you might see errors if you are using the *mgcld* daemon version 11.16.4. For solution, refer to https://www.intel.com/content/www/us/en/support/programmable/articles/000099465.html

Renewing the License

The software license expires 12 months after the date of purchase. To renew an expired license file, revisit the SSLC. You can renew a license only for the version that you purchased.

6.1.3. Intellectual Property (IP) Cores Licenses

You only need to purchase a full production license for licensed FPGA IP cores after completing hardware testing and you are ready to use the IP in production. The rest of the time, you can use the FPGA IP Evaluation Mode feature to evaluate IP cores.

You must purchase the license and generate a full production license key before you can generate an unrestricted device programming file. During FPGA IP Evaluation Mode, the Quartus Prime Compiler only generates a time-limited device programming file (`<project name>_time_limited.sof`) that expires at the time limit.

Altera licenses IP cores on a per-seat, perpetual basis. The license fee includes first-year maintenance and support. You must renew the maintenance contract to receive updates, bug fixes, and technical support beyond the first year.

Purchase a license through your local sales office or distributor. Altera FPGA partners can deliver third-party IP cores for evaluation with the appropriate license.

Important: IPs that require no-cost licenses are supported for license auto-fetching during Quartus Prime Pro Edition software compilation during synthesis stage. For more information, refer to Method 3: Auto-Fetching During Quartus Prime Pro Edition Compilation Process on page 55.

Related Information

- AN 320: Using Altera IP Cores Evaluation Mode
- AN 343: OpenCore Evaluation of AMPP Megafunctions
- Intellectual Property and Reference Designs
- Altera FPGA Self-Service Licensing Center

6.1.4. Siemens EDA* AXI Verification IP Suite License (Altera FPGA Edition)

The Siemens EDA* (formerly Mentor Graphics) Verification IP Suite (Altera FPGA Edition) provides bus functional models for simulation and verification. This software requires a license file.

If you are using a legacy version of Quartus II software (prior to version 12.1) and want to upgrade your software, you must regenerate your license file before continuing to use the Siemens EDA* AXI Verification IP Suite (Altera FPGA Edition) Bus Functional Models (BFMs) and Inline Monitor (Siemens EDA* Verification IP Suite (Altera FPGA Edition)) for AXI4.

Notice: The Siemens EDA* AXI Verification IP Suite License (Altera FPGA Edition) no longer supports the AXI3 BFM.

To access Siemens EDA* Verification IP Suite (Altera FPGA Edition) with the Quartus Prime Lite Edition software, you must upgrade to version 12.1 or higher and purchase seat licenses by contacting your Altera sales representative.

6.2. Evaluating the Quartus Prime Software

You can evaluate the Quartus Prime software at no cost for a period of time.

When you start the Quartus Prime software, if the software cannot detect a valid license file, the **License Setup Required** dialog box displays with the available licensing options:

Figure 5. License Setup Required Dialog

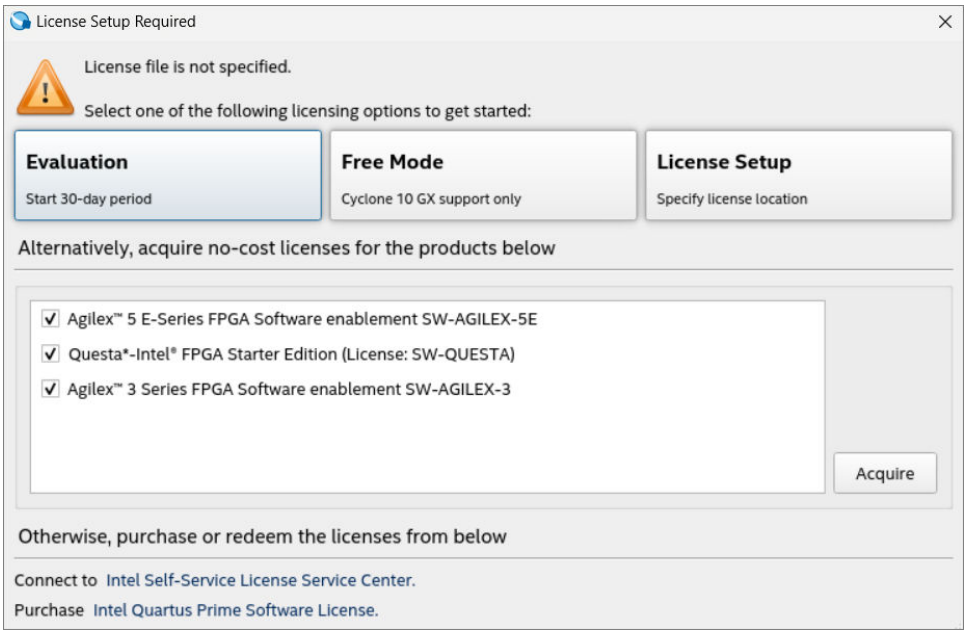


Table 12. Quartus Prime Software Licensing Options

Option	Description
Evaluation (Start 30-day period)	Allows using the Quartus Prime software for 30 days and generating bitstream. *Tip:* Select this option to evaluate the software before purchase.
Free Mode (Cyclone 10 GX support only)	Supports only Cyclone 10 GX devices. *Note:* This mode is available only for the Quartus Prime Pro Edition software.
License Setup (Specify License Location)	Displays the **License Setup** page of the **Options** dialog box where you can specify the location of your license file.
Agilex 5 E-Series FPGA Software Enablement **Questa* Altera FPGA Starter Edition** **Agilex 3 FPGA Software Enablement**	Depending on your selection and clicking **Acquire**, you can acquire a free license for Agilex 5 E-Series FPGA, Agilex 3 FPGA, and/or Questa* Altera FPGA Starter Edition software. Alternatively, once you launch the Quartus Prime Pro Edition software, you can obtain these free license files using the **Get no-cost licenses** button by navigating to **Tools ➤ License Setup**. For more information about these options, refer to Acquiring Free, No-Cost Licenses on page 51.
Connect to the Altera Self-Service Licensing Center	Launches the Altera FPGA Self-Service Licensing Center, which allows you to view, request, activate, and manage your software licenses.
Purchase Quartus Prime software license	Launches your default Internet browser and displays the Buy Design Software page, where you can view and purchase software.

Related Information

Agilex 5 E-Series No Cost License

6.3. Acquiring Free, No-Cost Licenses

Earlier, you had to log into the Altera FPGA Self-Service Licensing Center (SSLC), submit an online request, download the free license, and attach it to Quartus Prime Pro Edition software manually. With this no-cost licensing flow, you can now acquire eligible no-cost product licenses directly through the Quartus Prime Pro Edition software GUI without going through the Altera SSLC and registration process. The license is saved on your target system.

The no-cost licensing flow supports the following use cases:

- Access eligible no-cost products only through fresh Quartus Prime Pro Edition software installations.
- Access eligible no-cost products as additional devices or IPs with other licenses installed.

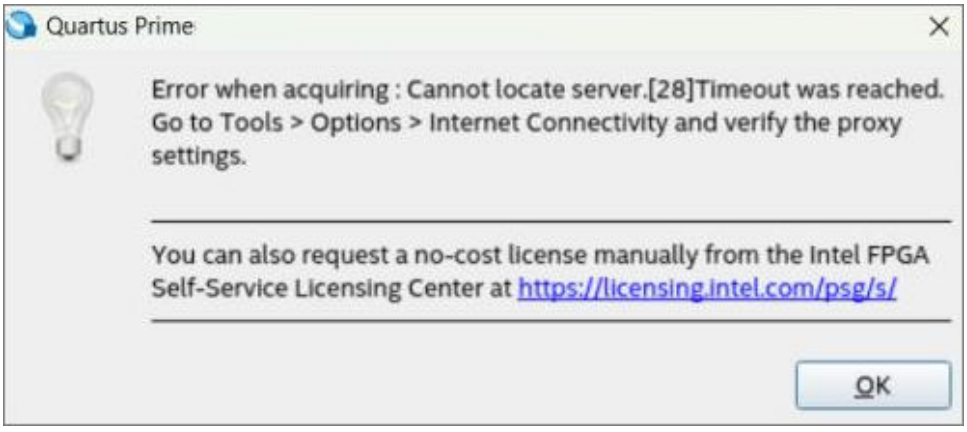
The no-cost licensing flow uses `libcurl` and web API calls to obtain one the licenses using the product ID and host ID (network card Mac address).

You can obtain the free, no-cost license in the following ways:

- Method 1: At the Quartus Prime Pro Edition Software Launch on page 52
- Method 2: License Setup Screen in the Quartus Prime Pro Edition Software GUI on page 53
- Method 3: Auto-Fetching During Quartus Prime Pro Edition Compilation Process on page 55

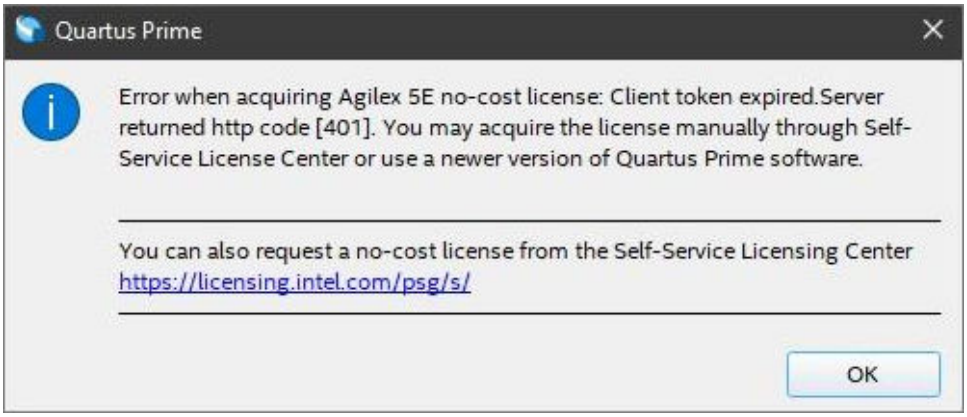
For Methods 1 and 2, if the license acquisition is successful, the Quartus Prime Pro Edition software launches in the licensed mode. If the license acquisition fails, you might see either the timeout or the token expiry error messages. Acknowledging the error launches the Quartus Prime Pro Edition software without the no-cost license.

Figure 7. Timeout Error



Depending on your network configurations, you may need to set up the proxy server's address and port number via the **Tools ➤ Options ➤ Internet Connectivity** page. You must use only HTTP protocol, using which the Quartus Prime Pro Edition software connects to the proxy server to reach the license server endpoint using HTTPS protocol.

Figure 8. Token Expired error

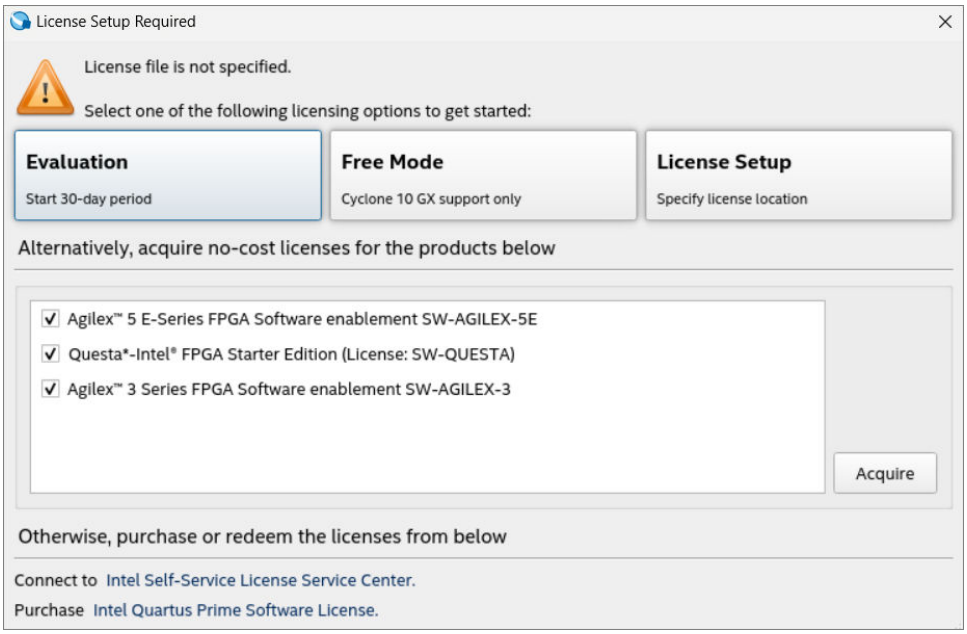


When the Quartus Prime Pro Edition software connects to the license server to acquire the no-cost license, it checks with the server if the token used to connect with the license server is still valid. If the token has expired, Quartus Prime Pro Edition software cannot connect to the license server to acquire any licenses. You can then acquire the licenses manually via the Altera FPGA Self-Service Licensing Center (SSLC) or install the latest Quartus Prime Pro Edition software. For additional information about this token, refer to Refreshing License Server Token Automatically on page 56.

Method 1: At the Quartus Prime Pro Edition Software Launch

Launching the Quartus Prime Pro Edition software without a license prompts you with the **License Setup Required** dialog.

Figure 9. License Setup Required Dialog



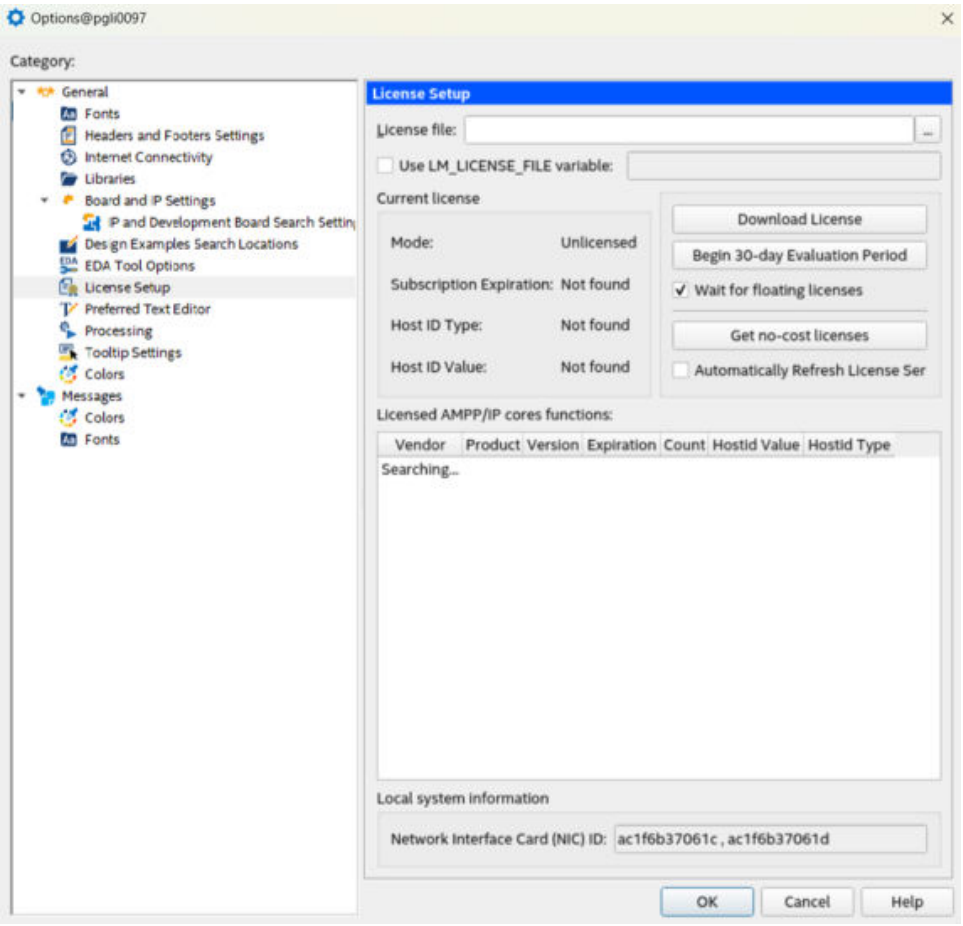
To obtain licenses for Agilex 5 E-Series FPGA, Agilex 3 FPGA, and Questa* Altera FPGA Starter Edition from this dialog and set up the license, perform these steps:

1. Select the desired licenses.
2. Click **Acquire**.
3. If you selected Questa* Altera FPGA Starter Edition, after acquiring the license, set the `SALT_LICENSE_SERVER` environment variable.
   - **On Linux**: Set `SALT_LICENSE_SERVER` to *<home directory>*`/.altera.quartus/quartus2_lic.dat`.
   - **On Windows**: Set `SALT_LICENSE_SERVER` to *<home directory>*`/quartus2_lic.dat`.
4. Click **OK**. Upon successful acquisition, the Quartus Prime Pro Edition software launches in the licensed mode.

Method 2: License Setup Screen in the Quartus Prime Pro Edition Software GUI

From within the Quartus Prime Pro Edition software GUI, navigate to **Tools ➤ License Setup**.

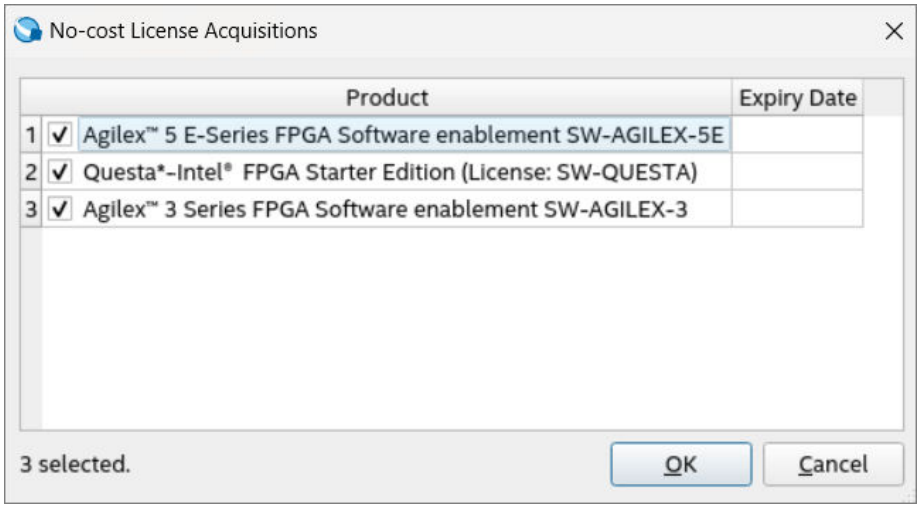
Figure 11. License Setup Options Dialog



To obtain licenses for Agilex 5 E-Series FPGA, Agilex 3 FPGA, and Questa* Altera FPGA Starter Edition from this dialog:

1. Click the **Get no-cost licenses** button. The **No-cost License Acquisitions** dialog appears.

Figure 12. No-cost License Acquisitions Dialog



2. Select the desired licenses.
3. Click **OK** to proceed with the license acquisition. During acquisition, a progress bar displays at the bottom of the screen.
4. After acquiring the Questa* Altera FPGA Starter Edition, set the `SALT_LICENSE_SERVER` environment variable.
   - **On Linux**: Set `SALT_LICENSE_SERVER` to *<home directory>*`/.altera.quartus/quartus2_lic.dat`.
   - **On Windows**: Set `SALT_LICENSE_SERVER` to *<home directory>*`/quartus2_lic.dat`.

Method 3: Auto-Fetching During Quartus Prime Pro Edition Compilation Process

The Quartus Prime Pro Edition software can now obtain no-cost licenses for the following IPs automatically during the compilation at the Synthesis stage:

- Nios V/m Microcontroller Altera IP
- Nios V/g General Purpose Processor Altera IP
- Nios V/c Compact Microcontroller Altera IP
- MIPI CSI 2 Altera IP
- MIPI DSI 2 Altera IP
- AXI Multichannel DMA Altera IP for PCI Express*

This means that although you can manually fetch their no-cost licenses from SSLC manually, the Quartus Prime Pro Edition software now has the capability to verify these IPs for licenses during compilation and automatically fetch them from SSLC if they are missing.

Note: On Linux, if you auto-fetched the licenses on one system and later switched to a new system, the auto-fetching does not work on the new system. To make auto-fetching work on the new system, you must delete the cached license file available at *<home>*/.altera.quartus/quartus2_lic.dat.

If the auto-fetching is successful, the Quartus **Message** pane displays the following message when compiling your design:

```
Info(<message_ID>): Successfully acquired license for <IP_name (ID)>.
```

If the auto-fetching is unsuccessful, the Quartus **Message** pane displays the following warning:

```
Warning(24850): Failed to acquire license for <IP_name (ID)>.
Error when acquiring <IP_name>: Cannot locate server. [28]Timeout was reached.
Go to Tools > Options > Internet Connectivity and verify the proxy settings.
You can also request a no-cost license manually from the Altera FPGA Self-Service
Licensing Center
```

6.3.1. Refreshing License Server Token Automatically

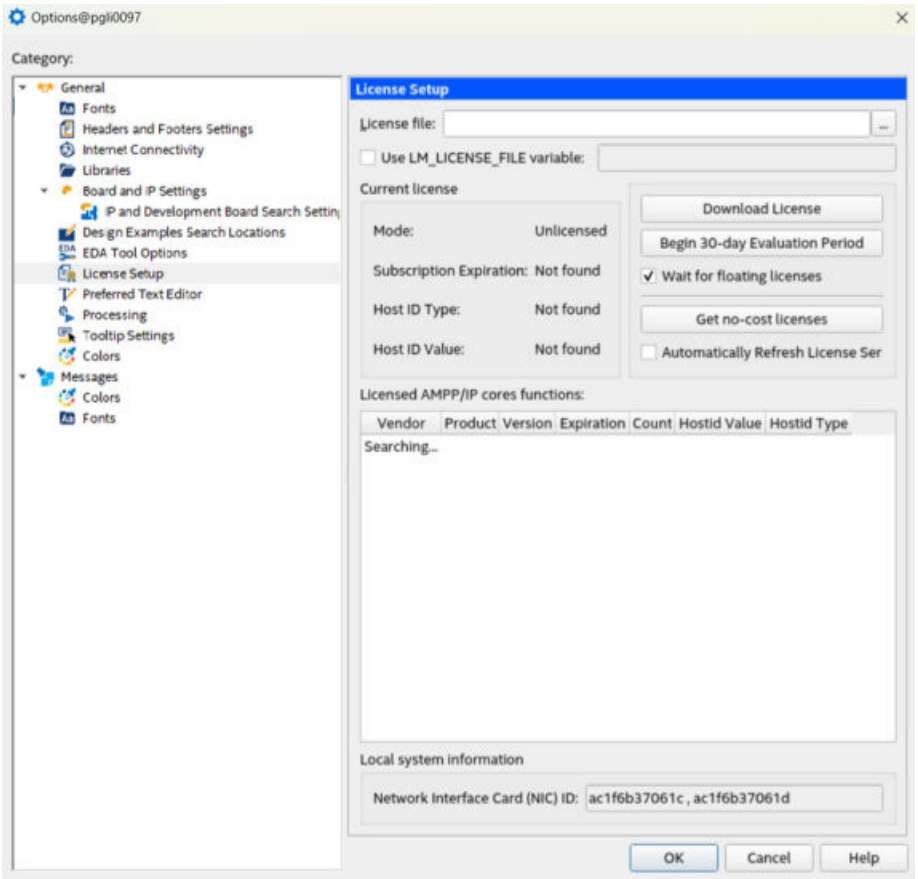
The automatic server token refresh flow works with the free, no-cost license for eligible products and ensures that your local Quartus Prime Pro Edition software installation is updated with the security token used to communicate with the licensing server. Refer to Acquiring Free, No-Cost Licenses on page 51 for more information about the no-cost licenses.

When acquiring the no-cost license, the Quartus Prime Pro Edition software connects to the license server and verifies whether the token used to connect with the license server is still valid. If a new token is available, the application accepts and saves it securely and uses it in the subsequent calls to the license server.

The automatic server token refresh flow is disabled by default. It is enabled when you meet the following use cases:

- You have successfully acquired a no-cost license for eligible products through the Quartus Prime Pro Edition software GUI.
- You have not manually disabled the **Automatically Refresh License Server Token** checkbox in the **License Setup** page.

Figure 13. License Setup Options Dialog



Note:

- Every Quartus Prime Pro Edition software release has a different token embedded within and each token has one year validity. For example, suppose the token was issued on February 1, 2024, it expires on Jan 31, 2025, which means the token expiry is not dependent on the software release date but on the date the token was issued.
- Every time you launch the Quartus Prime Pro Edition software, and you have the **Automatically Refresh License Server Token** option enabled in the **License Setup** page, the Quartus Prime Pro Edition software connects to the licensing server to verify if any new token is available for download. If a new token is available, it downloads the new token and saves it to your local installation. For example, on July 15, 2024, the Quartus Prime Pro Edition software connects to refresh a token that was generated on May 1, 2024, giving your local installation the ability to acquire a no-cost license valid until April 30, 2025. This means that the token validity does not start from the date it is refreshed, but the date on which it was generated or created.

6.4. Licensing Altera FPGA Software Walkthrough

To set up a license for Altera FPGA software, follow these steps:

1. Obtaining Necessary Hardware Information on page 58
2. Requesting a License File from the Altera FPGA Self-Service Licensing Center on page 59
3. Depending on the type of license that you use:
   - ☐ Setting up a Fixed License on page 60.
   - ☐ Setting up a License in a Network License Server on page 61.

6.4.1. Obtaining Necessary Hardware Information

To generate a `license.dat` file, Altera requires information about the computer on which you want to install the license.

The following table indicates what information you need, depending on the type of license and the license host's operating system.

Table 13. Hardware Information Necessary for Altera FPGA software licenses

License Type	OS	Hardware Information	Notes
Fixed license	Windows* or Linux	NIC ID or software guard ID	Legacy Quartus II software version 7.2 and earlier support software guards.
Floating license	Windows* or Linux license servers	NIC ID	Use the ID from a physical NIC card, not a virtual ID.
	UNIX license servers	Host ID	Equivalent to the NIC ID.

Important: If you use a floating license server for Altera FPGA software, you must upgrade to the latest license daemon software (v11.18.2.0 or later). Altera FPGA software version 22.1 and later does not work with older versions of the daemon software. To download the latest daemon software, visit the Flexlm License Daemons for Altera FPGA Software web page.

Refer to Getting Hardware Information for License on page 64 for instructions on obtaining the hardware information.

With the hardware information, you are ready to go to the Altera FPGA Self-Service Licensing Center.

Related Information

Getting Hardware Information for License on page 64

6.4.2. Requesting a License File from the Altera FPGA Self-Service Licensing Center

You must log into the Altera FPGA Self-Service Licensing Center to create and manage your licenses. From the **Altera FPGA Self-Service Licensing Center Home** page, use the new License Assistant. You can find this assistant on the bottom right corner of every page. For detailed instructions about how to obtain and manage licenses in the Altera FPGA Self-Service Licensing Center, refer to Using the Altera FPGA Self-Service Licensing Center on page 67.



For new purchases, select one of the following options and navigate through the guided steps to generate and receive the license file by email:

- Find and generate license by License Activation Code
- View your unassigned licenses and generate license

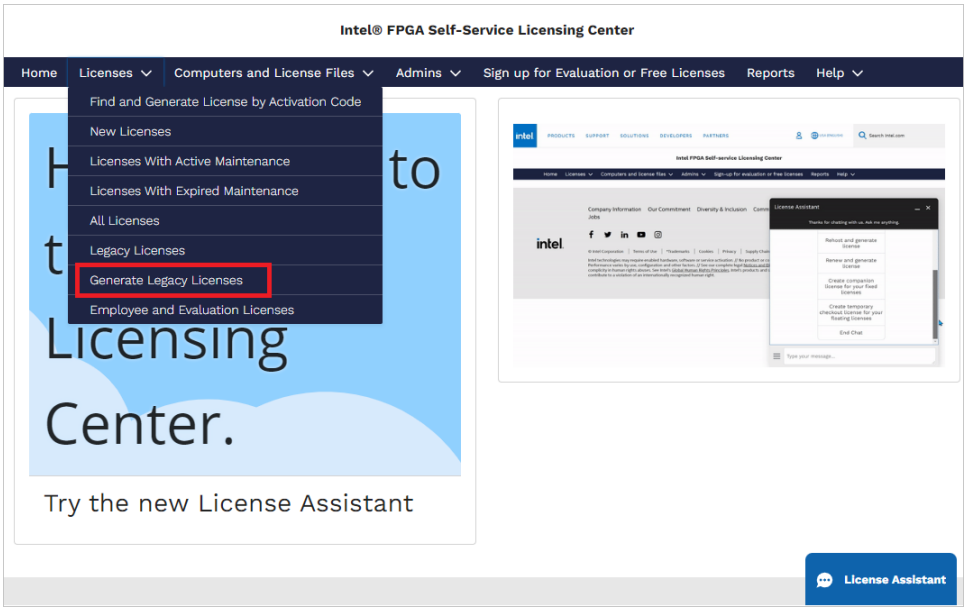
Altera sends the license file to the email address in your account profile.

Note: If you do not receive your license email within 12 hours of requesting a license, or if you do not know all the required information to complete the process, contact customer support via the Help option in the Altera FPGA Self-Service Licensing Center.

If you have a floating, multi-user license, the FLEXlm licensing scheme allows you to set up two redundant license servers to serve licenses. You must request a license file for redundant servers from the Altera FPGA Self-Service Licensing Center.

For software, IP, or Development Kit legacy licenses with a maintenance expiration date prior to January 1, 2009, you can now obtain a license file via the Altera FPGA Self-Service Licensing Center's **Generate Legacy Licenses** option.

Figure 14. Altera FPGA Self-Service Licensing Center



6.4.3. Setting up a Fixed License

To set up a fixed license file (single user, single computer):

1. Create a security copy of any existing `license.dat` file.
2. Save the `license.dat` file on your local hard drive. The preferred location is:
   - ☐ Linux:

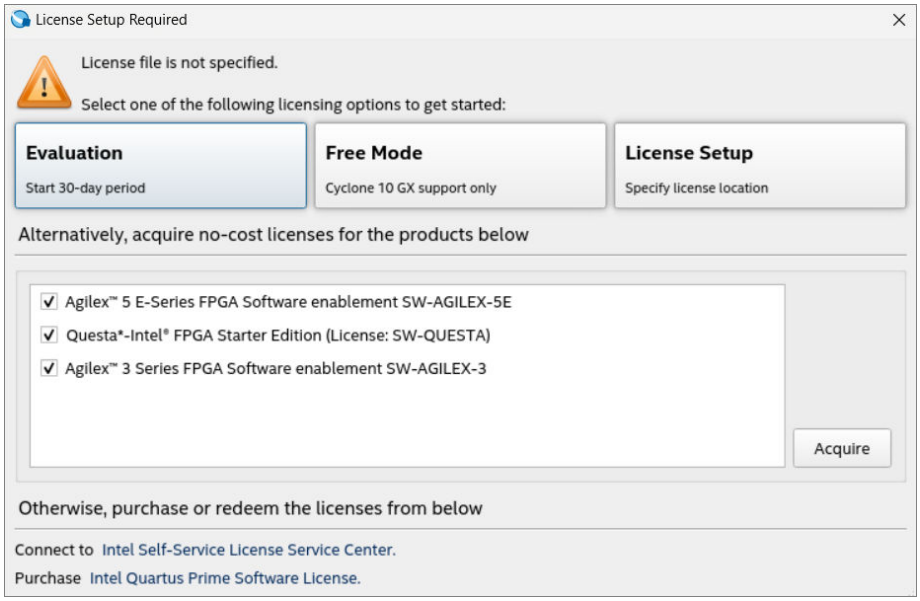
     ```
     /usr/local/flexlm/licenses
     ```

   - ☐ Windows:

     ```
     C:\licenses\flexlm
     ```

   Note: If you save the file in other location, make sure that the path contains only alphanumeric characters, and do not include any special characters or symbols, such as !$%^&*<>, empty spaces, or non-English characters.
3. Start the Quartus Prime software.
   If the Quartus Prime software cannot detect a valid license file, the **License Setup Required** dialog box prompts you to license, evaluate, or buy the software.

Figure 15. License Setup Required Dialog



4. If you have a valid license file, click **License Setup** to specify the location of your license file.
   The **License Setup** page in the **Options** dialog box appears.
5. In the **License file** box, specify the full path name of the `license.dat` file, and click **OK**.

Alternatively, you can specify the license file location by using the `LM_LICENSE_FILE` environment variable. However, the location that you specify in the **License Setup** dialog box supersedes `LM_LICENSE_FILE`.

Related Information

Setting Quartus Prime Environment Variables on page 36

6.4.4. Setting up a License in a Network License Server

6.4.4.1. Setting up a License File in the License Server

Perform the following steps in the license server:

1. Save the `license.dat` file. The preferred location is:
   - ☐ Windows*:

     ```
     <drive>:\flexlm
     ```

   - ☐ Linux:

     ```
     /usr/local/flexlm/licenses
     ```

2. Identify the hostname of the license server.
3. If the license server uses Windows*, specify the port number for the licensing manager.

- ☐ Choose a number outside of the `27000–28000` range, and unique on the machine.

In Linux systems, the FLEXlm license manager automatically chooses a free port between 27000 and 27009.

4. Modify the SERVER line in the `license.dat` file to include the hostname and port number of the license server.

```
SERVER <hostname> <8 or 12-character host or NIC ID> <host port number>
```

5. Identify the path to the `alterad` vendor daemon executable.[4]
   - ☐ Windows*:

     ```
     <installation_directory>\bin64\alterad.exe
     ```

   - ☐ Linux:

     ```
     <installation-directory>/linux64/alterad
     ```

6. Optionally, identify the user-defined port numbers for the `alterad` daemons.

   The FLEXlm software works with Internet firewalls that require you to specify port numbers in the license file on the SERVER line and each VENDOR or DAEMON line. The syntax to specify a port is `PORT=<number>`. Finally, you must allow firewall access to those port numbers.

7. Modify the VENDOR line in the `license.dat` file to include the path to the `alterad` daemon[5] and the daemon's port number.

```
VENDOR alterad <path/to/alterad> [port=<user-defined port number>]
```

8. If you are using a license file for the Questa Altera FPGA Edition software and/or Siemens EDA AXI BFMs, identify the path to the vendor daemon executable `saltd`. [4]
9. (Optional) Identify the user-defined port numbers for the `saltd` daemon.

   Allow firewall access to those port numbers.

10. Modify the VENDOR line in the `license.dat` file to include the path to the `saltd` daemon[5] and the daemon's port number.

```
VENDOR saltd <path/to/saltd> [port=<user-defined port number>]
```

Note: If the *VENDOR* line mentions *mgcld*, you must manually change it to `saltd`. `mgcld` is no longer supported in the Questa Altera FPGA Edition version 2024.1 and later.

When you complete all modifications, ensure the license file conditions are met.

Note: Do not change any of the INCREMENT lines. Leave the INCREMENT lines with the original vendor daemon name (`mgcld`).

---

(4) If the license server does not provide the necessary vendor daemon, copy the required daemon from another machine, save the file in a location on the license server, and specify the daemon location on the license server in the license file.

(5) If the server path has spaces in it, enclose the full path in quotation marks.

Example 2. Setting Up Floating Network Licenses on the License Server

The following example shows how to specify port numbers in a floating license file, where ports `1800`, `1801`, and `1802` provide access through a firewall.

```
SERVER myServer 0123456789ab 1800
VENDOR alterad ./alterad port=1801
VENDOR saltd ./saltd port=1802
```

With the modified `license.dat` file, you can set up the FLEXlm license manager on the license server, and finally start the license server.

6.4.4.2. Launching the Licensing Server

The Quartus Prime software administers licensing for single or multiple users in a network installation with the FLEXlm license manager software.

The requisites to configure a new license server are:

- System administration (Administrator) privileges.
- A valid `license.dat` license file.
- (Linux only) To run the FLEXlm `lmgrd` license server manager, make sure that the `/usr/tmp` directory exists.

To launch a new license server:

1. At a command prompt, type:
   - ☐ Windows*

     ```
     <installation-directory>\bin64\lmgrd -c \path\to\license.dat
     ```

   - ☐ Linux:

     ```
     <installation-directory>/linux/lmgrd -c /path/to/license.dat
     ```

For more information about using the license manager server, refer to *Configuring the License Manager Server*.

Related Information

- Selecting the Installation Path on page 30
- Configuring the License Manager Server on page 96
- Using the Altera FPGA Self-Service Licensing Center on page 67

6.4.4.3. Specifying a Network License Server in the Quartus Prime Software

1. Start the Quartus Prime software.
   If the **License Setup Required** dialog box opens, select **If you have a valid license file, specify the location of your license file**, and click **OK**.
2. Click **Tools ➤ License Setup**.
3. In the **License file** box, specify the port and location of the licensing server, and then click **OK**.

   Use the <*port*>@<*hostname*> notation, where <*port*> is the license port number and <*hostname*> is the server's host name.

Note: Separate multiple license servers and node locking license files with ":" (Linux) or ";" (Windows).

Alternatively, you can specify the license file location by using the `LM_LICENSE_FILE` environment variable. However, the location that you specify in the **License Setup** dialog box supersedes `LM_LICENSE_FILE`.

Note: If you are using the legacy Quartus II software version 7.2 or earlier, you must also use a software guard. Attach the software guard to a parallel or USB port on your computer.

6.5. Getting Hardware Information for License

This appendix shows how to obtain information about the computer on which you want to install the Altera FPGA software license. Most licenses require a network interface card (NIC) ID, which is the physical address of your Ethernet card.

Table 14. Hardware Information Necessary for Altera FPGA software licenses

License Type	OS	Hardware Information	Notes
Fixed license	Windows* or Linux	NIC ID or software guard ID	Legacy Quartus II software version 7.2 and earlier support software guards.
Floating license	Windows* or Linux license servers	NIC ID	Use the ID from a physical NIC card, not a virtual ID.
	UNIX license servers	Host ID	Equivalent to the NIC ID.

6.5.1. Identifying Host's NIC ID

A network interface card (NIC) ID is a 12-digit hexadecimal string embedded in the network card that uniquely identifies a computer. You must identify the NIC ID of the computer that hosts the software installation or the license server.

Note: If you have a triple redundant license server for floating licenses, the first server that you specify is the master. Use the master server's NIC ID or host ID.

6.5.1.1. Finding the NIC ID for Windows

Use the `ipconfig` utility to find the NIC ID on Windows.

- At a command prompt, type:

```
ipconfig /all
```

The command prints network information. The NIC Address appears as **Physical Address**.

Figure 17. NIC ID in Output of ipconfig Command



If the system has more than one network card, you can use the NIC ID of any network card connected to the computer.

6.5.1.2. Finding the NIC ID on Linux

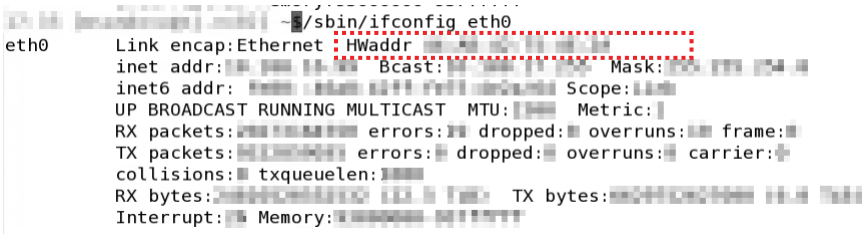
Use the `ifconfig` utility to find the NIC ID on Linux.

- At a command prompt, type:

```
/sbin/ifconfig eth0
```

The command prints network information. The NIC Address appears as **HWaddr**.

Figure 18. NIC ID in the Output of ifconfig Command

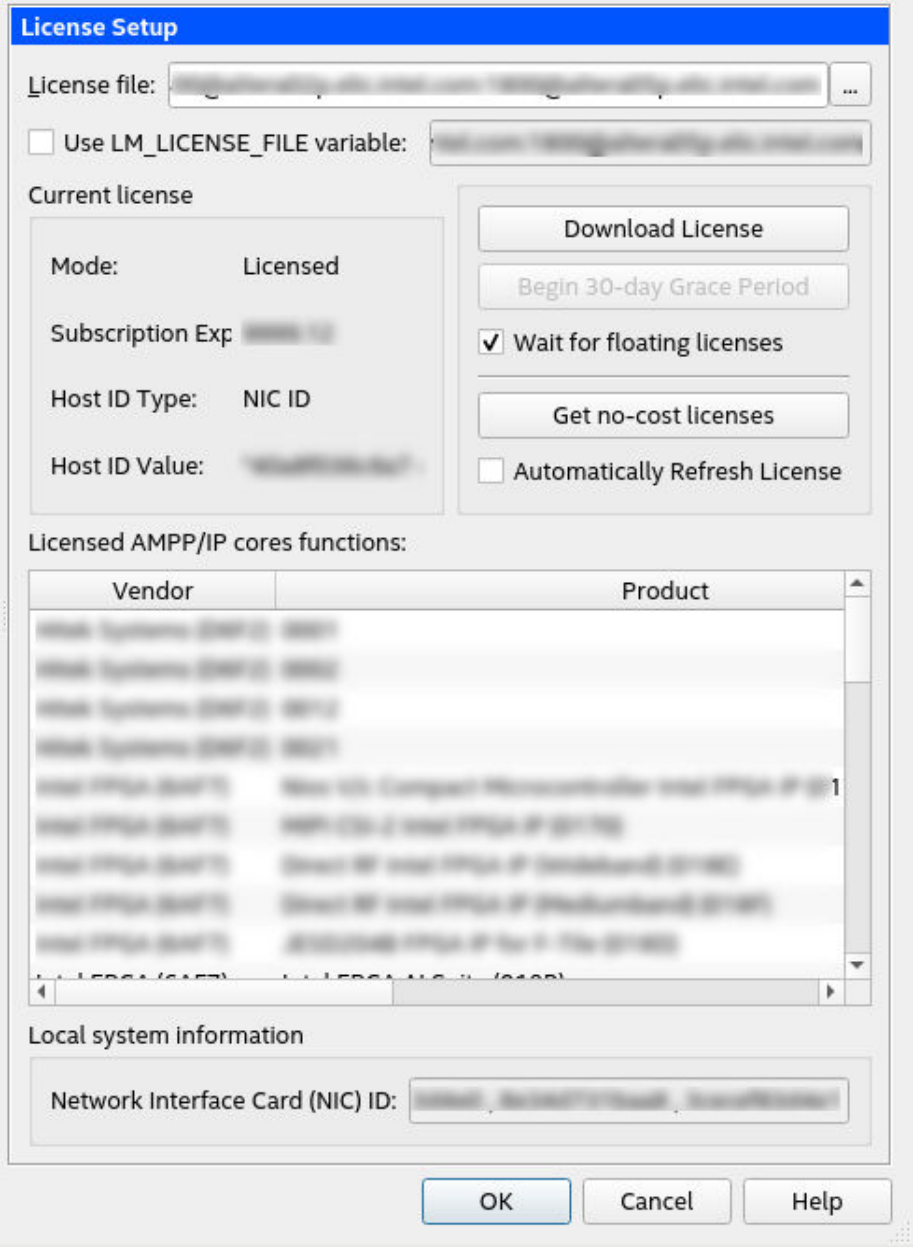


6.5.1.3. Finding the NIC ID in the Quartus Prime Software

Use the following method to find the NIC ID in the Quartus Prime software:

- Display the **License Setup** settings by clicking **Tools ➤ License Setup**.

Figure 19. NIC ID in License Setup Settings



The computer's NIC ID appears under **Local System info** in the **Network Interface Card (NIC) ID** field.

6.5.1.4. Finding the NIC ID on Windows* Using FLEXlm Utilities

- At a command prompt, type:

```
<installation-directory>\bin64\lmutil lmhostid
```

Related Information

Selecting the Installation Path on page 30

6.5.2. Identifying the UNIX Host ID

The license server host ID for UNIX is equivalent to the NIC ID.

6.5.3. Locating Your Hard-Disk Serial Number

Some versions of software use the hard disk serial number for licensing. A hard-disk serial number is an eight-character hexadecimal ID that identifies the PC with the Altera FPGA license. If the ID is not eight characters, include a leading zero.

6.5.3.1. Finding the Hard-disk Serial Number on a Windows* PC

- Type the following command at a command prompt:

  ```
  vol c:
  ```

  In the output of the command, the hard-disk serial number is the volume serial number.

6.5.3.2. Finding the Hard-disk Serial Number if the Quartus Prime Software is Installed on Your Computer

- Click **Tools ➤ License Setup**. The hard-disk serial number appears under **Local Systems info** in the **C: drive serial number** field. Verifying the hard-disk serial number with the Quartus Prime software ensures that you use the correct hard-disk serial number if your PC has multiple hard drives.

6.5.4. Identifying the USB Software Guard ID

Legacy Quartus II software version 7.2 and earlier support software guards. A USB software guard ID is a ten-character alphanumeric number beginning with the letter T.

- If the Quartus Prime software is installed and the guard is attached, click **Tools ➤ License Setup**. Your software guard ID appears under **Local System info** in the **Software Guard ID** field.
- Otherwise, find the software guard ID on the printed label on the guard.

6.6. Using the Altera FPGA Self-Service Licensing Center

The Altera FPGA Self-Service Licensing Center allows you to view, request, activate, and manage your software licenses. You must have an account to access the Self-Service Licensing Center.

Besides requesting licensing files and activating software, you can perform the following tasks with the Self-Service Licensing Center:

- **View existing licenses**: You can view all your existing licenses, including their expiration dates, available rehosts, and licensed users.
- **Renew licenses**: You can renew your existing licenses.
- **Rehost licenses**: You can transfer your existing licenses from one computer to another. You have a total of three rehosts for each license.
- **Add seats to a floating license**: You can add seats to your existing floating licenses.
- **Manage license users**: You can specify the licensed user for each of your existing licenses.
- **View license history**: You can view the license file history for each computer that you add to your account.

Additionally, you can also split a floating license, obtain a checkout license, or obtain a companion license with the Self-Service Licensing Center.

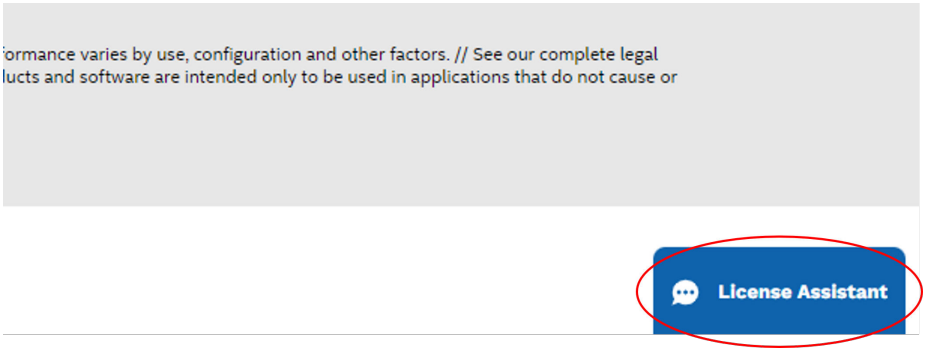
Refer to the following topics for detailed instructions:

Related Information

Altera FPGA Self-Service Licensing Center

6.6.1. Using the License Assistant

You can find the License Assistant link at the right corner of every page within the licensing center. Follow through the steps as guided by the License Assistant to manage your licenses.



You can perform the following tasks using the License Assistant:

Table 15. Performing Licensing Tasks with License Assistant

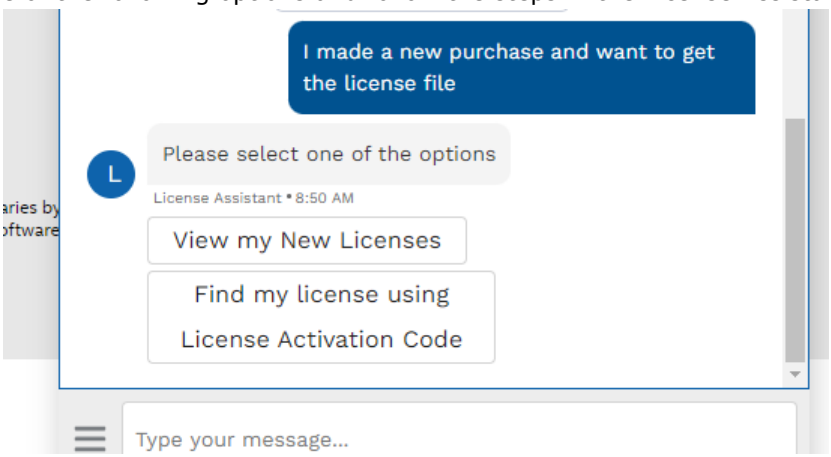
Task	Description
Find and generate license through the License Activation Code	The License Activation Code is the code that you receive in the delivery confirmation email from Altera or the distributor from whom you made the purchase.
View your unassigned licenses and generating the license	Unassigned licenses are your new licenses where you have not assigned a computer yet.

continued...

Task	Description
Regenerate license by Primary Computer	Use this option when you want to generate the latest license file for the licenses associated with your computer.
Rehost and generate a license	Using this option, you can change the computer on a specific license record, or you can edit the computer information (for example, changing primary computer ID, adding redundant or companion servers) and generate license.
Renew and generate a license	If you have recently made a renewal purchase and you want to apply the renewal purchase on your existing license to extend the dates and get the updated license file
Create companion license for your fixed licenses	A companion license grants license portability to a fixed product license. Each fixed product can be assigned up to three computers; an office, a lab, and a home computer if all are only used by the same person.
Create a temporary checkout license for your floating licenses	A checkout license enables portability for users with only floating seat licenses. The license administrator can issue a two-week fixed license, allowing one of the users to have access to the same product licenses from the network while traveling with a laptop.

6.6.2. Getting a License File with Your New Purchase

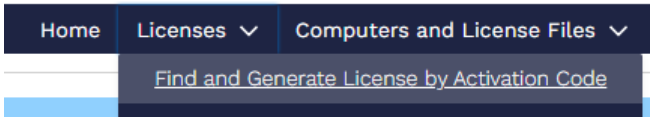
Select one of the following options and follow the steps in the License Assistance:



Option A: Finding Your License by Activation Code and Generating License

Follow these steps:

1. On the menu bar, select **Licenses ➤ Find and Generate License by Activation Code**.

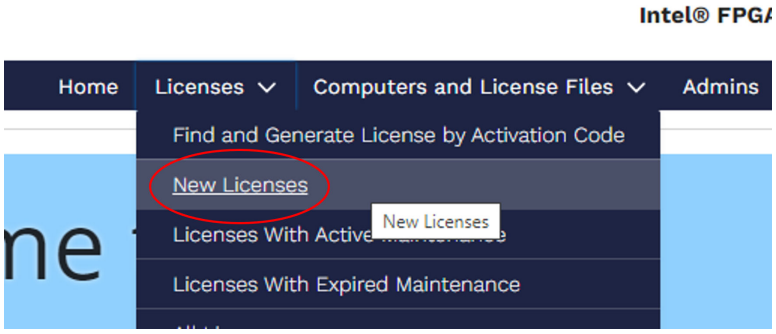


2. Enter the license activation code that you received in the email and click **Search**.
3. Select the license and click **Next**.
4. Assign a computer that you have previously added to your account or add a new computer.

Option B: Viewing Your New Licenses and Generating Licenses

Follow these steps:

1. On the menu bar, select **Licenses ➤ New Licenses**.



2. Click the **License Name** link.
3. On the actions bar, click **Generate License**.
4. Assign a computer that you have previously added to your account or add a new computer.

6.6.3. Viewing Licenses

1. On the menu bar, select **Licenses**. The following default license list view appears.

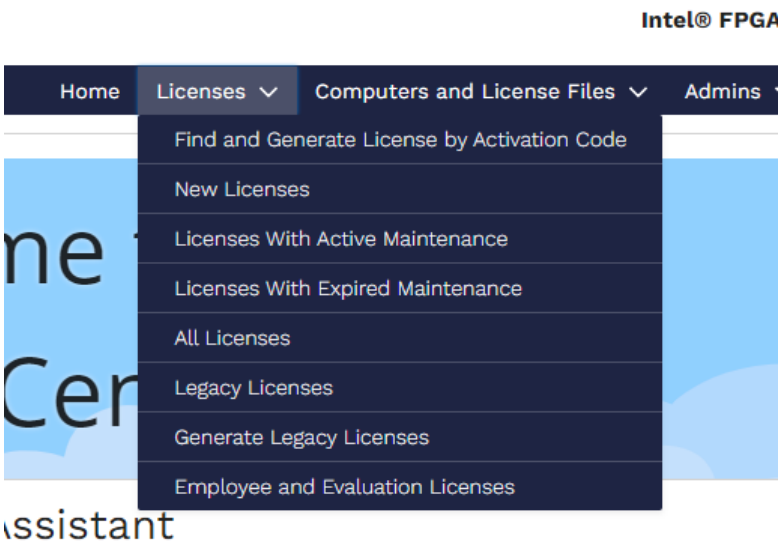


Table 16. Default Licenses

License Type	Description
New Licenses	Use this option to view licenses that do not have a computer assigned yet or your renewal and add seat licenses.
Licenses with Active Maintenance	Use this option to view all licenses that are under active maintenance.
Licenses with Expired Maintenance	Use this option to view all licenses that have not expired but maintenance support has ended.
All Licenses	Use this option to view all licenses in your account excluding the Legacy Licenses.
Legacy Licenses	Use this option to view all non-expired licenses with a maintenance expiration date prior to 1st, January 2009.

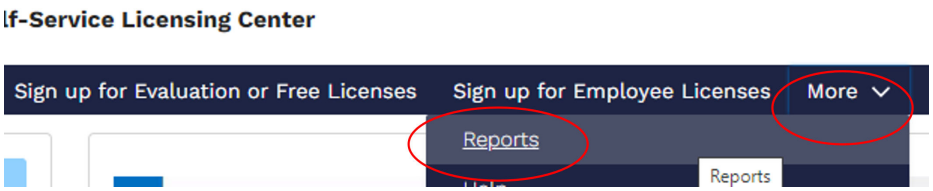
2. To search licenses, select one of the license types under the **License** menu and type the search keyword in the **Search this list** field.
3. Click **Printable View** to export the list of licenses.
4. To view a license detail, click the **License Name** link in any of the license list view as described in the table above .

6.6.4. Filtering Your Licenses

You can use one of the following methods to filter and view your licenses:

Filtering and Exporting Licenses Using Reports

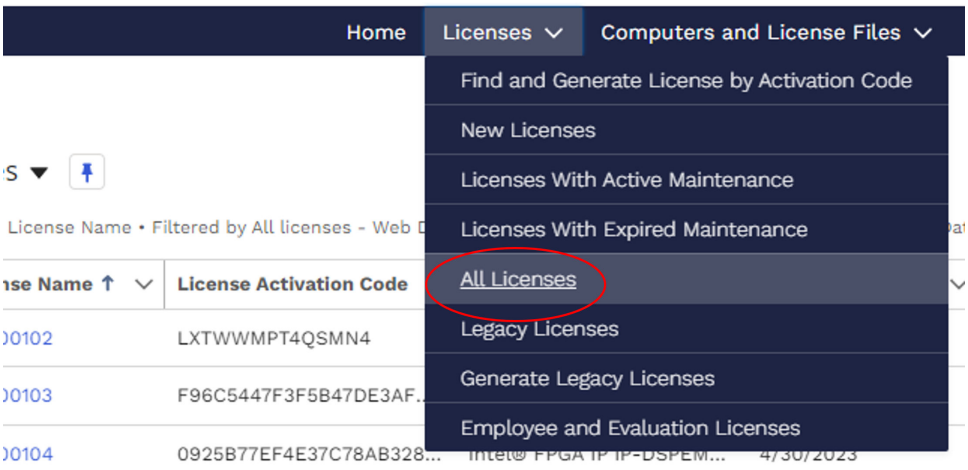
1. On the menu bar, select **Reports**.



2. Select **All Reports** in the **Reports** left navigation. The following license related reports appear:
   - Active Maintenance Licenses
   - All Licenses
3. Click the **All Licenses** report.
4. Click the **Filters** icon in the report to apply filters and view the report.
5. Click the **Export** icon in the report to export the list of licenses either as a formatted report or details only report in the excel format.

Filtering Licenses Using Custom List Views

1. On the menu bar, select **Licenses ➤ All Licenses**.



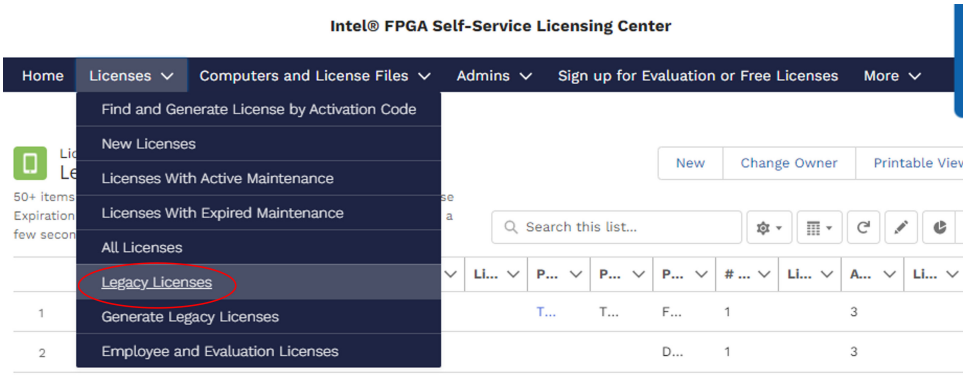
2. Click the **Gear** icon and select **Clone**.
3. Enter a name for your custom list view and click **Save**.
4. Click the **Filter** icon to modify the filter criteria to view the specific licenses that you desire.
5. Click the **Gear** icon and choose **Select Fields to Display** to modify the fields that you see in the list view.

6.6.5. Viewing and Generating a Legacy License

Note: Legacy licenses are non-expired licenses with a maintenance expiration date prior to 1st, January 2009.

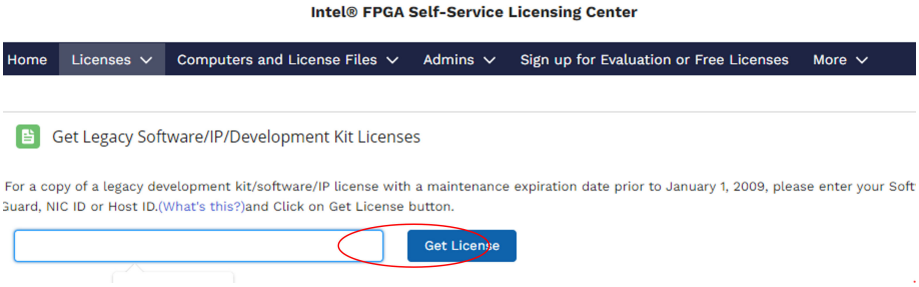
Viewing a Legacy License

On the menu bar, select **Licenses ➤ Legacy Licenses**.



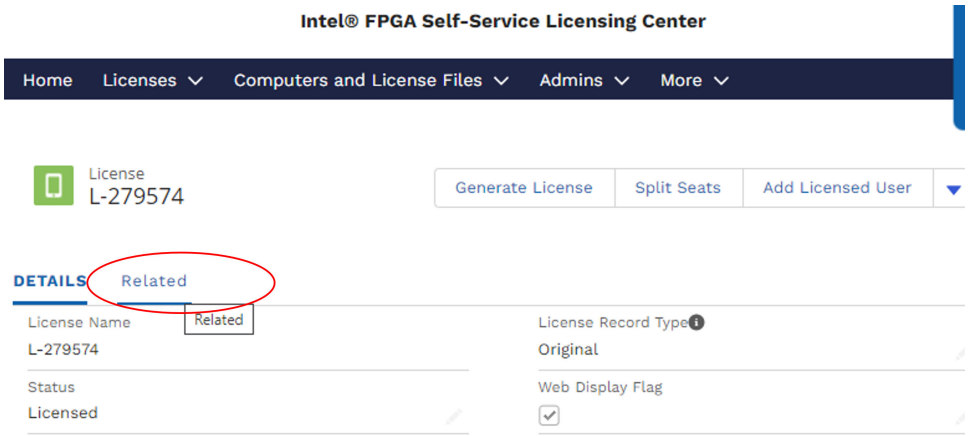
Generating a Legacy License

1. On the menu bar, select **Licenses ➤ Generate Legacy Licenses**.
2. For a copy of a legacy development kit, software, or IP license with a maintenance expiration date prior to January 1, 2009, enter your Software Guard, NIC ID, or Host ID. Refer to Getting Hardware Information for License on page 64 for more information.
3. Click **Get License**.



6.6.6. Viewing Licenses or License Files Associated with a Computer

1. On the menu bar, select **Computers and License Files ➤ Active Computers**.
2. Click the link in the Primary Computer ID column of your computer.
3. Click the **Entitlements** tab.



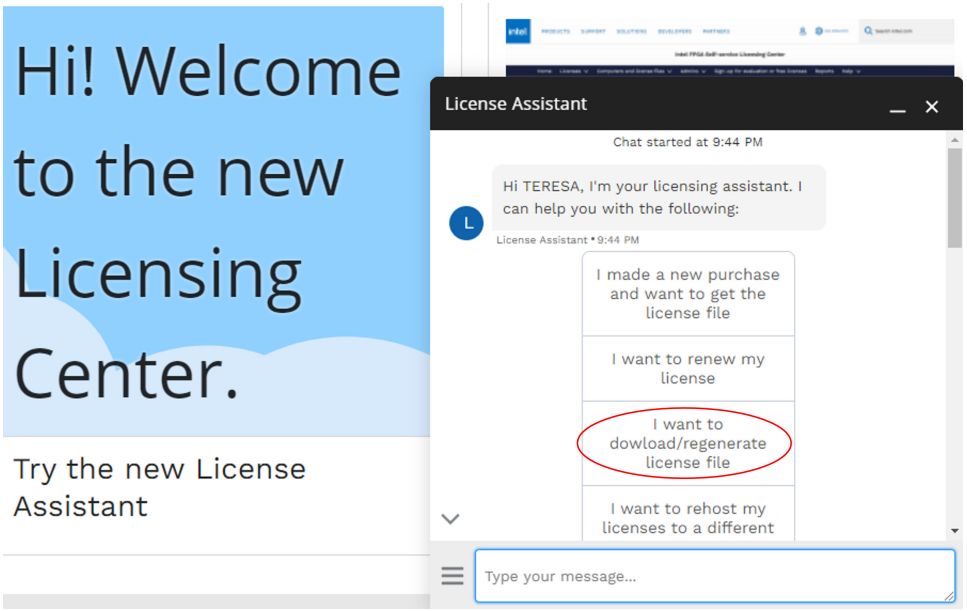
The **Licenses** section displays the licenses associated with the computer. The **License Generation Requests** section displays the previously generated license files for the computer.

6.6.7. Regenerating a License File

Use one of the following methods to regenerate a license file:

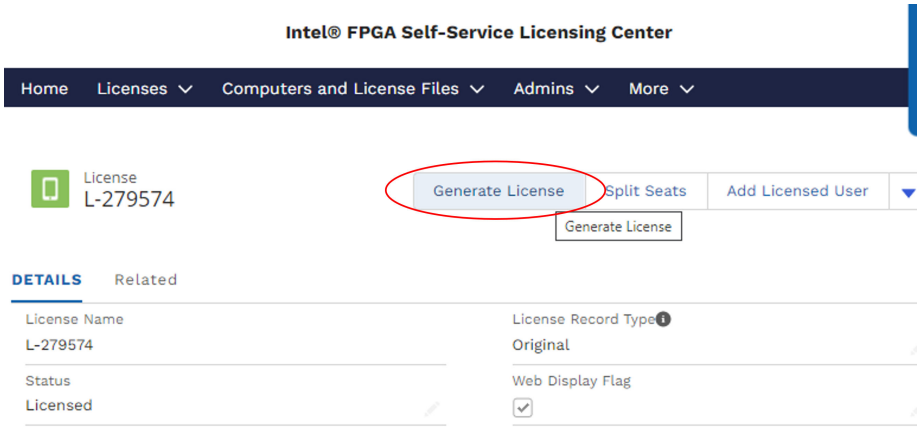
Regenerating License Using the License Assistant

Select the **Regenerate license by Primary Computer** option and follow the steps in the License Assistance.



Regenerating License Using Computers and License Files Menu

1. On the menu bar, select **Computers and License Files ➤ Active Computers**.
2. Click the link in the **Primary Computer ID** column of your computer.
3. On the actions bar, click **Generate License**.



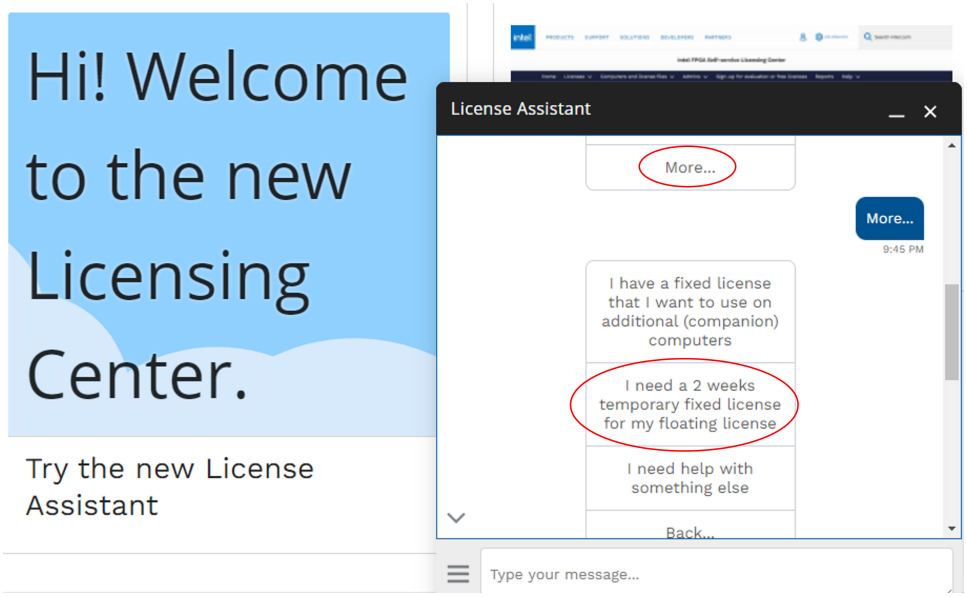
6.6.8. Generating a Temporary Checkout License

Use one of the following methods to generate a temporary checkout license:

Note: A checkout license enables portability for users with only floating seat licenses. The license administrator can issue a two-week fixed license, allowing one of the users to have access to the same product licenses from the network while traveling with a laptop.

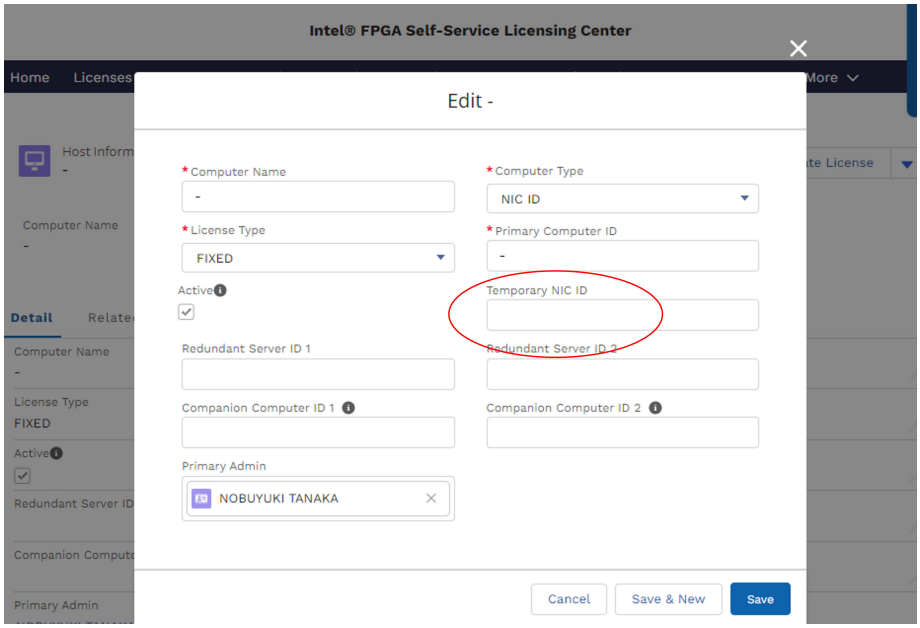
Generating a Temporary Checkout License Using the License Assistant

Select the **Create Temporary Checkout License for Your Floating License** option and follow the steps in the License Assistance.



Generating a Temporary Checkout License by Adding a Temporary NIC ID

1. On the menu bar, select **Computers and License Files ➤ Active Computers**.
2. Click the **Primary Computer ID** link of the computer for which you want to generate a temporary checkout license.
3. Click **Edit**.
4. Enter the Temporary NIC ID.



5. Click **Save**.
6. Click **Generate Temporary Checkout License**.

6.6.9. Generating a Companion License

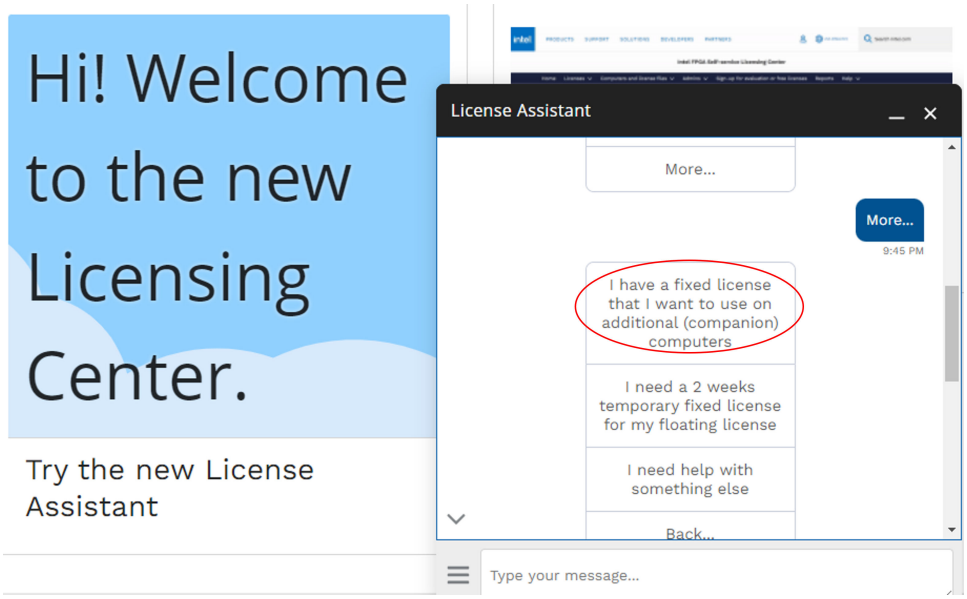
A companion license grants license portability to a fixed product license. Each fixed product can be assigned to up to three computers: an office, a lab, and a home computer if all are only used by the same person.

Use one of the following methods to generate a companion license:

Attention: The Questa* Altera FPGA Edition software does not support companion licenses. You must purchase additional licenses if you want to run the software on other computers.

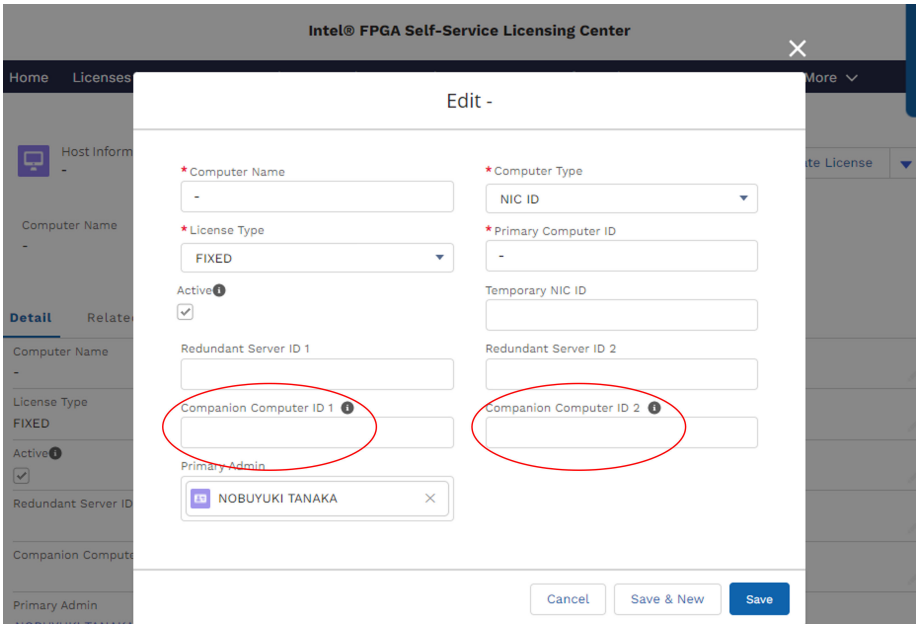
Generating a Companion License Using the License Assistant

Select the **Create companion license for your fixed licenses** option and follow the steps in the License Assistance.



Generating a Companion License by Adding Companion Computer ID Details

1. On the menu bar, select **Computers and License Files ➤ Active Computers**.
2. Click the **Primary Computer ID** link of the computer for which you want to generate a temporary checkout license.
3. Click **Edit**.
4. Enter the Companion Computer ID1 and/or Companion Computer ID2.



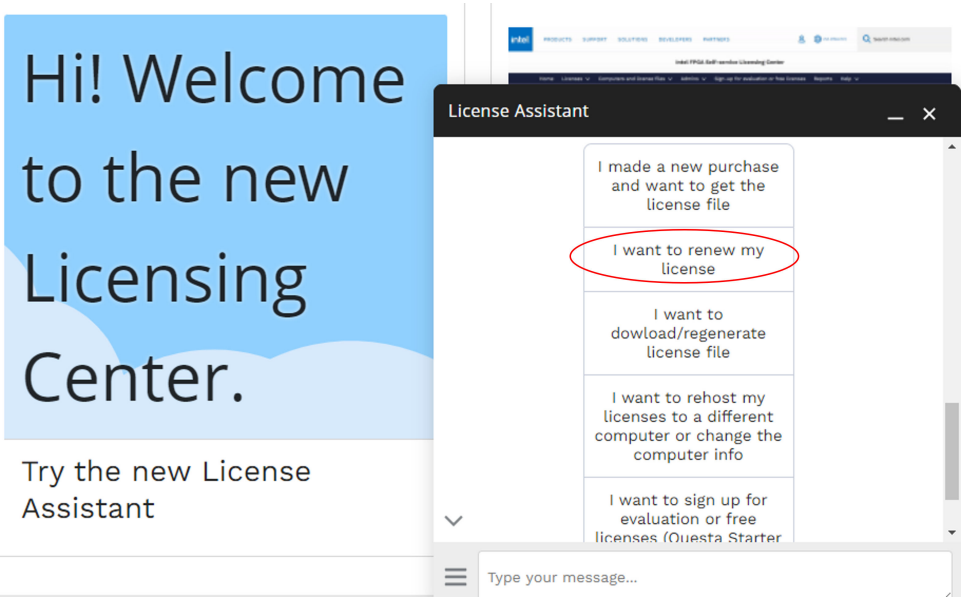
5. Click **Save**.
6. Click **Generate License**.

6.6.10. Renewing Your License

Use one of the following methods to renew your license:

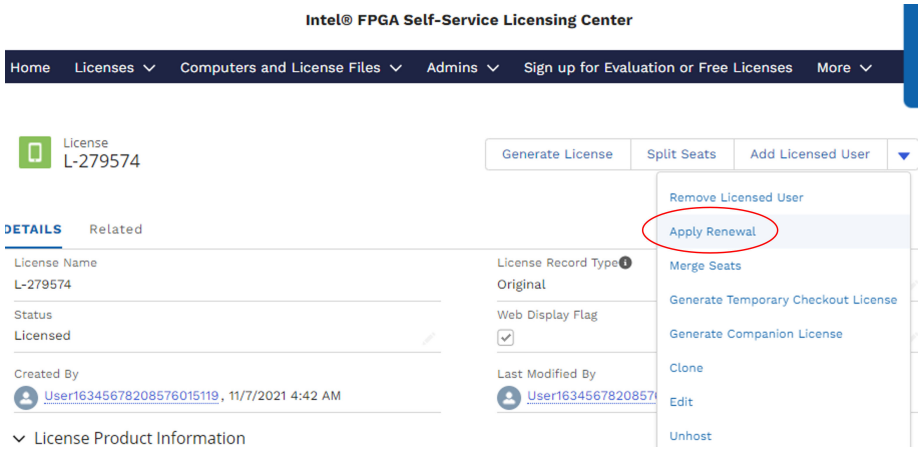
Renewing License Using the License Assistant

Select the **Renew and Generate License** option and follow the steps in the License Assistant.



Applying for Renewals Using the Licenses Menu

1. On the menu bar, select **Licenses ➤ New Licenses**.
2. Click the **License Name** link of the license you want to use to renew your existing license.
3. On the drop-down menu in the **Actions** bar, click **Apply Renewal**.



4. Enter the number of seats to renew.
5. Select your existing eligible license to renew.

Note: You can only view licenses that are already assigned to a computer.

6. Select your existing eligible license to renew.
7. Click **Confirm**.

6.6.11. Managing a Computer Profile

Create a computer profile in the Altera FPGA Self-Service Licensing Center to allow the assignment of licenses.

1. Log in to the Altera FPGA Self-Service Licensing Center.
2. On the menu bar, click **Computer and License Files ➤ All Computers**. Existing computer profiles, if any, display.
3. Click the **New** button to create a new computer profile.
4. Perform these steps in the **Create Computer** dialog box:
   a. Type your computer name in the **Computer Name** field. You can specify any name meaningful to you in this field.
   b. Select the **License Type** for your computer. Your software license type must match the license type (float or fixed) of any license you want to assign your computer.
   c. Select the primary **Computer Type** for your computer. The primary computer type is the NIC ID, Host ID, or software guard ID.
   d. Type your NIC ID, host ID, or software guard ID number in the **Primary Computer ID** field. For details about how to extract information about your computer hardware, refer to Getting Hardware Information for License on page 64 or Hardware Information Required When You Request a License.
   e. [FOR FIXED LICENSE] If you have a fixed license and want to use your license for multiple computers, you can specify up to two companion IDs in the **Companion ID** and **Companion ID 2** fields.

      Your companion ID is your NIC ID, Host ID, or software guard ID. You can use your license or associated companion license only on one computer at a time.
   f. [FOR FLOATING LICENSE] If you have a floating license, you can set up your license on up to two redundant license servers. Type the NIC ID or Host ID for redundant servers in the **Redundant Server ID 1** and **Redundant Server ID 2** fields.
   g. Click **Save** to save the profile.

To edit an existing computer profile:

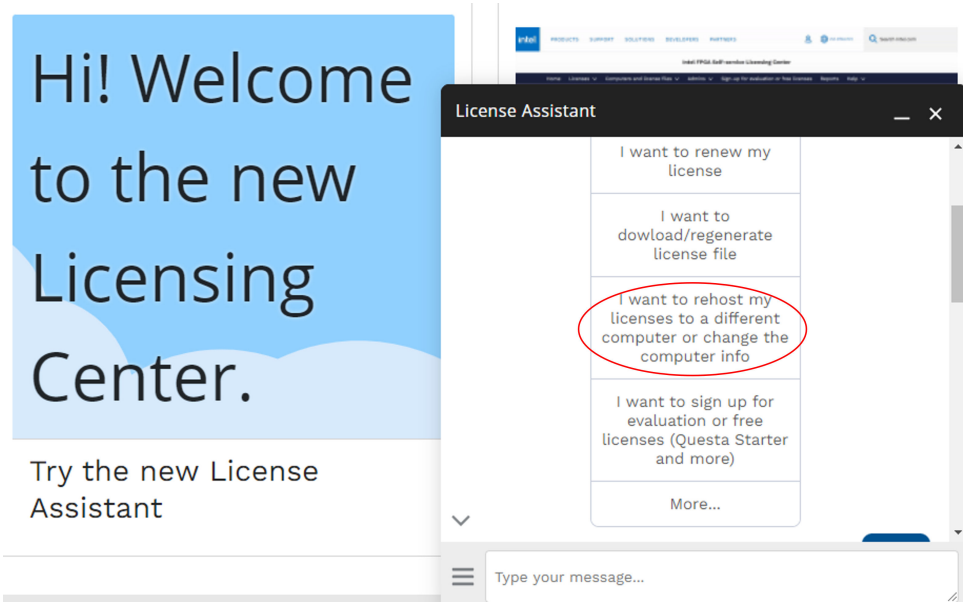
1. On the menu bar, click **Computer and License Files ➤ All Computers**. Existing computer profiles, if any, displays in the **All Computers** table.
2. Click the drop-down arrow icon at the end of a computer profile entry and select **Edit**. Edit <*profile_id*> dialog box display.
3. Modify the profile details and click **Save**.
4. If you want to change the owner of the profile, then click the drop-down arrow icon at the end of a computer profile entry again and select **Change Owner**. Change Owner dialog box displays.
5. Search the new owner name in the search users field and click **Submit**. This triggers an email notification to the new owner.

6.6.12. Rehosting a License on a Different Computer

Use one of the following methods to rehost a license on a different computer:

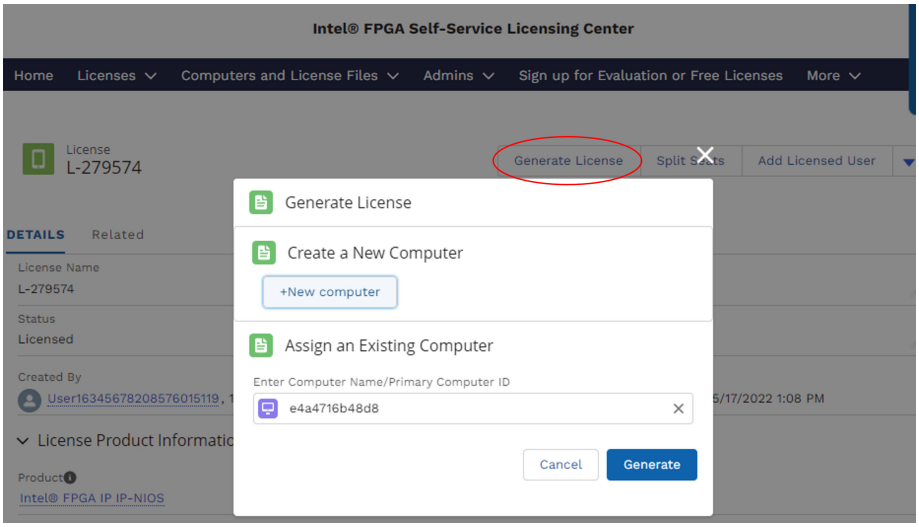
Rehosting a License Using the License Assistant

Select the **Rehost and generate license** option to move a license to a different computer. Follow the steps in the License Assistance.



Rehosting by Changing the Computer Associated With the License

1. On the menu bar, select **Licenses ➤ All Licenses**.
2. Click the **License Name** link of the license that you want to rehost on a different computer.
3. On the actions bar, click **Generate License**.



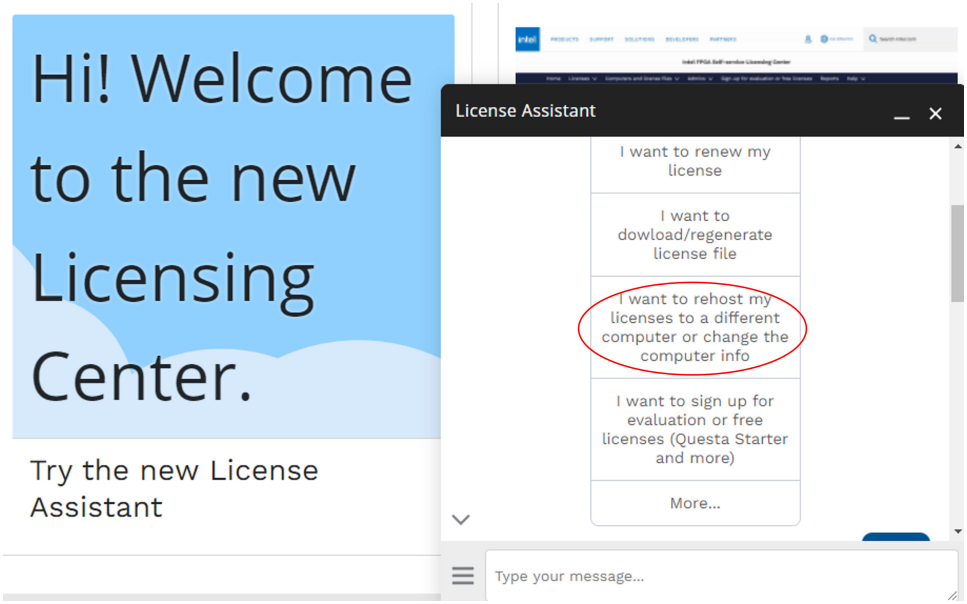
4. Create a new computer or use another existing computer.

6.6.13. Rehosting All Licenses from Current Computer to a Different Computer

Use one of the following methods to rehost all licenses:

Rehosting Licenses Using the Licenses Assistant

Select the **Rehost and generate license** option to change the computer information for all licenses. Follow the steps in the License Assistance.

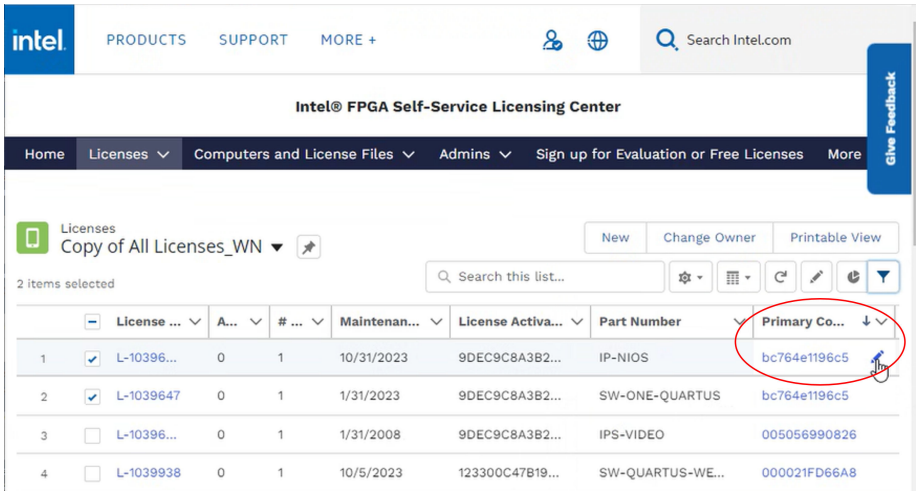


Rehosting Licenses by Changing Computer Details

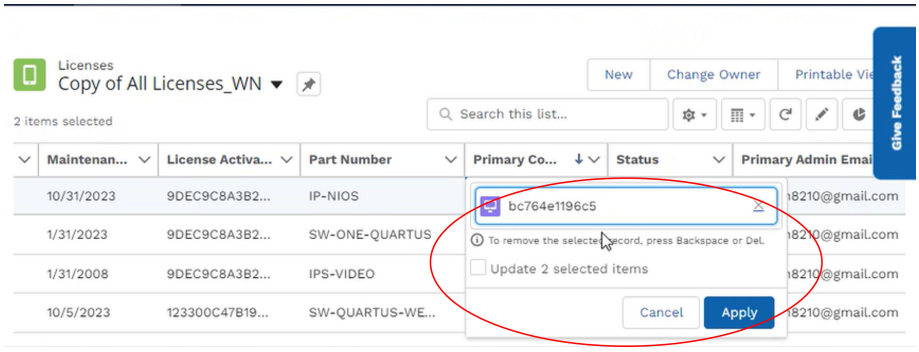
1. On the menu bar, select **Computers and License Files ➤ Active Computers**.
2. Click the **Primary Computer ID** link of the computer that you want to edit.
3. On the action bar, click **Edit**.
4. Modify the computer details.
5. Click **Save**.

6.6.14. Rehosting Multiple/Partial Licenses from Current Computer to a Different Computer

1. On the menu bar, select **Licenses ➤ All Licenses**.
2. Select the **Multiple/Partial Licenses Name** link on the licenses that you want to rehost from the current computer to a different computer.
3. In the **Primary Computer** column, click the pencil icon.



4. Change to the new host ID.



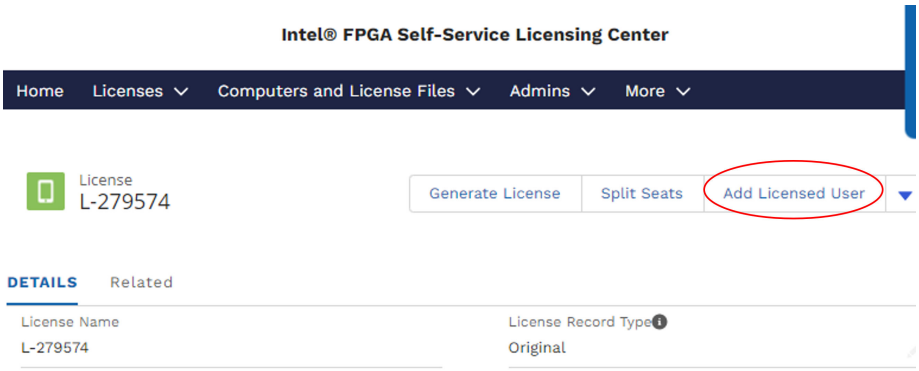
5. Select the **Update selected items** option and click **Apply**.

6.6.15. Sharing a License with Another User

You can add or remove licensed users for your existing licenses. Licensed users are granted access to their assigned licenses.

Adding a Licensed User

1. On the menu bar, select **Licenses ➤ All Licenses**.
2. Click the **License Name** link on the license that you want to share with another user.
3. On the actions bar, click **Add Licensed User**.

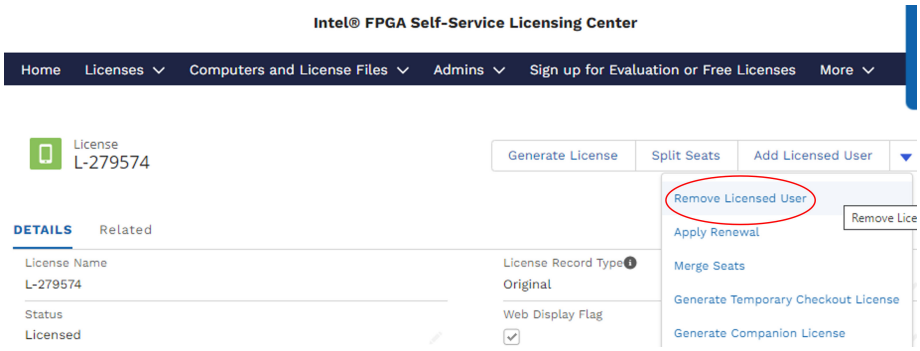


4. Enter the email address, first name, and last name of the user with whom you want to share this specific license.

   The licensed user receives an email notifying them that they have been added as a licensed user and now have access to the specific license that they have been assigned.
5. Click **Save**.

Removing a Licensed User

1. On the menu bar, select **Licenses ➤ All Licenses**.
2. Click the **License Name** link of the license for which you want to remove a user.
3. On the actions bar, click **Remove Licensed User** in the drop-down list.



Related Information

- Setting up a License in a Network License Server on page 61

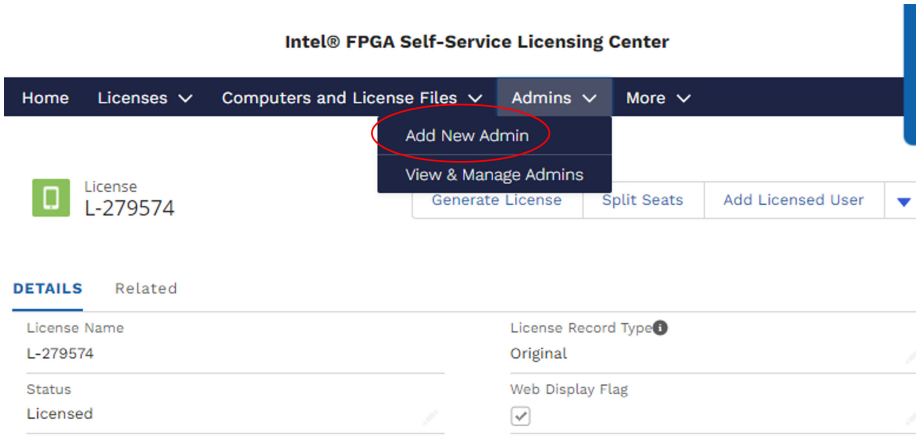
- Configuring the License Manager Server on page 96

6.6.16. Adding Delegate Administrators to Your Licenses and Computers

Note: Adding an administrator delegate provides access to all your licenses and computer.

Adding a New Administrator

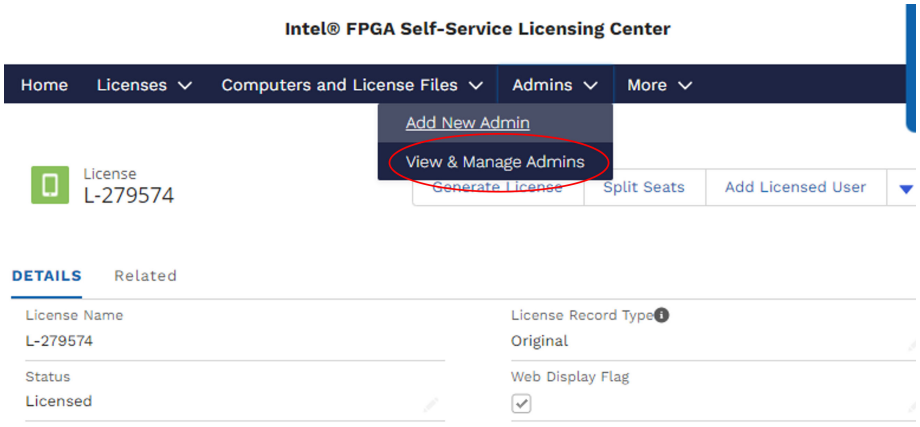
1. On the menu bar, select **Admins ➤ Add New Admin**.



2. Enter the first name, last name, and the email of the admin that you want to add.
3. Click **Save**.

Viewing the List of Administrators

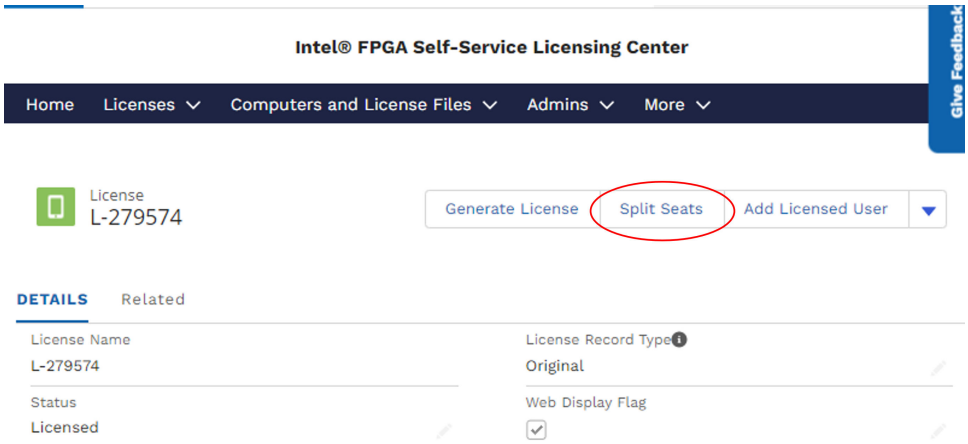
1. On the menu bar, select **Admins ➤ View & Manage Admins**. A list of active additional administrators appears.



2. To view all administrators, including previously active administrators, change the list view drop-down from **My Active Delegate Admins** to **My Delegate Admins**.

6.6.17. Splitting Seats on Your License and Generating Licenses

1. On the menu bar, select **Licenses ➤ All Licenses**.
2. Click the **License Name** link of the license where you want to split seats.
3. On the actions bar, click **Split Seats**.

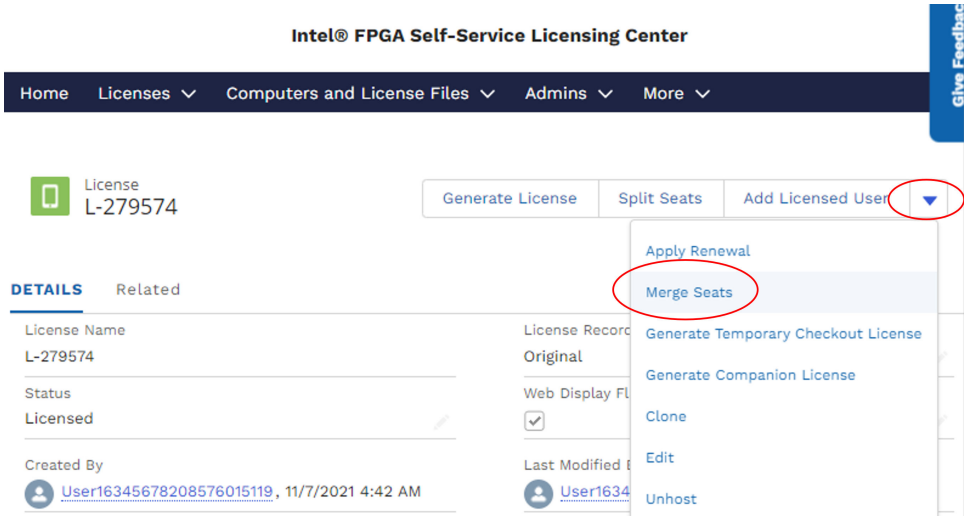


4. Enter the number of seats to move.
5. Select the number of servers to which you want to move the seat.
   - ☐ **Move the seat(s) to one server**: This means that your split seats are still residing in the same server.
   - ☐ **Move each seat to a unique server**: This means that your split seats are still residing in the same server, and you must perform a rehost to transfer them to another computer.

   The table shows the updated number of seats and products that you have.
6. Click **Confirm**.
7. Click **Generate License** on each of the split license to assign a computer and get license file.

6.6.18. Merging or Adding Seats on Your License and Generating License

1. On the menu bar, select **Licenses ➤ All Licenses**.
2. Click the **License Name** link for the license that you want to merge with another license.
3. On the actions bar, click the drop-down icon and select **Merge Seats**.



4. Enter the number of seats to merge.
5. Select the license that you want to merge these seats with.
6. Click **Save**.

6.6.19. Adding Floating Seats

You can add floating seats to an existing floating license to increase the number of users available. Adding additional seats to an existing floating license may affect its maintenance expiration date.

1. Log in to the Altera FPGA Self-Service Licensing Center.
2. On the menu bar, click **Licenses ➤ All Licenses**.
3. Click the license name for which you want to add or remove a licensed user. License details page displays.
4. On the drop-down menu in the Actions bar, click **Merge Seats**.
5. Enter the number of seats to add.
6. Select the license that you want to merge these seats into.
7. Click **Save**.

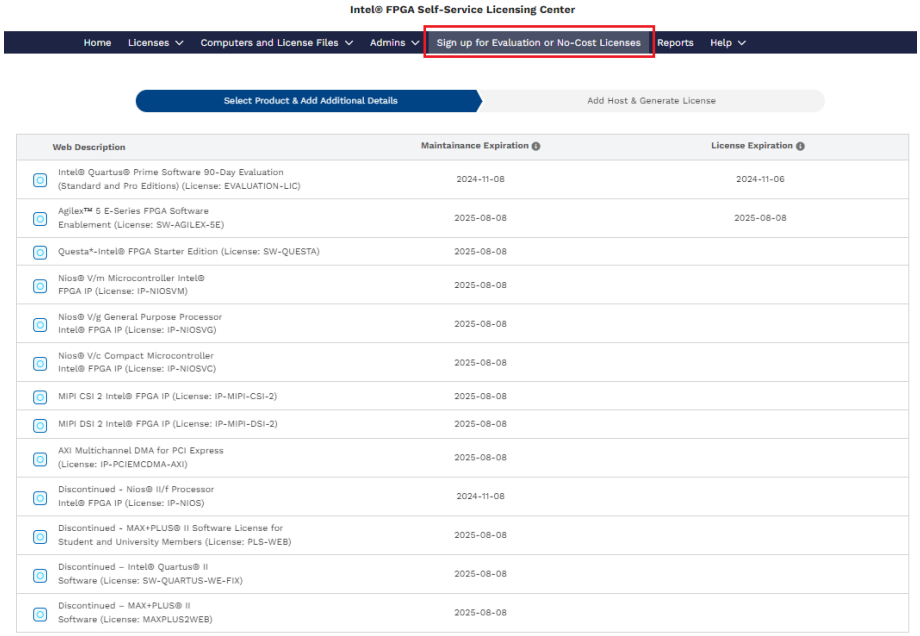
When you add additional seats to an existing floating license, the new maintenance subscription term of the floating license is calculated by adding the total number of maintenance months for the existing and new seats, and then dividing by the total number of seats. Any partial number is rounded up to the nearest whole number.

Example 3. Floating License Example

If you have an existing floating license with ten seats that expires in five months and you purchase two additional seats, your updated floating license maintenance subscription might expire in seven months for all twelve seats.

6.6.20. Signing Up for an Evaluation or No-Cost License

1. Select **Sign up for Evaluation or No-Cost Licenses**.



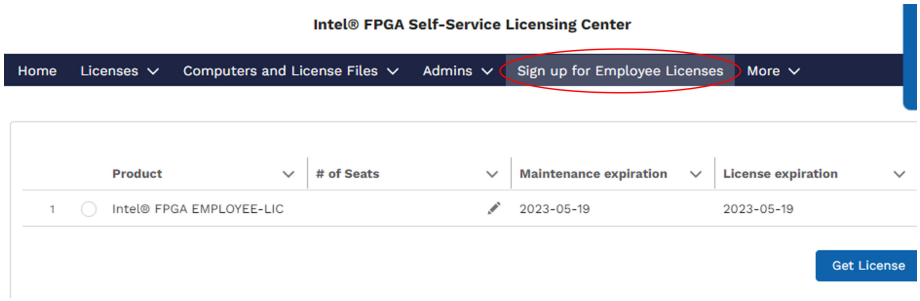
2. Select the product that you want to sign up.
3. Add one or more optional license features. Use **Select/Unselect All Options** checkbox if you want to select all optional products.

   Note: This option is only available when an evaluation or no-cost license has optional license features configured with it.
4. Enter the number of seats in the **# of Seats** field.
5. Click **Next**.
6. Assign a computer that you have previously added to your account or add a new computer using the **New Computer** button. Refer to Managing a Computer Profile on page 79 for more information.
7. Select the **Terms of Use** checkbox to accept the terms of use.
8. Optional: Select the **Feedback** checkbox if you want to provide feedback.
9. Click **Generate**.

6.6.21. Signing Up For an Employee License

Note: This feature is only available for Altera employees or Altera contingent workers.

1. Select **Sign up for Employee Licenses**.



2. Select the Employee license product that you want to sign up.
3. Enter the number of seats in the **# of Seats** column.
4. Select the **Terms of Use** checkbox to accept the terms of use.
5. Click **Get License**.
6. Assign a computer that you have previously added to your account or add a new computer.

6.7. About Altera FPGA Software License Files

This section contains details about the syntax of Altera FPGA Software License Files.

Related Information

- Configuring the License Manager Server on page 96
- Setting up a License File in the License Server on page 61

6.7.1. License File Components

This topic describes the different parts of the Altera FPGA license file (`license.dat`).

6.7.1.1. License File Header

Altera FPGA license files have headers that contain identifying information specific to the license, such as the type of license, the primary computer and companion IDs, issue and expiration dates, and a product license summary.

The following example shows the header of a floating server license file that contains a license for the Quartus Prime software and the Questa Altera FPGA Edition software:

```
# Altera Corporation Software and/or Intellectual Property License File
# Issued <dd MMMM YY _ date>
# Upgrade to these products will no longer be available after the Maintenance
Expiration
# date unless licenses are renewed.
# Floating Server License
# Primary Machine Name-xxxxxxxx
# Primary Machine ID-Host ID XXXXXXXXXXXX
# Redundant Server 2-N/A
# Redundant Server 3-N/A
# Product License Summary:
# QUARTUS PRIME PRO FLOAT, 1 Seat(s)
# - Maintenance Expiration of <YYYY.MM _ date>
# Quartus II SE Floating, 2 Seat(s)
# - Maintenance Expiration of <YYYY.MM _ date>
# Quartus II SE Floating, 1 Seat(s)
```

```
# - Maintenance Expiration of 1899.12
# Questa Altera FPGA Edition Software, 1 Seat(s)
# - Maintenance Expiration of <YYYY.MM _ date>
# - License Expires <dd-MMM-YYYY _ date>
...
```

6.7.1.2. SERVER, VENDOR, and USE_SERVER Lines

A floating license file starts with SERVER, VENDOR, and USE_SERVER lines to describe the license server. Fixed license files do not contain these entries. These license lines are structured as shown in the following example:

SERVER *<hostname> <8 or 12-character host or NIC ID> <port>*

VENDOR *<daemon> <path>*

USE_SERVER

The USE_SERVER line designates the license file as a floating license.

You can configure up to three redundant network license servers with the FLEXlm licensing scheme. In this case, there are three SERVER lines, one for each license server machine. The first server specified is the master.

Typically, a license file contains multiple VENDOR lines; one for each required license vendor daemon. Some license files might use the DAEMON keyword instead of VENDOR.

Table 17. Elements of the SERVER and VENDOR Lines for Floating Licenses

Element Name	Editable	Description
hostname	yes	The host name of the server in the license.dat file.
8- or 12-character host or NIC ID	no	The PC NIC or UNIX Host ID of the server. You must supply this ID when purchasing the license.
port	n/a	The port number.
daemon	no	The vendor daemon name: • alterad daemon: Quartus Prime software • saltd daemon: Questa Altera FPGA Edition software • armlmd daemon: ARM features of the ARM Development Suite (ADS)
path	yes	The path to the vendor daemon on the server. You must edit this entry.

6.7.1.3. FEATURE and INCREMENT Lines

All Altera FPGA license files contain FEATURE, and possibly INCREMENT lines, to describe the software features that are licensed by the license.dat file. Certain features use INCREMENT lines, which are structured similarly as FEATURE lines. The FEATURE line is structured as shown in the following example:

```
FEATURE <name> <daemon> <version> <expiration> <# of licenses> <authcode>
[<vendor_string>] [<hostid_lock>] [SIGN] [SUPERCEDE] [<issuer>]
```

A backslash (\) character indicates that the FEATURE or INCREMENT line continues on the next line in the text file. The last line of each FEATURE line does not require the backslash. Elements in brackets [ ] are present only in some license FEATURE or INCREMENT lines. The example below shows a FEATURE line, followed by an INCREMENT line.

```
# FEATURE START
# The following is a license file for the Quartus PRIME PRO Edition Fixed
# Number of seat licenses is 1
# No license expiration date
FEATURE maxplus2 alterad 2021.05 permanent uncounted B36133371D28 \
    HOSTID=e4a4716b48d8 SIGN="1A0F EECF 699E A64A 1E3A 83D1 BADC \
    7763 B15D 98FA EF7F A371 E02F 4D9F D9CF 0BCE 2A9C 2650 20FA \
    42CF ABD8 006A D037 DA2C 703B 9CFC 43E5 1A01 0C56 4D6D"
FEATURE quartus_pro alterad 2021.05 permanent uncounted 46499B9E717E \
    HOSTID=e4a4716b48d8 TS_OK SIGN="08EF B762 CF6A 18CC FF95 B124 \
    2FB0 F8F0 7A43 CAFE C88E DC28 5AC4 5F69 FFFB 1C1C DBF6 7A79 \
    87E7 FFEF D8C2 F96F 461A 0D38 DB4B 4E5B 840E F3F6 F314 4DE9"
# FEATURE END
############################################################################

INCREMENT alteramtivsim mgcld 2022.06 9-jun-2022 1 4F127AC9C4D11138B611 \
    \
  VENDOR_STRING=98B0746B ISSUER=Altera SN=175935529 SIGN2="0FEF B2F5 0646 \
    9F02 0405 423D D8B4 B072 3EFE 6CAA F66B 5E40 1C50 AAFE 3C0A 1869 269F \
    492D 2D25 1CFE 1AA1 060D F7E3 96A6 7BA8 6CE1 B7D1 AC24 2797 376B"

# FEATURE END
############################################################################
```

Table 18. Elements of the FEATURE and INCREMENT Code Lines

Element Name	Description
name	The feature name.
daemon	The name of the vendor daemon that manages the feature. This name corresponds to a VENDOR line in network/server licenses.
version	Any version of software released up to and including this date is licensed. Software with a later version date does not function using this license file.
expiration	The date after which the software or service pack cannot be used. The expiration date applies to trial licenses. For Altera FPGA license subscriptions, the expiration is permanent.
# of licenses	The number of concurrent users that can run each feature. For some license types, the entry is uncounted.
authcode	An authorization code generated by Altera .
vendor_strings	Specified for certain third-party license features.
hostid_lock	Specified if a feature is node-locked to a software guard, NIC ID, or hard-disk serial number. lectual property (IP) functions often contain node-locked feature lines in floating licenses. • Software guard: HOSTID=GUARD_ID=*<Software guard number>* • Network interface card (NIC): HOSTID=*<NIC ID>* • Hard-disk serial number: HOSTID=DISK_SERIAL_NUM=*<Hard-disk ID>*
issuer	The name of the QuestaSim* license issuer.
sign	An authorization code generated by Altera.
TS_OK	Enables remote log on for fixed license Quartus Prime software (includes FPGA IP, Nios II Software Build Tools for Eclipse, and DSP Builder).

The following table describes the available Altera FPGA license features. This table does not include license features for any lectual property (IP), development kits, or special license codes used for the Quartus Prime Lite Edition software.

Table 19. Available Altera FPGA License Features

Feature Name	Description
quartus	Quartus Prime Standard Edition software
quartus_pro	Quartus Prime Pro Edition software
alteraqsim	Questa Altera FPGA Edition Simulator
qsimstarter	Questa Altera FPGA Simulator Starter Edition

6.7.2. Using a FLEXlm Options File

If the license file contains FEATURE and INCREMENT lines for the same software but different maintenance dates, the license server may grant newer licenses from the FEATURE line to users running older versions of software, which leaves fewer licenses for users running newer versions of software.

By default, the license server grabs licenses from the first matching FEATURE or INCREMENT line in the license file. If all the licenses in the first FEATURE or INCREMENT line are in use, the license server attempts to grab licenses from the next FEATURE or INCREMENT lines, until the end of the license file. A FLEXlm options file allows you to control which users can grab which licenses by creating pools of licenses for each FEATURE and INCREMENT line.

To create a FLEXlm options file that controls a floating license server:

1. In a new text file, add a GROUP line for each user group you want to create, with a list of user names, separated by spaces for each group, as follows:

   ```
   GROUP <group name> <username 1> <username 2>
   ```

   Users can be members of more than one group as shown in the following example:

   ```
   GROUP quartus2010 kjones bknight root administrator
   ```

   ```
   GROUP quartus2009 bknight cface root administrator
   ```

2. Add an INCLUDE line for each INCREMENT and FEATURE line in the license file for the product you want to control.

   Set the VERSION keyword to the maintenance date or version date on the INCREMENT or FEATURE line from the license file. The GROUP field controls which group or groups can use the license line as follows:

   ```
   INCLUDE <product name>:VERSION=<maintenance date> GROUP <group name>
   ```

   For example:

   ```
   INCLUDE quartus:VERSION=2010.12 GROUP quartus2010
   INCLUDE quartus:VERSION=2009.06 GROUP quartus2009
   INCLUDE ip_base:VERSION=2010.12 GROUP quartus2010
   INCLUDE ip_base:VERSION=2009.06 GROUP quartus2009
   ```

3. Save the options file <*filename*>.dat.
4. Add the options file as the last option on the VENDOR line in the license file, as shown in the following example:

```
VENDOR alterad "C:\lic\alterad.exe" "C:\lic\alteraOptions.dat"
```

5. Restart the FLEXlm license server, or reread the license file.

Related Information

Rereading an Existing License File on a License Server on page 99

6.7.3. License.dat Example Files

This topic includes several example license.dat files. License files vary, depending on the type of licenses and the software that is enabled. License examples in this section should be used only for reference; they are not valid for actual licensing.

6.7.3.1. Fixed PC Software Guard License Example

The following example shows a license.dat file that contains a license for the IP Base Suite and the Quartus Prime Pro Edition software. This license file works on a PC that has the host ID e4a4716b48d8.

```
############################################################################
FEATURE START# FEATURE START
# The following is a license file for the IP Base Suite Package part number for
IP
# Base Suite (NCO, FFT, FIR Compiler II)
# Number of seat licenses is 1
# No license expiration date
PACKAGE ip_base alterad COMPONENTS="6AF7_00D8 6AF7_0014 6AF7_0034" \
    OPTIONS=SUITE SIGN="153B EB57 5B09 B585 D1D5 3EA8 1CC3 CB52 \
    DC01 2214 9CDA 4A26 598F 76C2 C126 00FE 5122 A135 BCB7 9D0A \
    2BAD 5F03 BE38 FE97 EDF8 5B2A 55F2 99E3 1468 72B8"
INCREMENT ip_base alterad 2021.05 permanent uncounted CBFAF3E3783C \

VENDOR_STRING="iiiiiiiihdLkhIIIIIIIIUPDuiaaaaaaaa11X38DDDDDDDDpjz5cd
ddddddtmGzGJJJJJJJJbqIh0uuuuuuuugYYWiVVVVVVVVbp0FVHHHHHHHHBUEakffff
ffffD2FFRkkkkkkkkWL$84" \
    HOSTID=e4a4716b48d8 TS_OK SIGN="0571 823B B38A 3D50 F9F5 B760 \
    77A7 08A3 5195 999C A11D 901B 54A6 AB40 4438 0137 FD4A 6625 \
    51C9 3A43 7C37 DADE 84D8 6FC6 1A9B E31C 1972 A291 8EBF A56B"
# FEATURE END
############################################################################
# FEATURE START
# The following is a license file for the Quartus PRIME PRO Edition Fixed
# Number of seat licenses is 1
# No license expiration date
FEATURE maxplus2 alterad 2021.05 permanent uncounted B36133371D28 \
    HOSTID=e4a4716b48d8 SIGN="1A0F EECF 699E A64A 1E3A 83D1 BADC \
    7763 B15D 98FA EF7F A371 E02F 4D9F D9CF 0BCE 2A9C 2650 20FA \
    42CF ABD8 006A D037 DA2C 703B 9CFC 43E5 1A01 0C56 4D6D"
FEATURE quartus_pro alterad 2021.05 permanent uncounted 46499B9E717E \
    HOSTID=e4a4716b48d8 TS_OK SIGN="08EF B762 CF6A 18CC FF95 B124 \
    2FB0 F8F0 7A43 CAFE C88E DC28 5AC4 5F69 FFFB 1C1C DBF6 7A79 \
    87E7 FFEF D8C2 F96F 461A 0D38 DB4B 4E5B 840E F3F6 F314 4DE9"
# FEATURE END
############################################################################
# End of Altera Corporation Software and/or Intellectual Property
License File. Issued 06/09/2021
```

6.7.3.2. Floating Network License Example

The example below shows a `license.dat` file that contains a license for the Quartus Prime Pro Edition software and the IP Base Suite. This license specifies that a single user can run the software and works when the PC with NIC ID `09876543` is set up as a license server and the user points to the license location.

```
##############################################################################
SERVER <hostname> 09876543 <port number>
VENDOR altera<d> <path to daemon executable>
VENDOR saltd <path to daemon executable>
USE_SERVER
##############################################################################
# FEATURE START
# The following is a license file for the IP Base Suite Package part number for
IP
# Base Suite (NCO, FFT, FIR Compiler II)
# Number of seat licenses is 1
# No license expiration date
PACKAGE ip_base altera<d> COMPONENTS="6AF7_00D8 6AF7_0014 6AF7_0034" \
    OPTIONS=SUITE SIGN="153B EB57 5B09 B585 D1D5 3EA8 1CC3 CB52 \
    DC01 2214 9CDA 4A26 598F 76C2 C126 00FE 5122 A135 BCB7 9D0A \
    2BAD 5F03 BE38 FE97 EDF8 5B2A 55F2 99E3 1468 72B8"
INCREMENT ip_base altera<d> <date> permanent 1 54C8E8A1FBA8 \

VENDOR_STRING="iiiiiiiihdLkhIIIIIIIIUPDuiaaaaaaaa11X38DDDDDDDDpjz5cd
ddddddtmGzGJJJJJJJJbqIh0uuuuuuuugYYWiVVVVVVVVbp0FVHHHHHHHHBUEakffff
ffffD2FFRkkkkkkkWL$84" \
    DUP_GROUP=UHD SIGN="02CC FF71 2A69 9432 0127 5793 5D3D C5B9 \
    B2D4 F31C 63E4 5735 3333 0156 7A6F 0A06 683D E4DB EC19 EF83 \
    DFCA CDBB A994 042B F35B ABE7 4215 391B 4ADB 0A70"
# FEATURE END
##############################################################################
# FEATURE START
# The following is a license file for the Quartus PRIME PRO Edition Floating
# Number of seat licenses is 1
# No license expiration date
FEATURE quartus_pro altera<d> <date> permanent 1 5BF77FD6D23D \
    SIGN="03E9 8852 506A 279B EEB5 D51B 7019 13CB 14EF 386E 5A1A \
    3270 4A3B 3289 E028 0CBA CF01 DF36 30CC CAEE A561 AB7D 5FEE \
    4B91 E405 0923 2FE0 51F7 D3D2 7DE7"
# FEATURE END
##############################################################################
# End of Altera Corporation Software and/or Intellectual Property
License File. Issued <date>
```

6.7.4. Syntax of license.dat License File

The `license.dat` file must meet these conditions to ensure license integrity:

- The text editor does not append `.txt` or any other file extension to the file name, for example, `license.dat.txt`, otherwise the software cannot find the license.
- The last `FEATURE` line ends with a carriage return (new line).
- Any `FEATURE` line that wraps to a second or third line must have a backslash (\) at the end of each line to indicate that the statement continues. However; `VENDOR_STRING` statements that wrap to multiple lines do not need a backslash (\). The backslash should only be added outside the line with double quotation marks (").
- The license file does not have hidden control characters:
  - ☐ Opening the license file with any software other than a plain text editor may add hidden characters.

    For example, WYSIWYG editors such as Microsoft Excel, Word, or WordPad, may insert special control characters such as a tab or carriage return. Pasting special control characters into another plain text document can corrupt the license, even if those characters are invisible in a plain text editor.
  - ☐ If you edit the license file in one operating system, then copy the license file in another operating system, and then copy the license file to the Windows* operating system, the second operating system may insert unwanted control characters into the license file. Make sure that you correctly convert the file.

For information about the contents of a license file and example licenses, refer to *About Altera FPGA Software License Files*.

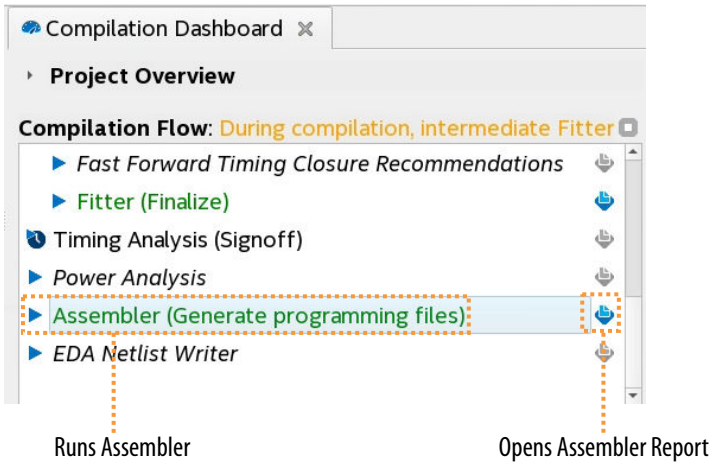
6.8. Checking the IP License Status

You can check the license status of all IP in a Quartus Prime project by viewing the Assembler report.

To generate and view the Assembler report in the GUI:

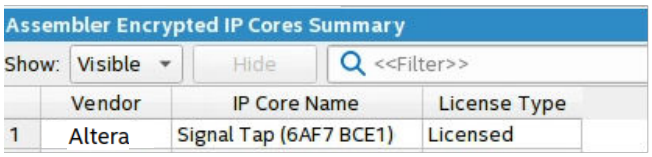
1. Click **Assembler** on the Compilation Dashboard.
2. When the Assembler (and any prerequisite stages of compilation) complete, click the **Report** icon for the Assembler in the Compilation Dashboard.

Figure 20. Assembler Report Icon in Compilation Dashboard



3. Click the **Encrypted IP Cores Summary** report.

Figure 21. Encrypted IP Cores Summary Report



To generate and view the Assembler report at the command line:

1. Type the following command:

```
quartus_asm <project name> -c  <project revision>
```

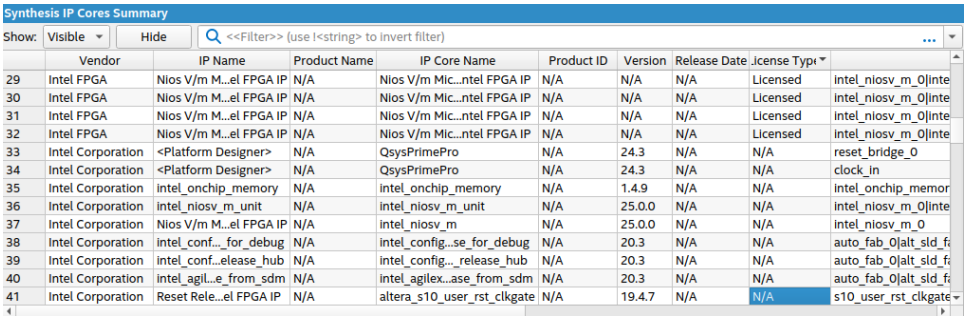
2. View the output report in `/output_files/<project_name>.asm.rpt`.

```
+-----------------------------------------------------------------------+
; Assembler Encrypted IP Cores Summary                                  ;
+--------+--------------------------------------------+---------------+
; Vendor ; IP Core Name                               ; License Type ;
+--------+--------------------------------------------+---------------+
; Altera  ; PCIe SRIOV with 4-PFs and 2K-VFs (6AF7 00FB) ; Unlicensed   ;
; Altera  ; Signal Tap (6AF7 BCE1)                     ; Licensed     ;
; Altera  ; Signal Tap (6AF7 BCEC)                     ; Licensed     ;
+--------+--------------------------------------------+---------------+
```

6.9. Viewing IP Core License Data

Following Analysis & Synthesis, you can view the details about the IP in your design in the **Synthesis IP Cores Summary** report that generates automatically. This report lists the following about each IP instance discovered during Analysis & Synthesis, allowing you to determine the license status of each IP in your design:

Figure 22. Synthesis IP Cores Summary Report



Synthesis IP Cores Summary

Show: Visible | Hide | <<Filter>> (use !<string> to invert filter)

	Vendor	IP Name	Product Name	IP Core Name	Product ID	Version	Release Date	License Type	
29	Intel FPGA	Nios V/m M...el FPGA IP	N/A	Nios V/m Mic...ntel FPGA IP	N/A	N/A	N/A	Licensed	intel_niosv_m_0|inte
30	Intel FPGA	Nios V/m M...el FPGA IP	N/A	Nios V/m Mic...ntel FPGA IP	N/A	N/A	N/A	Licensed	intel_niosv_m_0|inte
31	Intel FPGA	Nios V/m M...el FPGA IP	N/A	Nios V/m Mic...ntel FPGA IP	N/A	N/A	N/A	Licensed	intel_niosv_m_0|inte
32	Intel FPGA	Nios V/m M...el FPGA IP	N/A	Nios V/m Mic...ntel FPGA IP	N/A	N/A	N/A	Licensed	intel_niosv_m_0|inte
33	Intel Corporation	<Platform Designer>	N/A	QsysPrimePro	N/A	24.3	N/A	N/A	reset_bridge_0
34	Intel Corporation	<Platform Designer>	N/A	QsysPrimePro	N/A	24.3	N/A	N/A	clock_in
35	Intel Corporation	intel_onchip_memory	N/A	intel_onchip_memory	N/A	1.4.9	N/A	N/A	intel_onchip_memor
36	Intel Corporation	intel_niosv_m_unit	N/A	intel_niosv_m_unit	N/A	25.0.0	N/A	N/A	intel_niosv_m_0|inte
37	Intel Corporation	Nios V/m M...el FPGA IP	N/A	intel_niosv_m	N/A	25.0.0	N/A	N/A	intel_niosv_m_0
38	Intel Corporation	intel_conf..._for_debug	N/A	intel_config...se_for_debug	N/A	20.3	N/A	N/A	auto_fab_0|alt_sld_fa
39	Intel Corporation	intel_conf...elease_hub	N/A	intel_config..._release_hub	N/A	20.3	N/A	N/A	auto_fab_0|alt_sld_fa
40	Intel Corporation	intel_agil...e_from_sdm	N/A	intel_agilex...ase_from_sdm	N/A	20.3	N/A	N/A	auto_fab_0|alt_sld_fa
41	Intel Corporation	Reset Rele...el FPGA IP	N/A	altera_s10_user_rst_clkgate	N/A	19.4.7	N/A	N/A	s10_user_rst_clkgate

- **Vendor** name, such as *Altera Corporation*.
- **IP Name** in IP Catalog, such as *Reset Release FPGA IP*.
- **Product Name** displays the product name.
- **Product ID** displays the product HEX ID.
- **IP Core Name** as represented in underlying software code, such as `altera_s10_user_rst_clkgate`.
- **Version** of the IP, such as `19.4.7`.
- **Release Date** of the IP, such as `09.30.2024`.
- **License Type** indicates whether an IP is **Licensed** or not (**N/A**).
- **Entity Instance** name, such as `s10_user_rst_clkgate_0`.
- **IP Include File** displays the IP include file associated with each IP.

6.10. Configuring the License Manager Server

6.10.1. Installing the FLEXlm License Manager Server Software on Another License Server

To install the FLEXlm license manager server software on an additional license server:

1. On the additional license server, create a directory that mimics the location of the Altera FPGA software:
   - ☐ Windows*

     ```
     <installation-directory>\bin64
     ```

   - ☐ Linux

     ```
     <installation-directory>/linux64
     ```

2. From your local computer, copy the following files from the location of the Altera FPGA software to the new directory on the licensing server:
   - ☐ `lmgrd`
   - ☐ `lmutil`
   - ☐ `alterad`
   - ☐ `lmtools` (Windows* only)

6.10.2. Upgrading the FLEXlm License Manager Server Software

To support network licensing, the Altera FPGA software requires the FLEXlm license manager server software version 11.19.5.0 or later. The installation process for Quartus Prime software installs the FLEXlm software version 11.19.5.0. However, you should also verify that the FLEXlm software version of the license server is 11.19.5.0.

On the license server running the FLEXlm software, type at a command prompt:

Windows*	*<FLEXlm system directory>*\lmgrd -v *<FLEXlm system directory>*\alterad -v
Linux	/*<FLEXlm system directory>*/lmgrd -v /*<FLEXlm system directory>*/alterad -v

Starting from the Quartus Prime software version 24.1, version of the `lmgrd` or `alterad` daemon is 11.19.5.0. If the `lmgrd` or `alterad` daemons on your system is not version 11.19.5.0, you must upgrade both daemons with the versions that the Quartus Prime software provides.

Note: You can only have one vendor daemon running at a time on a single system. For example, `alterad` and `saltd` can co-exist, but you cannot have two running daemons of `alterad`. If you have an Altera FPGA software license and Questa Altera FPGA Edition software license, and also have other Siemens EDA* QuestaSim* software license. You can either merge all of them into a single file or put all Siemens EDA* QuestaSim* software licenses in a single file. If you put all Siemens EDA* QuestaSim* software licenses in a single file, you must remove the `VENDOR saltd` line from the Altera FPGA software license file. You cannot have Siemens EDA* licenses across two files for the same license server.

If a soft reload through the `lmutil lmreread` command failed, shut down the current license daemons before reloading them again.

Warning: If your FLEXlm software version is too old, you might observe errors. Refer to https://www.intel.com/content/www/us/en/support/programmable/articles/000099465.html for more information.

To upgrade an older version of the FLEXlm software:

1. Make a backup copy of your current `lmgrd` and `alterad` daemons.
2. Copy the new versions of the files to the license server over your current daemons.

   If you installed the FLEXlm software using the Quartus Prime installer, the current versions of the `lmgrd` and `alterad` daemons are in:

Windows*	*<installation-directory>*\quartus\bin64\alterad.exe *<installation-directory>*\quartus\bin64\lmgrd.exe
Linux	/*<installation-directory>*/quartus/linux64/alterad *<installation-directory>*/quartus/linux64/lmgrd

3. Restart the FLEXlm license server by typing at a command prompt:

Windows*	`<FLEXlm system directory>\lmutil lmdown -c <license file path>` `<FLEXlm system directory>\lmgrd -c <license file path>[-l <optional log path>]`
Linux	`/<FLEXlm system directory>/lmutil lmdown -c <license file path>` `/<FLEXlm system directory>/lmgrd -c <license file path> [-l <optional log path>]`

Related Information

- Rereading an Existing License File on a License Server on page 99
- License Daemon Downloads
- Flexlm License Daemons for Altera FPGA Software

6.10.3. (Windows Only) Starting and Stopping the License Server

After changing the license configuration, you must restart the license server.

Note: Before you start or stop the license server, you must configure a new license server as a Windows* service.

To start or stop the license server in the **LMTOOLS** dialog box:

1. Type the following command at a command prompt:
   `<installation-directory>\bin64\lmtools`
2. In the **LMTOOLS** dialog box, click **Configuration using Services**.
3. Select the name of the license server, usually **Flexlm License Server**.
4. Click the **Start/Stop/Reread** tab.
5. Click **Start Server** or **Stop Server**.

6.10.4. (Windows Only) Starting the License Server Automatically

Before you set up the license server to start automatically at startup, you must configure the license server as a Windows* service.

To start the license server automatically at startup:

1. At a command prompt, type:
   ```
   <installation-directory>\bin64\lmtools
   ```
2. In the **LMTOOLS** dialog box, click **Configuration using Services**.
3. Select the name of the license server.
   Usually, **Flexlm License Server**.
4. Click the **Config Services** tab.
5. Turn on **Use Services**.
6. Turn on **Start Server at Power Up**.

6.10.5. Rereading an Existing License File on a License Server

If you change the license file, you must reread the license file or restart the license server before you can run the Quartus Prime software again. If you have an existing FLEXlm license server with an existing license file for the MAX+PLUS® II software, or software from another vendor, and the FLEXlm license manager server software is version 11.11.1, you can copy and paste the `FEATURE` lines from your Quartus Prime `license.dat` file into your existing license file. Make sure you modify the FEATURE lines for your server.

- At a command prompt, type:

Windows*	`<Quartus Prime system directory>/linux64/lmutil lmreread`
Linux	`<Quartus Prime system directory>\bin64\lmutil lmreread`

Note: You can see a list and description of the available FLEXlm options by typing `lmgrd -help` at a command prompt.

Related Information

Setting up a License File in the License Server on page 61

6.11. Troubleshooting License File Issues

Generally, to troubleshoot licensing issues, perform these steps:

1. Check that the license file has not expired.
2. Regenerate the license from the SSLC.
3. Verify that the `LM_LICENSE_FILE` environment variable points to the location of the `license.dat` file. The setup of the environment variable depends on the OS on your PC. For more information, refer to

Visit the following links for additional help with troubleshooting Altera FPGA software license files:

- Quartus Prime Software Licensing Questions and Answers
- Altera FPGA Software License Troubleshooter
- Altera FPGA Licensing Support Center

The following articles carry solutions to some of the frequently asked questions:

- Does the Questa* Altera FPGA Starter Edition software require a license file?
- Why doesn't my Quartus Prime software license work on RedHat 7.x?
- Why does the Quartus Prime software fail to detect a floating license?
- How do I determine whether the Intellectual Property (IP) that I used in my design is licensed?
- How can I stop older versions of the Quartus Prime software or IP from using newer floating network licenses?
- When is a license checked out by the Quartus Prime software, IP cores, and Questa Altera FPGA Edition software?

- Is there a specific environment variable that I can use for Quartus Prime software licensing, such as ALTERAD_LICENSE_FILE?
- How do I get my license if it does not show in the self-service licensing center because the maintenance expired before January 1, 2009?
- How do I determine whether the Intellectual Property (IP) that I used in my design is licensed?
- Why does the Quartus Prime software fail to detect a floating license?

Tip: If you are facing any issue not listed in this topic, explore Altera FPGA Knowledge Base articles or post your query on Altera FPGA Software Installation & Licensing forum.

7. Next Steps After Installing and Licensing the Software

Once you complete installing and licensing the required Altera FPGA development software, your system should now be ready to begin FPGA design development. Refer to the following topics for additional resources and training to help you get started:

- Additional Resources on page 101
- Training on page 102

Navigating Content Through Tasks

Use the following table to navigate this guide through user-tasks:

				
Determine Hardware and Software Requirements	Download Altera FPGA Software through the installer or manually	Install Altera FPGA Software	• Evaluate the Quartus Prime Software • Obtain License from SSLC	Review Next Steps

7.1. Additional Resources

Resource	Description
Altera Community - Altera FPGA Software Installation & Licensing Forum	Allows you to post queries and get responses to Altera FPGA software installation and licensing issues.
Altera FPGA Software Download and Installation Support page	Provides high-level help to get started with the Altera FPGA software installation and download.
Altera FPGA Knowledge Base	Provides links to applicable articles that span a variety of FPGA related issues.
Altera FPGA Self-Service Licensing Center	Provides support for licensing Altera FPGA software.
Altera FPGA Software License Troubleshooter	Answers frequently asked questions about licenses.
AN 320: Using Altera FPGA IP Evaluation Mode	Describes the FPGA IP Evaluation Mode that allows you to evaluate licensed FPGA IP cores in simulation and hardware, before deciding to purchase a full production IP core license.
How to Setup Quartus Prime Licenses	YouTube* video that demonstrates how to setup Quartus Prime licenses.
Setting up and debugging Quartus Prime floating licenses	YouTube* video that demonstrates how to set up and debug Quartus Prime floating licenses.
Free IP Base Suite Licenses	Provides a list of such free IP Base Suite licenses.

continued...

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

Resource	Description
	To help shorten your design time, Altera provides full-production licenses for some of the most popular IP cores in the FPGA IP Base Suite, which is free with the Quartus Prime software.
Altera FPGA Licensing Support Center	Provides information on license types, getting a license file, setting up a license file, and resolving license-related issues.
Quartus Prime Pro and Standard Software User Guides	Each user guide in the Pro Edition and Standard Edition collection covers a specific topic and is designed to help you easily and efficiently find the information you need to see your design through to completion.
Power Thermal Analyzer User Guide	Describes the Power Thermal Analyzer (PTC). The current version of the PTC supports Agilex 3, Agilex 5, Agilex 7, and Stratix 10 devices.
Advanced Link Analyzer User Guide	Describes the Advanced Link Analyzer that is a state-of-the art jitter/noise eye link analysis tool. This tool allows you to quickly and easily evaluate high-speed serial link performance.
Questa* Altera FPGA Edition Quick-Start: Quartus Prime Pro Edition User Guide	Demonstrates how to simulate a Quartus Prime Pro Edition design in the Questa* Altera FPGA Edition simulator.
Questa* Altera FPGA Edition Quick-Start: Quartus Prime Standard Edition User Guide	Demonstrates how to simulate a Quartus Prime Standard Edition design in the Questa* Altera FPGA Edition simulator.
DSP Builder (Advanced Blockset): Handbook	Describes DSP Builder, which is a high-level synthesis technology, that optimizes the high-level, untimed netlist into low-level, pipelined hardware for your target FPGA device and desired clock rate.
Get Started with the Altera oneAPI Base Toolkit for Linux*	Provides Linux-specific getting started instructions for oneAPI Base Toolkit.
FPGA AI Suite: Getting Started Guide	Provides an overview and installation instructions of FPGA AI Suite, and a process-walkthrough of running inference on a Resnet50 graph, including performance and area estimation.
Altera SoC Embedded Design Suite (SoC EDS) User Guide	Describes the Altera SoC Embedded Design Suite (SoC EDS) comprehensive tool suite for embedded software development on Altera SoC devices.
Ashling* RiscFree* IDE for Altera User Guide	Describes the RiscFree* IDE for Altera FPGAs Arm-based HPS and Nios V processors.

7.2. Training

You can take up the following training to learn more about the development tools and software you have installed:

- The Quartus Prime Software: Foundation (Standard Edition) (Online Training)
- Instructor-Led Training: Using Quartus Prime Software
- Using the Quartus Prime Standard Edition Software: An Introduction
- Verilog HDL Basics
- Verilog HDL Advanced
- VHDL Basics
- SystemVerilog with the Quartus Prime Software
- How to Begin a Simple FPGA Design
- University Self-Guided Lab: Become an FPGA Designer in 4 Hours
- Power Thermal Analyzer for Altera FPGA Devices
- Arm Development Studio for Altera SoC FPGAs Ask an Expert

- DSP Builder Advanced Blockset: Getting Started
- Introduction to FPGA AI Suite with OpenVINO Toolkit
- Using FPGAs with the oneAPI Toolkits
- Using oneAPI with Altera FPGAs Ask an Expert
- Introduction to Optimizing FPGAs with the oneAPI Toolkits
- Instructor-Led Training: Altera SoC FPGA Basics
- University Self-Guided Lab: Introduction to Simulation and Debug of FPGAs

For more Altera FPGA trainings, refer to the Altera FPGA Technical Training Catalog and Altera FPGA channel on YouTube.

MNL-1065 | 2026.02.27

Send Feedback

A. Altera FPGA Software Installation and Licensing Archives

For the latest and previous versions of this guide, refer to *Altera FPGA Software Installation and Licensing*. If a software version is not listed, the guide for the previous software version applies.

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

B. Document Revision History for Altera FPGA Software Installation and Licensing

Document Version	Quartus Prime Version	Changes
2026.02.27	25.3	Updated *Setting up a License File in the License Server* topic with an additional note.
2025.11.28	25.3	• Corrected Questa* Altera FPGA Edition licensing environment variable name. The correct environment variable name is `SALT_LICENSE_SERVER`.
2025.09.29	25.3	• Updated *Quartus Prime Design Suite Overview* topic for Quartus Prime Pro Embedded Edition and Quartus Prime Driver Installer. • Updated *Software Available in the Altera FPGA Software Download Center* topic for Quartus Prime Pro Embedded Edition. • Updated *Downloading Design-Specific Software Components* for latest behavior and Quartus Prime Pro Embedded Edition • Updated *Selecting the Installation Path* for latest default settings. • Updated *Getting Started With the Quartus Prime Installer GUI* topic to reflect automated download of drivers by default. • Updated *Installing Programming Cable Drivers* topic to reflect automated download of drivers by default. • Rebranded "Questa Intel FPGA Edition" as "Questa Altera FPGA Edition." • Rebranded "Intel SoC FPGA Embedded Design Suite" as "Altera SoC FPGA Embedded Development Suite" (SoC EDS)" • Rebranded "Intel Simics Simulator for Intel FPGAs" as "Simics Simulator for Altera FPGAs" • Rebranded "Intel FPGA Self-Service Licensing Center" to "Altera FPGA Self-Service Licensing Center." • Replaced "Power and Thermal Calculator" with "Power Thermal Analyzer."
2025.04.14	25.1	• Added support for Agilex 3 devices. • Rebranded "Intel FPGA Power and Thermal Calculator" as "Power and Thermal Calculator." • Rebranded "Ashling* RiscFree* IDE for Intel FPGA" as "Ashling* RiscFree IDE for Altera." • Rebranded "Intel Advanced Link Analyzer" as "Advanced Link Analyzer." • Replaced all instances of `LM_LICENSE_FILE` with `SALT_LICENSE_FILE`. • Renamed *Downloading Docker Containers for Quartus Prime Pro Edition FPGA Design Software* as *Downloading Docker Containers for Quartus Prime FPGA Design Software*. • Added `libsm6` for Ubuntu 22.04 OS in *Operating System Requirements*.
2025.01.27	24.3.1	• Changed instances of Intel to Altera. • Changed instances of "DSP Builder for Intel FPGAs" to "DSP Builder." • Changed instance of "Intel FPGA IP" to "FPGA IP."

continued...

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera reserves the right to make changes to any products and services at any time without notice. Altera assumes no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera. Altera customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.
*Other names and brands may be claimed as the property of others.

Document Version	Quartus Prime Version	Changes
		• Removed *Intel High Level Synthesis Compiler Software Requirements*. • Added *Downloading Docker Containers for Quartus Prime Pro Edition FPGA Design Software*. • Added support for the new USB Blaster III product in *Installing Drivers on RHEL 8/Ubuntu 18 and Above Versions*.
2024.09.30	24.3	• Renamed *Disk Space Requirements* as *Disk Space and Memory Requirements* and revised the topic to include memory requirements. • Added a note to *Intellectual Property (IP) Cores Licenses*. • Revised the existing information and added a new section for no-cost IP licenses in *Acquiring Free, No-Cost Licenses*. • In several topics, switched the *mgcld* daemon to *saltd* daemon and the related environment variable from *MGLS_LICENSE_FILE* to `SALT_LICENSE_FILE`. • Added a note in *Upgrading the FLEXlm License Manager Server Software* and *Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License* about the errors observed when using the old *mgcld* daemon. • Enhanced the instructions in *Using CLI Commands*. • Revised the images in the following topics: ☐ *Getting Started With the Quartus Prime Installer GUI* ☐ *Modifying the Installer Settings* ☐ *Installing Altera FPGA Software Through Quartus Prime Installer* ☐ *Setting up a Fixed License* • Added *Viewing IP Core License Data*. • Revised the requirements in *Questa* Altera FPGA Edition Software Requirements*. • Revised the tools list in *Software Available in the FPGA Software Download Center*.
2024.07.08	24.2	• Added *Acquiring Free No-Cost License*. • Added *Refreshing License Server Token Automatically*. • Renamed *Downloading Individual Executable Files* as *Downloading Software and Device Support Files* • Removed the following topics and merged their information in *Downloading Software and Device Support Files*. ☐ Downloading `.tar` Files ☐ Downloading Device Support ☐ Downloading Additional Software • Revised the information in *Installing Programming Cable Drivers*. • Renamed *Installing Drivers on RHEL 8 and Above and Ubuntu Versions 18 and 20 OS* to *Installing Drivers on RHEL 8/Ubuntu 18 and Above Versions*. • Revised the information in *Installing Drivers on RHEL 8/Ubuntu 18 and Above Versions*. • Revised *Operating System Requirements* to include Ubuntu 22.04 requirements.
2024.04.09	24.1	• Removed the support for AXI3 BFM in *Siemens EDA* AXI Verification IP Suite License (Intel FPGA Edition)*.
2024.04.01	24.1	• Removed `libnsl` from *Operating System Requirements*. • Updated FLEXlm license manager software version to 11.19.5.0 in *Upgrading the FLEXlm License Manager Server Software*. • Added support for generating bitstream with "Start the 30-day evaluation period with no license file" option in *Evaluating the Quartus Prime Software*. • Enhanced *Quartus Prime Software License* to include information about the 90-day evaluation license.

continued...

Document Version	Quartus Prime Version	Changes
		• Revised the information in the following topics: ☐ *Downloading Software Using the Quartus Prime Installer* ☐ *Accessing the Quartus Prime Installer* ☐ *Getting Started With the Quartus Prime Installer GUI* ☐ *Modifying the Installer Settings* ☐ *Using CLI Commands* • Removed *Nios II Embedded Design Suite License* topic as Nios II has been deprecated. • Removed in Nios II EDS License from *Summary of Altera FPGA Software Licenses Required*. • Removed the requirements for Ubuntu Enterprise Linux in *Operating System Requirements*. • Removed *Windows* Subsystem for Linux* (WSL 1) on Windows Requirements for Nios II EDS*.
2023.12.04	23.4	• Updated the number of default parallel download threads and its range in *Modifying the Installer Settings* and *Using CLI Commands*. • Added *Installing and Uninstalling a Software Patch*. • Updated the instructions in *Accessing the Quartus Prime Installer*.
2023.10.02	23.3	• Major reorganization of topics done in all chapters. • Removed references to Altera FPGA SDK for OpenCL™. • Split the *Downloading and Installing Intel FPGA Software* into separate chapters as *Downloading the Intel FPGA Software* and *Installing the Intel FPGA Software*. • Merged *About Intel FPGA Software License Files* and *Using the Intel FPGA Self-Service Licensing Center* chapters with *Licensing Intel FPGA Software* chapter. • Renamed *Types of Intel FPGA Software Licenses* as *Summary of Altera FPGA Software Licenses Required*. • Renamed *Windows* Subsystem for Linux* (WSL) on Windows Requirements* as *Windows* Subsystem for Linux* (WSL 1) on Windows Requirements for Nios II EDS*. • Merged *Intel Quartus Prime Software License* and *Intel Quartus Prime Lite Edition Software License* topics. • Revised the instructions in *Signing Up for an Evaluation or No-Cost License*.

continued...

Document Version	Quartus Prime Version	Changes
		• Removed the topics *Development Kits Containing the Intel Quartus Prime Software* and *University Program Software Licenses* and merged their information with *Summary of Altera FPGA Software Licenses Required*. • Removed the occurrences of RHEL 7 as Intel no longer supports this version. • Added the following new topics: ☐ *Before You Begin* ☐ *Navigating Content* ☐ *Downloading Design-Specific Software Components* ☐ *Prerequisite Knowledge and Training* ☐ *Acronyms* ☐ *Quartus Prime Design Suite Overview* ☐ *Verifying the Quartus Prime Release* ☐ *Other Intel FPGA Software Requirements* ☐ *Determining the Download Type* ☐ *Downloading Software Using the Quartus Prime Installer* ☐ *Accessing the Installer* ☐ *Getting Started With the Quartus Prime Installer GUI* ☐ *Modifying the Installer Settings* ☐ *Using CLI Commands* ☐ *Downloading Copyleft Licensed Software* ☐ *Downloading Copyleft Licensed Software* ☐ *Downloading Other Intel FPGA Software* ☐ *Installing Intel FPGA Software Through Quartus Prime Software Installer* ☐ *Next Steps After Installing and Licensing the Software* ☐ *Additional Resources* ☐ *Training* ☐ *Troubleshooting License File Issues* ☐ *Troubleshooting Installation Issues*
2023.07.05	23.2	Made minor corrections to the script in *Setting Quartus Prime Environment Variables*.
2023.06.26	23.2	• Revised the instructions in *Quartus Prime Software License* and *Questa* Altera FPGA Edition and Questa* Altera FPGA Starter Edition Software License*. • Updated the image in *Signing Up for an Evaluation or Free License*.
2023.04.03	23.1	• Updated product family name to "Intel Agilex 7." • In *Questa*-Intel FPGA Edition and Questa*-Intel FPGA Starter Edition Software License*, added a note about Questa*-Intel FPGA Edition software not supporting companion licenses.
2023.01.31	22.4	• Added a new appendix to provide detailed instructions for using the Intel FPGA Self-Service Licensing Center. • Moved few licensing center related topics from the *Licensing FPGA Software* chapter to the *Using the Intel FPGA Self-Service Licensing Center* appendix. • Reorganized few topics within the *Licensing FPGA Software* chapter to improve the logical flow.
2022.12.19	22.4	• Maintenance release.

continued...

Document Version	Quartus Prime Version	Changes
2022.09.26	22.3	• Added Ashling RiscFree* IDE for Intel FPGAs in *Software Available in the FPGA Software Download Center*. • Added frequently asked questions on the front page of this guide. • Revised the requirements in *Questa*-Intel FPGA Edition Software Requirements*.
2022.06.21	22.2	• Added a note about upgrading the daemon software for floating license server in *Obtain Necessary Hardware Information*. • Updated an image in the *Request a License File from the Self Service Licensing Center*. • Updated the FLEXlm license manager server software version to 11.18.2.0 in *Upgrading the FLEXlm License Manager Server Software*. • Removed Microsoft Internet Explorer browser and updated its deprecation notice in *Software Requirements*. • Merged the topics *Installing Drivers on Red Hat Enterprise Linux Systems* and *Installing on Red Hat Enterprise Linux 7 or Above* and updated the topic title. • Updated the instructions in *Installing on RHEL 7 or Above and Ubuntu Versions 18 and 20 OS*. • Removed the `lib32ncurses5` software prerequisite package for Ubuntu and replaced it with `libncurses5-dev:i386` in *Software Requirements*. • Added information about specifying Questa*-Intel FPGA Edition software license using the `mgcld` license daemon and included a note about using Questa*-Intel FPGA Edition software license with Remote Desktop in *Questa*-Intel FPGA Edition and Questa*-Intel FPGA Starter Edition Software License*.
2022.03.28	22.1	• Updated the instructions to access memory recommendations in *Minimum Hardware Requirements*. • Added a deprecation notice for Microsoft Internet Explorer version 11.0 in *Software Requirements*. • Updated the instructions to download software from the FPGA Software Download Center in the following topics: ☐ *Downloading and Installing Intel FPGA Software* ☐ *Downloading Individual Executable Files* ☐ *Downloading .tar Files* ☐ *Adding Device Support and Other Intel FPGA Software to Existing Installation* ☐ *Updating Intel FPGA Software* • Made minor command syntax change in the file names in *Installing Intel FPGA Software at the Command Prompt*. • Updated the image of FPGA Self-Service Licensing Center page in *Request a License File from the Self Service Licensing Center*. • Added instructions to generate the license and set up the license in *Questa*-Intel FPGA Edition and Questa*-Intel FPGA Starter Edition Software License*. • Updated the following topics entirely: ☐ *Managing Licensed Users* ☐ *Renewing a License* ☐ *Managing a Computer Profile* ☐ *Transferring a License File to Another Computer* ☐ *Adding Floating Seats* • Updated the license setup image in *Finding the NIC ID in the Intel Quartus Prime Software*. • Updated the software requirements for RHEL 8 in *Software Requirements*.

continued...

Document Version	Quartus Prime Version	Changes
2021.12.13	21.4	• Modified the topic titles from *Questa*-Intel FPGA Edition Software License* and *Intel Quartus Prime Lite Edition and Questa*-Intel FPGA Starter Edition Software License* to *Questa*-Intel FPGA Edition and Questa*-Intel FPGA Starter Edition Software License* and *Intel Quartus Prime Lite Edition Software License*, and merged some of their information. • Updated *Software Requirements* and *Installing Intel FPGA Software at the Command Prompt* topics to suggest that the Intel Quartus Prime software needs administrator privileges on Windows for installation.
2021.10.04	21.3	• Added a note about additional information available for setting up JTAG server connection over SSH in *Installing and Configuring jtagserver on Windows* and *Installing and Configuring a Local JTAG Server (jtagd) on Linux*. • Changed all occurrences of "ModelSim - Intel FPGA Edition" to "Questa - Intel FPGA Edition". • Changed all occurrences of Mentor Graphics to Siemens EDA. • Renamed "Mentor Graphics AXI Verification IP Suite (Intel FPGA Edition)" to "Siemens EDA AXI Verification IP Suite (Intel FPGA Edition)". • Renamed all occurrences of "ModelSim - Intel FPGA Starter Edition" to "Questa*-Intel FPGA Starter Edition". • In *Software Available in the Download Center*, changed MD5 Message Digest Algorithm to SHA-1. • Updated *Intel Quartus Prime Environment Variables* topic with additional information for PATH variable and added a sample script to setup the environment. • In *Questa*-Intel FPGA Edition Software Requirements*, updated the OS requirements. • Updated the licensing details in *Licensing Altera FPGA Software*. • Under *Chapter 4, Licensing FPGA Software Walkthrough* section, removed *Activate Product* topic and updated the instructions in *Request a License File from the Self-service Licensing Center*. • Updated the Ubuntu OS prerequisites in *Software Requirements*. • Updated the details for Questa in *Specifying the License for the Questa*-Intel FPGA Edition Software*. • Changed `alteramtivsim` and `alteramtivlog` to `intelqsim` and `intelqsimstarter` and removed a note about `alteramtivsim` and `alteramtivlog` license subscriptions in *FEATURE and INCREMENT Lines*.
2021.06.21	21.2	• Updated the Red Hat Enterprise Linux versions to 7 and 8 in *Software Requirements* and *Questa* Altera FPGA Edition Software Requirements*. • Added Questa*-Intel FPGA Edition to the list in *Software Available in the Download Center*. • Added a note about the Quartus Prime software installer in *Installing Altera FPGA Software at the Command Prompt*. • Added a note about the `--install-lic` option for `--accept_eula` in *Command-Line Options*. • Added Windows 10 method of starting the Altera FPGA software in *Starting the Quartus Prime Software*. • Removed the topic *Installing on Red Hat Enterprise Linux 6* as RHEL 6 is no longer supported. • Added Windows 10 method of adding device support in *Adding Device Support and Other Altera FPGA Software to Existing Installation*. • Added Windows 10-specific instruction to *Uninstalling on Windows**. • Made minor updates to requirements in *Intel High-Level Synthesis Compiler Software Requirements*. • Updated the license sample in *FEATURE and INCREMENT Lines*.

continued...

Document Version	Quartus Prime Version	Changes
		• Updated the license sample in *Floating Network License Example*. • Updated the license sample in *Fixed PC Software Guard License Example*. • Removed Windows 7 support.
2021.03.29	21.1	• Updated *Installing Windows* Subsystem for Linux* (WSL) on Windows* to include an additional package that is necessary when building a custom library with Nios II EDS. • Updated the requirements in *Minimum Hardware Requirements* and *Software Requirements* topics.
2020.12.14	20.4	• Updated the list of Ubuntu prerequisite packages in *Software Requirements*. • Renamed the topic *Installing and Configuring a JTAG Server* and its subtopics. • Updated the topics *Installing and Configuring jtagserver on Windows* and *Installing and Configuring a Local JTAG Server on Linux* completely. • Document maintenance updates.
2020.09.28	20.3	• Removed the occurrences of SoC EDS from the guide. • Added additional instructions in *Installing Windows Subsystem for Linux (WSL) on Windows*. • Made the following updates in *Software Available in the Download Center*: ☐ Added Intel High Level Synthesis Compiler and Power Thermal Analyzer to the software list. ☐ Included Intel Agilex and eASIC™ N5X devices. ☐ Added a note about discontinuation of older releases. • Removed the topic *Installing on Red Hat Enterprise 4 or Earlier Versions*. • Changed the topic title *Installing on Red Hat Enterprise 5 and 6* to *Installing on Red Hat Enterprise Linux 6* since Red Hat Enterprise Linux 5 version is no longer supported. • Modified description about the license entitlements and removed the mention of Intel FPGA SDK for OpenCL in the additional software list in *Licensing Altera FPGA Software*. • Updated the HLS requirements in *Intel HLS Synthesis Compiler Software Requirements*. • Added SUSE Linux Enterprise Server 12 to the list in *Questa* Altera FPGA Edition Software Requirements*. • Updated the `$QSYS_ROOTDIR` directory to `${installdir}/qsys/bin` in *Quartus Prime Environment Variables*. • Removed Red Hat Linux Enterprise 5 in *Questa* Altera FPGA Edition Software Requirements* and *Software Requirements* topics. • Added an entry for Windows in *Software Requirements*.
2020.07.13	20.2	• Added a note about having multiple license versions running on a system in *Upgrading the FLEXlm License Manager Server Software*. • Removed all occurrences of the `ALTERAD_LICENSE_FILE` environment variable since it is deprecated and should no longer be used.
2020.06.22	20.2	• Updated the processor requirements in *Minimum Hardware Requirements*.

continued...

Document Version	Quartus Prime Version	Changes
2020.04.13	20.1	• Added CPU requirements to *Minimum Hardware Requirements*. • Removed a step about Akamai DLM3 in *Downloading Individual Executable Files* since it is no longer supported. • Updated step 6 (which described download options) completely in *Downloading .tar Files*. • Updated step 5 (which described download options) completely in *Adding Device Support and Other Altera FPGA Software to Existing Installation*. • Updated step 5 (which described download options) completely in *Updating Altera FPGA Software*.
2019.12.16	19.4	• Updated the *Upgrading the FLEXlm License Manager Server Software* topic completely. • Corrected typo in *Installing Windows Subsystem for Linux (WSL) on Windows.* • Minor correction in the title of *Installing Programming Cable Drivers* . • Removed some repetitive topics under *Starting the Quartus Prime Software*.
2019.09.30	19.3	• Added "Checking the IP License Status" topic. • Added footnote for Nios II EDS changes in *Software Available in the Download Center* and added cross-reference links. Added the same information in *Nios II Embedded Design Suite License* topic. • In *Environment Variables* topic, added `PATH` variable section and moved text related to `$QSYS_ROOTDIR` beneath this section. • Updated a note in *FEATURE and INCREMENT lines* about mixed language support. • In *Upgrading the FLEXlm License Manager Server Software* topic, added a link to the License Daemon Downloads page and updated the FLEXlm software version to 11.16.1.0. • Added a new topic *Installing Windows Subsystem for Linux (WSL) on Windows.*
2019.06.27	19.1	• Removed the topic *Set Up a Floating Network License on a Local Computer* since it was outdated. • Renamed "Mentor Graphics* Verification IP (VIP) Altera Edition" to "Mentor Graphics AXI Verification IP Suite (Intel FPGA Edition)". • In *License File Header* topic, updated the license header sample to remove references to Altera. • In *Obtain Necessary Hardware Information*, added Linux OS support for fixed licenses. • Added footnote about Remote Desktop limits to "Specifying the License for the Questa* Altera FPGA Edition Software."
2019.04.01	19.1	• In *Quartus Prime Environment Variables*, added `LC_ALL` variable to the list to explain the Perl warning observed if there is a mismatch in the locale setting. • In *Software Available in the Download Center*, added footnote about RSA Data Security, Inc. MD5 Message-Digest Algorithm. • In *Software Requirements* topic, updated RHEL, Ubuntu, and SUSE package details. • Global changes: ☐ Rebranded Altera URLs to Intel URLs. ☐ Rebranded myAltera to My Intel along with the related screenshots.
2018.09.24	18.1.0	• Added statement that the Quartus Prime software installer does not support spaces in the installation path. • Renamed *Installation Directory* topic to *Installation Path*.

continued...

Document Version	Quartus Prime Version	Changes
2018.05.07	18.0.0	• Reorganized content. • Moved content about getting hardware information to Appendix. • Added how to install USB drivers for Red Hat Enterprise systems. • Added software requirements for Intel High Level Synthesis Compiler. • Added information about specifying more than one licensing server. • Removed references to automatic web license retrieval. • Removed instructions to set environment variables on Linux and Windows. • Created a Licensing Intel FPGA Software Walkthrough topic. • Updated the supported Red Hat Linux version number (5, 6 and 7 instead of 5 and 6).
2018.04.16	17.1.0	Updated licensing information for ModelSim - Intel FPGA Edition software
2017.11.06	17.1.0	• Fixed outdated links from topic: "Using the Self-Service Licensing Center" • Added topics: "Installing Intel FPGA Software at the Command Prompt" and "Downloading and Installing Intel FPGA Software on Multiple Systems" • Removed command-line information from "Additional Installation Procedures"
2017.07.19	17.0.0	• Removed "How to Contact Us" topic
2017.05.08	17.0.0	• Included new free licensing mode for Intel Quartus Prime Pro Edition software, which supports Intel Cyclone 10 GX devices only. • Removed all references to Solaris, which is no longer supported. • Added make as a required Linux library.
2016.10.31	16.1.0	• Removed all references to Talkback feature. • Changed instances of Altera to Intel FPGA.
2016.05.02	16.0.0	Removed instructions on downloading `.iso` files, which are no longer supported.
2015.11.02	15.1.0	Updated to reflect the new Quartus Prime Lite, Standard, and Pro Edition software.
2015.05.04	15.0.0	• Reorganized the document. • Added a quick start for fixed, single user licenses.
2015.03.09	14.1.1	• Updated the supported Red Hat Linux version number (5 and 6 instead of 6 and 7) • Changed recommended screen resolution from 768 x 1024 to 1024 x 768
2014.12.15	14.1.0	• Updated RPM package prerequisite information for Red Hat Linux Enterprise 7 and ModelSim-Altera Edition software and SoC EDS on systems running Red Hat Linux Enterprise 7 • Added Microsoft redistributable package requirements for ModelSim-Altera Edition • Updated FLEXlm license server version to 11.11.1 • Updated available software in the "Altera Software" topic • Removed references to Windows XP and Windows Server 2003 • Updated terminology in the "Adding Floating Seats" topic • Updated instructions for the "Setting Windows Environment Variables" topic • Added a firewall example to the "Modifying the Network License File" topic

continued...

Document Version	Quartus Prime Version	Changes
2014.18.08	14.0a10.0	• Updated RPM package prerequisite information for Red Hat Linux Enterprise 6 and ModelSim-Altera Edition software and SoC EDS on systems running Red Hat Linux Enterprise 6
June, 2014	14.0.0	• Updated "Cables and Ports" topic. • Updated "Upgrading or Setting Up a License Manager Server" topic. • Updated "Quartus II General Environment Variables" topic. • Updated "Download and Installation Prerequisites" topic. • Removed references to 32-bit Quartus II software. • Added "Downloading and Installing Altera Software Updates" topic. • Updated "Rehosting a License" topic. • Updated 32-bit libraries in "Download and Installation Prerequisites" topic. • Updated "Modifying the Network License File" topic.
Nov, 2013	13.1.0	• Added references to Altera SDK for OpenCL in "Altera Software" topic. • Updated information about downloading .tar files in "Installation Options" topic. • Removed note about installing software separately from "Installing Device Family Support" topic.
May, 2013	13.0.0	• Removed references to the TCP/IP protocol in "More Download, Installation, and Setup Procedures" topic. • Removed references to SOPC Builder in "Supported Software Subscriptions" topic. • Removed the "Individual Software Files" topic, and "Using the 64-Bit Version of the Quartus II Software for Linux" topic. • Updated Altera Complete Design Suite DVD information in "Altera Software" topic. • Removed reference to Windows Vista in "Cables and Ports" topic. • Updated downloading, installing, and uninstalling Altera software information in "Altera Download Center" topic. • Updated information about installing the FLEXlm software in "Upgrading the FLEXlm License Manager Server Software" topic.